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CHAPTER 3

DESCRIPTIVE INORGANIC CHEMISTRY: HALOGENS, GROUP IV AND D-BLOCK ELEMENTS

SUGGESTED TIME: 20 PERIODS

IN THIS CHAPTER

This chapter explores the descriptive inorganic chemistry of three important families of the periodic table: the halogens (Group VII), Group IV elements from carbon to lead, and the d-block transition metals of the first series. Learners will master trends in physical and chemical properties, key compounds and their reactions, and the characteristic features of transition metals — variable oxidation states, coloured complexes, catalysis and magnetism — all essential for GCE Advanced Level success.

- By the end of this chapter I can describe the physical states, colours and trends in physical properties of the halogens and relate them to their uses.
- By the end of this chapter I can compare the chemical reactivity of the halogens and explain it using oxidizing power and electronegativity trends.
- By the end of this chapter I can write balanced equations for the reactions of halogens and describe the preparation, properties and tests for hydrogen halides and halide ions.
- By the end of this chapter I can name the oxoacids of the halogens and compare their acid strength, oxidizing strength and thermal stability.
- By the end of this chapter I can describe the trends in physical properties of Group IV elements and explain the unique behaviour and allotropy of carbon.
- By the end of this chapter I can write equations for the reactions of Group IV elements with oxygen, acids and alkalis, and describe the formulae, structures and properties of their hydrides, oxides and chlorides.

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CHAPTER OBJECTIVES

- By the end of this chapter I can describe the physical states, colours and trends in physical properties of the halogens and relate them to their uses.
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- By the end of this chapter I can write equations for the reactions of Group IV elements with oxygen, acids and alkalis, and describe the formulae, structures and properties of their hydrides, oxides and chlorides.
- By the end of this chapter I can write the electronic configurations (spdf and electron-

in-boxes) of the first transition series elements and their ions.

- By the end of this chapter I can explain the characteristic properties of transition metals: variable oxidation states, complex formation, colour, catalytic activity and magnetic behaviour.
- By the end of this chapter I can name coordination complexes, predict their shapes and identify structural and stereoisomerism in complexes.

PREREQUISITES

- Atomic structure and electronic configuration (s, p, d orbitals) from Lower Sixth
- Periodic table organization and periodic trends (atomic radius, ionization energy, electronegativity)
- Chemical bonding: ionic, covalent, dative (coordinate) bonding and molecular shapes (VSEPR)
- Oxidation numbers and redox reactions (oxidizing and reducing agents)
- Acids, bases and qualitative tests for ions from Ordinary Level

1 THE HALOGENS: INTRODUCTION, PHYSICAL PROPERTIES AND COMPARATIVE CHEMISTRY

At the municipal water treatment station in Douala, chlorine is added to kill germs, yet the same element that disinfects our drinking water was once used as a deadly war gas. In the markets of Kribi and Bamenda, iodised table salt is sold to prevent goitre. Why do elements in the same family behave so similarly yet differ so dramatically — fluorine a pale yellow gas, iodine a shiny grey solid? This chapter decodes the chemistry of the halogens, the Group IV elements and the d-block metals that surround us daily. We begin with the halogens, the most reactive family of non-metals in the periodic table.

1.1 The Halogens: Position in the Periodic Table and Electronic Configuration

Look at the right-hand side of the periodic table, one column before the noble gases. There you find five elements that chemists group together because they behave like members of one family: fluorine (F), chlorine (Cl), bromine (Br), iodine (I) and astatine (At). The name halogen comes from Greek words meaning "salt-former", because these elements combine directly with metals to give salts — sodium chloride, our common table salt, being the most famous.

What makes them a family? Write out their outer electronic configurations:

Element	Symbol	Outer electronic configuration
Fluorine	F	$2s^2\,2p^5$
Chlorine	Cl	$3s^2\,3p^5$
Bromine	Br	$4s^2\,4p^5$
Iodine	I	$5s^2\,5p^5$
Astatine	At	$6s^2\,6p^5$

Every halogen is one electron short of a stable noble gas configuration. This single fact explains almost everything about their chemistry: each halogen atom readily gains one electron (forming the ion X^-) or shares one electron (forming a single covalent bond). It also explains why the free elements exist as **diatomic molecules** X_2: two atoms share a pair of electrons in a single covalent bond, and each atom then achieves a complete octet.

DEFINITION **The halogens**

A **halogen** is an element of Group VII (Group 17) of the periodic table, with outer electronic configuration $ns^2\,np^5$. The free elements exist as diatomic molecules X_2 ($X = F, Cl, Br, I$) held together by a single covalent bond. The halogens are fluorine, chlorine, bromine, iodine and astatine.

WARNING **Astatine**

Astatine is radioactive and extremely rare; its chemistry is studied only in tracer amounts. In GCE questions, "the halogens" almost always means F, Cl, Br and I. Do not waste time discussing At unless the question demands it.

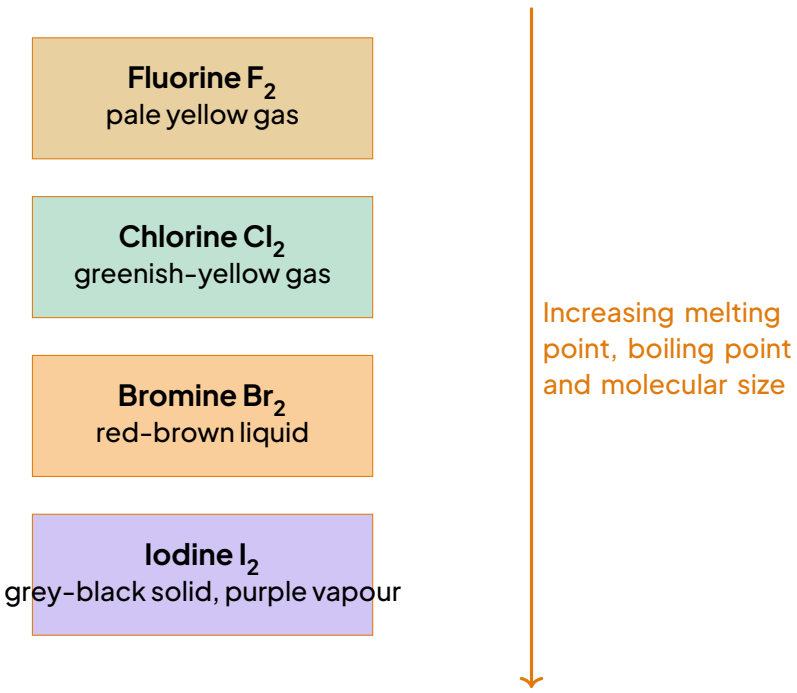
1.2 Colours and Physical States at Room Temperature

If you could line up samples of the four common halogens on a bench (do not try this — they are all toxic!), you would see a beautiful gradation: a pale yellow gas, a greenish-yellow gas, a dark red-brown liquid that fumes, and a grey-black crystalline solid which, when gently warmed, sublimes into a striking purple vapour.

Halogen	Appearance at room temperature	State
F_2	Pale yellow	Gas
Cl_2	Greenish-yellow (yellow-green)	Gas
Br_2	Red-brown (dark red liquid, orange-brown vapour)	Liquid
I_2	Grey-black shiny solid; purple vapour on warming	Solid (sublimes)

WARNING **Colour of element vs colour of solution**

Do not confuse the colour of the halogen vapour with the colour of its solution. Iodine gives a brown solution in water (or in aqueous KI, where it dissolves by forming the tri-iodide ion I_3^-) but a purple/violet solution in organic solvents such as cyclohexane or tetrachloromethane. Similarly, bromine water is orange-yellow, while bromine in cyclohexane is orange-red. Examiners love this trap.



Group VII at room temperature: state and colour change progressively down the group, while melting point, boiling point and molecular size increase.

1.3 Trends in Physical Properties Down Group VII

Melting and boiling points. The change of state from gas (F_2, Cl_2) to liquid (Br_2) to solid (I_2) tells us immediately that melting and boiling points increase down the group. Why? The molecules X_2 are non-polar, so the only attractions between them are **van der Waals forces** (instantaneous dipole–induced dipole forces). Down the group, the number of electrons per molecule increases, the electron cloud becomes larger and more easily polarised, so the temporary dipoles become stronger and the van der Waals attractions increase. More energy is therefore needed to separate the molecules, and the melting and boiling points rise.

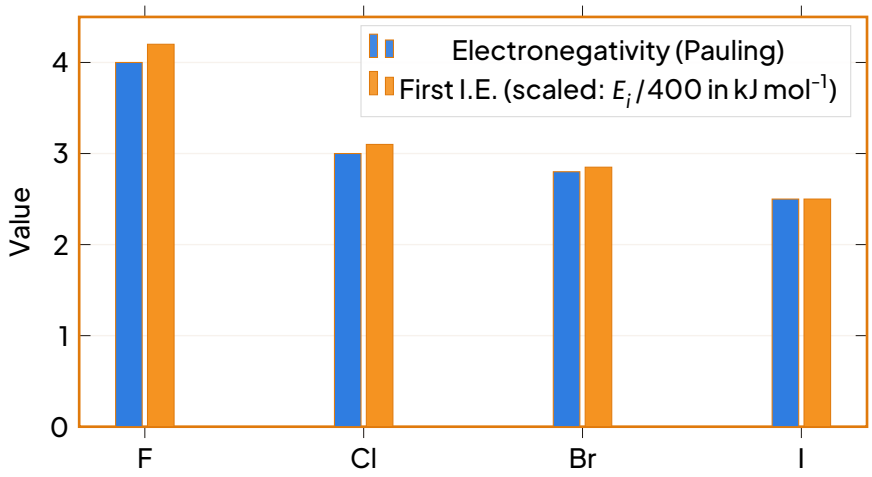
PROPERTY **Trend in melting and boiling points**

Down Group VII, melting and boiling points increase because the strength of the van der Waals forces between the X_2 molecules increases with the size (and number of electrons) of the molecule. (Explanation examinable — learn the three steps: more electrons → more polarisable cloud → stronger van der Waals forces.)

Atomic radius. Down the group each element has one more filled electron shell, so the atomic radius increases steadily: F < Cl < Br < I.

Electronegativity and first ionization energy. Electronegativity (the power of an atom in a molecule to attract the bonding electrons) decreases down the group: fluorine is the most electronegative element of the whole periodic table. The first ionization energy also decreases down the

group, because the outermost electron is further from the nucleus and better shielded by inner shells, so it is held less tightly and is easier to remove.



Electronegativity and first ionization energy both decrease down Group VII (first ionization energies, about 1681, 1251, 1140 and 1008 kJ mol^{-1} for F, Cl, Br, I, are scaled by 1/400 so both quantities fit on one axis).

KEY POINTS **Summary of trends down Group VII**

- Atomic radius: increases (extra electron shells).
- Melting and boiling points: increase (stronger van der Waals forces).
- Electronegativity: decreases.
- First ionization energy: decreases.
- Oxidizing power (reactivity): decreases.

1.4 Comparative Chemistry of the Halogens

Trend in oxidizing power. A halogen reacts by gaining an electron: $X_2 + 2\,e^- \longrightarrow 2\,X^-$. It is therefore an **oxidizing agent**. The more easily the atom gains the electron, the stronger the oxidizing agent. Down the group the atomic radius increases, so the nucleus attracts the incoming electron less strongly (more shielding, greater distance), and the oxidizing power decreases: $F_2 > Cl_2 > Br_2 > I_2$.

PROPERTY **Oxidizing power down the group**

Oxidizing power decreases down Group VII because the atom's ability to attract and gain

an electron weakens as the atomic radius increases and shielding by inner electron shells grows. Fluorine is the strongest oxidizing agent of all the elements.

Evidence 1: displacement reactions. A more reactive (more oxidizing) halogen displaces a less reactive one from a solution of its halide. Bubble chlorine through potassium bromide solution and the colourless solution turns orange-brown — bromine has been set free:

$$Cl_2(aq) + 2\,KBr(aq) \longrightarrow 2\,KCl(aq) + Br_2(aq) \qquad \text{ionic:} \qquad Cl_2 + 2\,Br^- \longrightarrow 2\,Cl^- + Br_2$$

Similarly, chlorine displaces iodine from potassium iodide (solution turns brown, and gives a blue-black colour with starch):

$$Cl_2(aq) + 2\,KI(aq) \longrightarrow 2\,KCl(aq) + I_2(aq)$$

and bromine displaces iodine:

$$Br_2(aq) + 2\,KI(aq) \longrightarrow 2\,KBr(aq) + I_2(aq)$$

But iodine cannot displace bromine or chlorine, and bromine cannot displace chlorine. The order of displacement exactly matches the order of oxidizing power.

METHOD **Comparing halogen reactivity by displacement**

1. Take aqueous solutions of the halides (e.g. KCl, KBr, KI) in separate test tubes.
2. Add a few drops of the halogen water (e.g. chlorine water or bromine water) to each.
3. Add a little cyclohexane, shake, and observe the colour of the organic layer: purple indicates I_2, orange-red indicates Br_2.
4. Conclusion: a colour change means displacement has occurred, so the added halogen is more reactive than the halogen of the halide. No change means it is less reactive.

Evidence 2: reactions with hydrogen. All halogens combine with hydrogen to give hydrogen halides, $H_2 + X_2 \longrightarrow 2\,HX$, but the vigour decreases dramatically down the group:

Halogen	Reaction with H_2
F_2	Explodes violently, even in the dark and in the cold.
Cl_2	Explodes in sunlight (or on ignition); slow in the dark.
Br_2	Combines on heating; moderate speed.
I_2	Reacts only on strong heating, and the reaction is reversible (partial): $H_2 + I_2 \rightleftharpoons 2\,HI$.

The decreasing vigour, and the decreasing thermal stability of the hydrogen halides formed (HF is very stable; HI decomposes easily on heating), confirm that reactivity falls down the group.

Evidence 3: reactions with metals and with phosphorus. Heated metals react with halogens to give halides, the vigour again decreasing down the group. Sodium burns brightly in chlorine to give white fumes of sodium chloride, $2\,Na(s) + Cl_2(g) \longrightarrow 2\,NaCl(s)$. With iron, the comparison is especially instructive: chlorine oxidises iron to iron(III), $2\,Fe(s) + 3\,Cl_2(g) \longrightarrow 2\,FeCl_3(s)$, whereas the weaker oxidant iodine gives only iron(II), $Fe(s) + I_2(s) \longrightarrow FeI_2(s)$. With white phosphorus, excess chlorine gives PCl_5 ($P_4 + 10\,Cl_2 \longrightarrow 4\,PCl_5$) while iodine can only manage PI_3 — again showing chlorine's greater oxidizing strength.

Reaction with water and with alkalis: disproportionation. Chlorine dissolves in water to give "chlorine water", a mixture of two acids:

$$Cl_2(aq) + H_2O(l) \rightleftharpoons HCl(aq) + HOCl(aq)$$

Here chlorine is simultaneously reduced (to Cl^- in HCl, oxidation state −1) and oxidised (to HOCl, oxidation state +1). A reaction in which the same element is both oxidised and reduced is called a **disproportionation**. The hypochlorous acid HOCl is the active germ-killer in water treatment.

With cold dilute sodium hydroxide, chlorine gives chloride and hypochlorite — this is the basis of household bleach:

$$Cl_2(g) + 2\,NaOH(aq) \longrightarrow NaCl(aq) + NaOCl(aq) + H_2O(l) \quad \text{(cold, dilute)}$$

With hot concentrated sodium hydroxide, the products are chloride and chlorate(V):

$$3\,Cl_2(g) + 6\,NaOH(aq) \longrightarrow 5\,NaCl(aq) + NaClO_3(aq) + 3\,H_2O(l) \quad \text{(hot, concentrated)}$$

WORKED EXAMPLE 1 **Displacement and oxidizing power**

Statement. A student adds chlorine water to a solution of potassium iodide, then shakes the mixture with cyclohexane. (a) Describe what is observed. (b) Write the ionic equation for the reaction. (c) Explain, in terms of electron transfer, what this shows about the relative oxidizing powers of chlorine and iodine. (d) Would a reaction occur if iodine water were added to potassium chloride solution? Justify.

SOLUTION

Solution. (a) The aqueous layer turns brown (formation of I_2); after shaking with cyclohexane, the upper organic layer becomes purple/violet.
(b) $Cl_2(aq) + 2\,I^-(aq) \longrightarrow 2\,Cl^-(aq) + I_2(aq)$.
(c) Each iodide ion loses one electron (oxidation): $2\,I^- \longrightarrow I_2 + 2\,e^-$. Each chlorine molecule gains two electrons (reduction): $Cl_2 + 2\,e^- \longrightarrow 2\,Cl^-$. Since chlorine takes electrons from iodide, chlorine is the stronger oxidizing agent; iodine is the weaker one. This agrees with the decrease of oxidizing power down Group VII.
(d) No reaction. Iodine is a weaker oxidizing agent than chlorine, so it cannot remove electrons from Cl^- ions. The mixture would simply show the brown colour of iodine, unchanged.

1.5 Uses of the Halogens

- **Chlorine:** sterilisation of drinking water and swimming pools (as at the Douala treatment station); manufacture of bleach (NaOCl), of hydrochloric acid, of plastics such as PVC, and of solvents.
- **Fluorine compounds:** fluorides (e.g. NaF) in toothpaste to prevent tooth decay; CFCs formerly used as refrigerants and propellants (now phased out because they destroy the ozone layer); the non-stick polymer PTFE (Teflon).
- **Bromine:** silver bromide AgBr is the light-sensitive salt in photographic film; bromine compounds are used as flame retardants.
- **Iodine:** tincture of iodine (iodine in ethanol) as an antiseptic for wounds; iodine added to table salt (as KIO_3 or KI) to prevent goitre — a public-health measure visible in every Cameroonian market.

DID YOU KNOW? **Chlorine: saviour and killer**

Chlorination of drinking water is one of the greatest public-health advances in history, saving millions from cholera and typhoid. Yet chlorine was also the first chemical weapon used on a large scale (Ypres, 1915). The same oxidizing power that destroys bacteria in your water attacks lung tissue when inhaled — a sobering reminder that in chemistry, dose and context decide whether a substance is medicine or poison.

WARNING **Exam tip**

GCE examiners very often ask: "Explain the trend in melting points of the halogens." Always answer in terms of the increasing strength of van der Waals forces as the molecules become larger (more electrons, more polarisable). Never say "the covalent bond gets stronger" — melting I_2 does not break any covalent bond; it only separates the molecules from one another.

EXERCISE 1 **Trends and displacement**

1. Arrange F_2, Cl_2, Br_2, I_2 in order of increasing boiling point and justify your answer.
2. Predict and explain what happens when bromine water is added to (a) potassium chloride solution, (b) potassium iodide solution. Write ionic equations where reaction occurs.
3. Explain why the first ionization energy decreases from fluorine to iodine.

SOLUTION **Exercise 1**

1. $F_2 < Cl_2 < Br_2 < I_2$. Down the group the X_2 molecules contain more electrons and have larger, more polarisable electron clouds, so the van der Waals forces between molecules strengthen and more energy is needed to separate them.
2. (a) No reaction with KCl: bromine is a weaker oxidizing agent than chlorine and cannot displace it. (b) With KI the solution turns brown: $Br_2 + 2\,I^- \longrightarrow 2\,Br^- + I_2$; bromine, being the stronger oxidant, takes electrons from the iodide ions.
3. Down the group the outer electron occupies a shell further from the nucleus and is more strongly shielded by inner shells; the effective attraction on it is weaker, so less energy is needed to remove it.

2 VARIABLE OXIDATION STATES AND REACTIONS OF THE HALOGENS

In the previous section we saw that the halogens are powerful oxidising agents whose strength decreases down the group: fluorine is the most ferocious electron-grabber known, while iodine is comparatively gentle. We now ask two deeper questions. First, if halogens love electrons so much, how can compounds such as $HClO_4$ exist, in which chlorine appears to have lost electrons? Second, how do the halogens actually behave when they meet hydrogen, metals, water or alkali? These two questions — the **variable oxidation states** of the halogens and their **reactions** — form the heart of this section and are examined almost every year at the GCE Advanced Level.

2.1 Oxidation States Shown by the Halogens

The puzzle. In sodium chloride, NaCl, chlorine has clearly gained one electron: its oxidation state is −1. Yet in perchloric acid, $HClO_4$, chlorine is bonded to four oxygens, each more electronegative than itself, and its oxidation state is +7. The same element swings from −1 to +7 — a span of eight units. Understanding why is the key to all halogen oxo-chemistry.

METHOD **Assigning oxidation numbers in oxo-compounds**

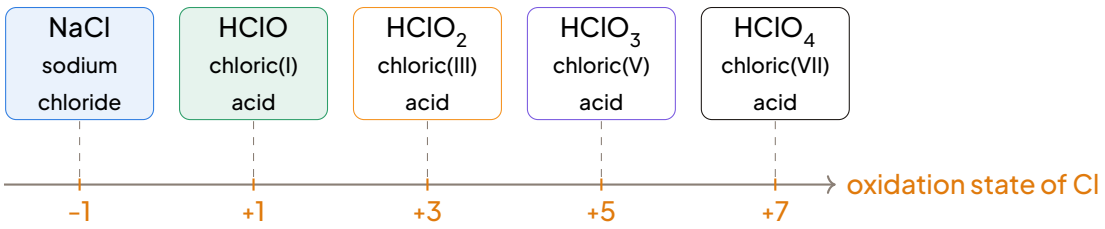
To find the oxidation state of a halogen in an oxo-compound:

1. Assign oxygen its usual value −2 and hydrogen +1.
2. Write that the sum of all oxidation states equals the overall charge of the species (0 for a neutral molecule, the ionic charge for an ion).
3. Solve for the halogen.

Example with $HClO_3$: $+1 + x + 3(-2) = 0$, so $x = +5$. Chlorine is in oxidation state +5. Example with ClO_2^-: $x + 2(-2) = -1$, so $x = +3$.

Applying this method to the chlorine oxoacids and their anhydrides gives the full ladder of oxidation states:

Compound	Oxidation state of Cl	Name (Stock notation)
NaCl	−1	Sodium chloride
HClO (or Cl_2O)	+1	Chloric(I) acid (hypochlorous acid)
$HClO_2$	+3	Chloric(III) acid (chlorous acid)
$HClO_3$	+5	Chloric(V) acid (chloric acid)
$HClO_4$	+7	Chloric(VII) acid (perchloric acid)



The ladder of chlorine oxidation states, from −1 in NaCl to +7 in $HClO_4$.

2.2 Why Fluorine Shows Only −1

PROPERTY **Fluorine: only −1**

Fluorine never shows positive oxidation states. Two reasons combine:

1. It is the **most electronegative element** in the Periodic Table ($\chi = 4.0$), so in every compound the bonding electrons are assigned to fluorine.
2. Being in Period 2, it has no accessible d-orbitals; its octet cannot be expanded, so it can never form more than one covalent bond and can never be oxidised beyond −1 (its only states are 0 in F_2 and −1 in compounds).

Why Cl, Br and I can be positive. Chlorine ($[Ne]3s^2 3p^5$), bromine and iodine possess **empty d-orbitals** of the valence shell (3*d* for Cl, 4*d* for Br, 5*d* for I). When bonded to a more electronegative element such as oxygen or fluorine, paired *ns* and *np* electrons can be promoted into these empty *nd* orbitals, unpairing up to seven electrons and allowing 1, 3, 5 or 7 covalent bonds. Hence the oxidation states +1, +3, +5, +7 appear. The energy cost of promotion is repaid by the

extra bonds formed — but only with very electronegative partners, which is why the positive states occur in oxides, oxoacids and interhalogens, never with metals.

WARNING **Common mistake**

Do not write that fluorine shows +7 "like chlorine". Fluorine has no *d*-orbitals (Period 2) and is more electronegative than oxygen, so FO bonds still assign electrons to F. Its maximum oxidation state in any compound is −1.

2.3 Reaction of the Halogens with Hydrogen

All halogens combine with hydrogen to give the hydrogen halides HX, but the vigour of the reaction falls dramatically down the group — a beautiful illustration of decreasing oxidising power:

$H_2(g) + F_2(g) \longrightarrow 2\,HF(g)$	explosive, even in the dark and cold
$H_2(g) + Cl_2(g) \longrightarrow 2\,HCl(g)$	explosive in sunlight (photochemical), slow in the dark
$H_2(g) + Br_2(g) \rightleftharpoons 2\,HBr(g)$	needs heat (~200 °C) and a Pt catalyst; equilibrium
$H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)$	slow, reversible, even on heating; equilibrium lies far left

Because the H – X bond weakens as the halogen atom grows larger (bond enthalpy: H – F 562, H – Cl 431, H – Br 366, H – I 299 kJ mol^{-1}), the **thermal stability** of the hydrogen halides decreases down the group: HF and HCl resist heat, HBr decomposes partially on strong heating, and HI decomposes appreciably into H_2 and I_2 (notice the violet fumes) when a hot glass rod is held in the gas.

2.4 Reaction with Metals and Non-metals

Metals. Halogens oxidise metals to **ionic halides**; the more vigorous the halogen, the higher the oxidation state of the metal product:

$2\,Na(s) + Cl_2(g) \longrightarrow 2\,NaCl(s)$	
$2\,Fe(s) + 3\,Cl_2(g) \longrightarrow 2\,FeCl_3(s)$	excess chlorine
$2\,Fe(s) + I_2(g) \longrightarrow 2\,FeI_2(s)$	iodine is weaker, stops at Fe(II)
$2\,Al(s) + 3\,Cl_2(g) \longrightarrow 2\,AlCl_3(s)$	sublimes as dimer Al_2Cl_6

Note that chlorine, a strong oxidant, drives iron to its higher state, iron(III). With the weaker iodine, iron gives only FeI_2 — direct evidence of the oxidising-power trend.

Non-metals. With phosphorus, chlorine gives PCl_3 (limited chlorine) or PCl_5 (excess chlorine); sulphur gives SCl_2:

$$2\,P(s) + 3\,Cl_2(g) \longrightarrow 2\,PCl_3(l) \qquad 2\,P(s) + 5\,Cl_2(g) \longrightarrow 2\,PCl_5(s) \qquad S(s) + Cl_2(g) \longrightarrow SCl_2(l)$$

2.5 Chlorine and Water: the Bleaching Reaction

Chlorine dissolves slightly in water (~0.09 mol dm^{-3} at 298 K, 1 atm) and reacts reversibly:

$$Cl_2(g) + H_2O(l) \rightleftharpoons HCl(aq) + HClO(aq) \quad K_c \approx 4 \times 10^{-4}$$

Look at the oxidation states: chlorine goes from 0 in Cl_2 to −1 in HCl and to +1 in HClO. One and the same element is reduced and oxidised in the same reaction.

DEFINITION **Disproportionation**

A **disproportionation** is a redox reaction in which atoms of the same element are simultaneously oxidised and reduced.

The chloric(I) acid HClO is the active **bleaching agent**: it oxidises coloured organic dyes to colourless products, itself being reduced:

$$HClO(aq) + \text{dye} \longrightarrow HCl(aq) + (\text{dye} + O)\,(\text{colourless})$$

This is why moist litmus paper is bleached by chlorine (first red, then white), and why chlorine water disinfects the drinking water of many Cameroonian towns.

2.6 Reaction with Alkali: Cold Dilute versus Hot Concentrated

Chlorine disproportionates in sodium hydroxide, but the product depends on conditions:

$$Cl_2(g) + 2\,NaOH(aq) \longrightarrow NaCl(aq) + NaClO(aq) + H_2O(l) \qquad \text{cold, dilute}$$
$$3\,Cl_2(g) + 6\,NaOH(aq) \longrightarrow 5\,NaCl(aq) + NaClO_3(aq) + 3\,H_2O(l) \qquad \text{hot, concentrated}$$

In cold alkali, chlorine goes from 0 to −1 (NaCl) and +1 (NaClO); in hot alkali the chlorate(I) itself disproportionates further ($3ClO^- \rightarrow 2Cl^- + ClO_3^-$), giving chlorate(V), $NaClO_3$ (+5).

Bromine and iodine behave similarly, but the intermediate hypohalites (BrO^-, IO^-) are unstable even at room temperature and disproportionate rapidly to halide and halate(V):

$$3\,Br_2 + 6\,OH^- \longrightarrow 5\,Br^- + BrO_3^- + 3\,H_2O \quad \text{(even cold)}$$

WARNING **Hot alkali trap**

A very common mistake is to write $Cl_2 + 2\,NaOH \longrightarrow NaCl + NaClO + H_2O$ for hot concentrated alkali. Hot conditions give the chlorate(V): $3\,Cl_2 + 6\,NaOH \longrightarrow 5\,NaCl + NaClO_3 + 3\,H_2O$. Read the question's conditions before writing the equation!

KEY POINTS **Exam tip**

Always balance disproportionation equations carefully and show the oxidation-state changes. GCE examiners award separate marks for correct stoichiometry and for identifying the species oxidised and reduced.

2.7 Displacement of Halides and the Thiosulfate Reaction

A more reactive halogen displaces a less reactive one from its halide solution — the classic test for halide ions and the basis of the redox series:

$$Cl_2(aq) + 2\,KBr(aq) \longrightarrow 2\,KCl(aq) + Br_2(aq)$$ orange/brown colour appears in organic layer

$$Cl_2(aq) + 2\,KI(aq) \longrightarrow 2\,KCl(aq) + I_2(aq)$$ brown solution / black solid

$$Br_2(aq) + 2\,KI(aq) \longrightarrow 2\,KBr(aq) + I_2(aq)$$

The liberated iodine can be estimated quantitatively by titration with sodium thiosulfate — a classic practical reaction:

$$I_2(aq) + 2\,Na_2S_2O_3(aq) \longrightarrow 2\,NaI(aq) + Na_2S_4O_6(aq)$$

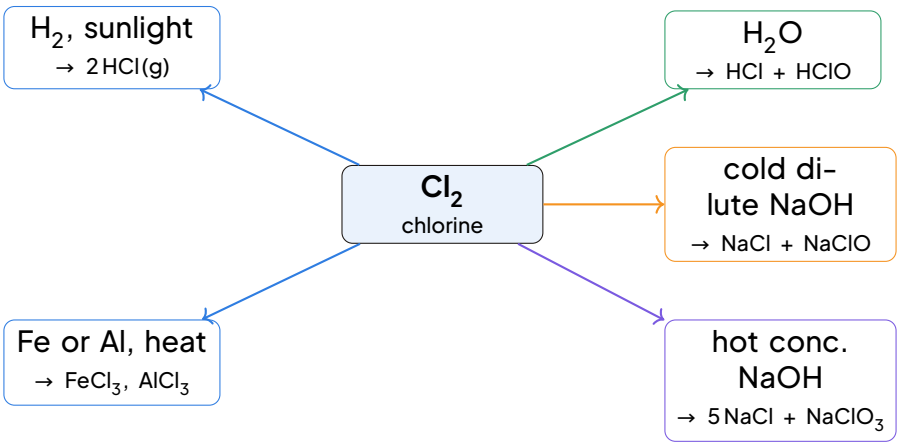
Here iodine is reduced to iodide (0 → -1) while thiosulfate is oxidised to tetrathionate (+2 → +2.5 average). Starch indicator (deep blue-black with I_2) is added near the end-point, which is the sharp disappearance of the blue colour.

WORKED EXAMPLE 2 **Chlorine and hot alkali**

Statement. Chlorine is bubbled into hot concentrated sodium hydroxide. (a) Write the balanced equation. (b) State the oxidation number of chlorine in each chlorine-containing species. (c) Identify the type of redox reaction.

SOLUTION

Solution. (a) $3\,Cl_2(g) + 6\,NaOH(aq) \longrightarrow 5\,NaCl(aq) + NaClO_3(aq) + 3\,H_2O(l)$.
(b) In Cl_2: 0. In NaCl: -1. In $NaClO_3$: $+1 + x + 3(-2) = 0$, so $x = +5$.
(c) Since chlorine is simultaneously reduced (0 → -1) and oxidised (0 → +5), this is a disproportionation. Check the electron balance: five Cl atoms gain one electron each ($5e^-$), one Cl atom loses five electrons ($5e^-$) — consistent.



Summary flow diagram of the principal reactions of chlorine.

EXERCISE 2 **Oxidation states and displacement**

1. Determine the oxidation state of the halogen in: KBr, Br_2O, KIO_3, HIO_4.
2. Explain, in terms of electronic structure, why iodine can show +7 but fluorine cannot.
3. Predict and explain what is observed when chlorine water is added to potassium iodide solution, and write the ionic equation.
4. Balance: $Br_2 + NaOH \longrightarrow NaBr + NaBrO_3 + H_2O$ (hot conditions).

SOLUTION **Exercise 1**

1. KBr: −1; Br_2O: +1; KIO_3: +5; HIO_4: +7.
2. Iodine ($[Kr]5s^2 5p^5$) has empty 5*d* orbitals into which paired electrons can be promoted, unpairing up to seven electrons for bonding with oxygen. Fluorine (Period 2) has no *d*-orbitals and is the most electronegative element, so it can never be assigned a positive oxidation state.
3. The solution turns brown (black solid on standing): chlorine oxidises iodide to iodine. $Cl_2(aq) + 2\,I^-(aq) \longrightarrow 2\,Cl^-(aq) + I_2(aq)$.
4. $3\,Br_2 + 6\,NaOH \longrightarrow 5\,NaBr + NaBrO_3 + 3\,H_2O$.

3 HYDROGEN HALIDES, HALIDE TESTS AND OXOACIDS OF THE HALOGENS

In the previous section we saw that the halogens are powerful oxidising agents whose strength decreases down the group, and that they display variable oxidation states in their compounds. We now turn to two important families of halogen compounds: the **hydrogen halides** HX, in which the halogen has oxidation state −1, and the **oxoacids**, in which the halogen shows positive oxidation states up to +7. These compounds are not only examination favourites; they are also the basis of the qualitative analysis tests for halide ions that you will meet in the practical paper.

3.1 Preparation of the Hydrogen Halides

Action of concentrated sulphuric acid on metal halides

The standard laboratory preparation of a hydrogen halide is the action of concentrated sulphuric acid on a solid metal halide. Concentrated H_2SO_4 is chosen because it is a non-volatile acid (boiling point about 338 °C), so it displaces the more volatile hydrogen halide from its salt:

$$NaCl(s) + H_2SO_4(l) \longrightarrow NaHSO_4(s) + HCl(g)$$

On heating with excess NaCl:

$$NaCl(s) + NaHSO_4(s) \longrightarrow Na_2SO_4(s) + HCl(g)$$

$$CaF_2(s) + H_2SO_4(l) \longrightarrow CaSO_4(s) + 2\,HF(g)$$

The misty fumes of HCl or HF are collected by upward delivery (they are denser than air) or dissolved in water to give the aqueous acid. Note that the preparation of HF is carried out in a lead or platinum apparatus, because HF attacks glass (it reacts with SiO_2 to form SiF_4 and H_2SiF_6).

This method, however, **fails for HBr and HI**. The reason is chemical, not practical: concentrated sulphuric acid is itself an oxidising agent, and bromide and iodide ions are good enough reducing agents to be oxidised by it. With sodium bromide, the HBr first formed is partly oxidised:

$$NaBr(s) + H_2SO_4(l) \longrightarrow NaHSO_4(s) + HBr(g)$$

$$2\,HBr(g) + H_2SO_4(l) \longrightarrow Br_2(g) + SO_2(g) + 2\,H_2O(l)$$

The product is therefore contaminated with brown Br_2 vapour and colourless, choking SO_2. With sodium iodide the oxidation goes even further, because I^- is the strongest reducing agent of the halides; a mixture of reduction products (SO_2, S and even H_2S) is obtained:

$$8\,HI(g) + H_2SO_4(l) \longrightarrow 4\,I_2(s) + H_2S(g) + 4\,H_2O(l)$$

WARNING **A classic trap**

Concentrated H_2SO_4 cannot be used to prepare pure HBr or HI, because it oxidises Br^- and I^- to Br_2 and I_2. In the examination, if you are asked to "prepare pure hydrogen bromide", do not write the sulphuric acid method.

Alternative preparations of HBr and HI

To avoid oxidation we replace concentrated sulphuric acid by **phosphoric(V) acid**, H_3PO_4, which is non-volatile but not an oxidising agent:

$$NaBr(s) + H_3PO_4(l) \longrightarrow NaH_2PO_4(s) + HBr(g)$$

$$NaI(s) + H_3PO_4(l) \longrightarrow NaH_2PO_4(s) + HI(g)$$

Alternatively, HBr and HI can be prepared by the hydrolysis of phosphorus trihalides, made in situ from red phosphorus and the halogen:

$$2\,P(s) + 3\,Br_2(l) \longrightarrow 2\,PBr_3(l)$$
$$PBr_3(l) + 3\,H_2O(l) \longrightarrow H_3PO_3(aq) + 3\,HBr(g)$$

$$2\,P(s) + 3\,I_2(s) \longrightarrow 2\,PI_3(s)$$
$$PI_3(s) + 3\,H_2O(l) \longrightarrow H_3PO_3(aq) + 3\,HI(g)$$

3.2 Physical Properties and Acidic Strength of the Hydrogen Halides

All the hydrogen halides are **colourless gases** at room temperature which fume in moist air (the gas dissolves in atmospheric moisture to form tiny droplets of acid). They are **extremely soluble in water**; this is beautifully demonstrated by the **fountain experiment**.

EXPERIMENT **The fountain experiment**

A dry round-bottomed flask is filled with HCl gas and fitted with a jet tube dipping into water containing blue litmus. A few drops of water are introduced into the flask; the HCl dissolves so readily that the pressure inside drops sharply, and water is driven up through the jet as a spectacular fountain. The litmus turns red, showing that an acidic solution (hydrochloric acid) has formed. The same experiment works with HBr and HI.

In water the hydrogen halides behave as acids: $HX(aq) + H_2O(l) \longrightarrow H_3O^+(aq) + X^-(aq)$. Their strengths differ strikingly:

PROPERTY **Acid strength of the hydrogen halides**

Aqueous acid strength increases down the group:

$$HF \ll HCl < HBr < HI$$

HF is a weak acid ($K_a \approx 6.8 \times 10^{-4}$); HCl, HBr and HI are strong (fully ionised). The explanation is the strength of the H – X bond: as the halogen atom becomes larger down the group, the bonding pair is further from the hydrogen nucleus, the bond becomes longer and weaker, and H^+ is released more easily. The H – F bond is short and very strong (bond dissociation energy $\approx$ 565 kJ mol^{-1}), so HF ionises only slightly. (This proof is qualitative and is frequently demanded in "explain" questions.)

WARNING **Do not confuse the two trends**

Down Group VII, the oxidising power of the halogens X_2 decreases, but the acid strength and the reducing power of the hydrogen halides HX increase. Writing "HF is the strongest acid because fluorine is the most reactive halogen" is a serious error that costs marks every year.

Reducing power of the hydrogen halides

The ease with which X^- loses electrons increases down the group, so the reducing power increases: HF < HCl < HBr < HI. This is exactly what we exploited above: HI reduces concentrated H_2SO_4 all the way to H_2S (sulphur reduced from +6 to −2), HBr reduces it only to SO_2 (+6 to +4), while HCl and HF do not reduce it at all.

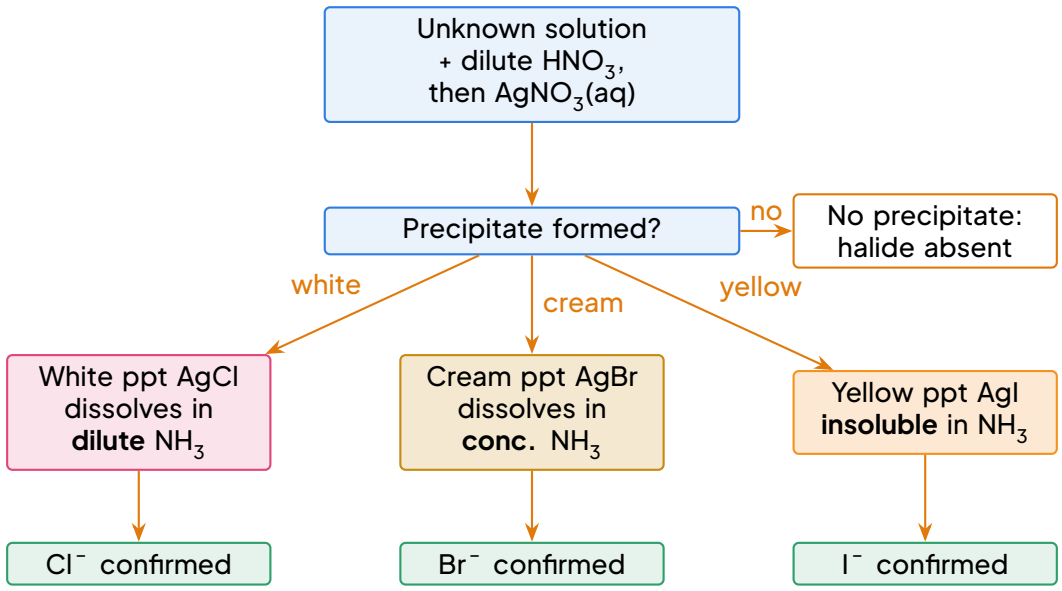
3.3 Qualitative Tests for Halide Ions

Identifying Cl^-, Br^- and I^- in an unknown solution is a compulsory practical skill. The key reagent is **silver nitrate solution**, $AgNO_3(aq)$, because the three silver halides have characteristic colours and different solubilities in ammonia solution.

METHOD **Testing for halide ions**

1. To about 2 cm^3 of the unknown solution, add dilute nitric acid, HNO_3, to destroy interfering ions such as CO_3^{2-} (which would also give a precipitate with Ag^+).
2. Add a few drops of $AgNO_3(aq)$. A precipitate indicates a halide:
   - white precipitate ⇒ Cl^-: $Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$
   - cream precipitate ⇒ Br^-: $Ag^+(aq) + Br^-(aq) \longrightarrow AgBr(s)$
   - yellow precipitate ⇒ I^-: $Ag^+(aq) + I^-(aq) \longrightarrow AgI(s)$

3. Confirm by adding ammonia solution: AgCl dissolves in dilute $NH_3(aq)$, AgBr dissolves only in concentrated $NH_3(aq)$, and AgI is insoluble even in concentrated ammonia. Dissolution is due to formation of the soluble complex $[Ag(NH_3)_2]^+$.



Flowchart for the identification of Cl^-, Br^- and I^- using silver nitrate and ammonia solutions.

Confirmatory tests

- **Concentrated H_2SO_4 on the solid salt.** A chloride gives only misty fumes of HCl; a bromide gives misty fumes plus brown Br_2 vapour; an iodide gives violet I_2 vapour and the rotten-egg smell of H_2S. The deeper the oxidation, the heavier the halide.
- **Test for chlorine gas.** Chlorine bleaches damp litmus paper (it turns red first, then white), because Cl_2 forms the bleaching agent HOCl with the moisture: $Cl_2(g) + H_2O(l) \longrightarrow HCl(aq) + HOCl(aq)$. Note: the paper must be damp; dry chlorine does not bleach.

WORKED EXAMPLE 3 **Identifying an unknown halide**

A white solid P dissolves in water. Addition of dilute HNO_3 followed by $AgNO_3(aq)$ gives a cream precipitate which does not dissolve in dilute ammonia but dissolves in concentrated ammonia. When concentrated H_2SO_4 is added to solid P, brown fumes are evolved. Identify the anion in P and write equations for the reactions observed.

SOLUTION

Solution. The cream precipitate with $AgNO_3$ suggests AgBr, hence the anion is the bro-

mide ion, Br^-. Confirmation: AgBr is insoluble in dilute NH_3 but soluble in concentrated NH_3 (formation of $[Ag(NH_3)_2]^+$), exactly as observed. The brown fumes with concentrated H_2SO_4 are Br_2, formed by oxidation of Br^-.
Equations:

$$Ag^+(aq) + Br^-(aq) \longrightarrow AgBr(s)$$
$$AgBr(s) + 2\,NH_3(aq) \longrightarrow [Ag(NH_3)_2]^+(aq) + Br^-(aq)$$
$$2\,Br^- + 2\,H_2SO_4 \longrightarrow Br_2 + SO_2 + SO_4^{2-} + 2\,H_2O$$

Conclusion: P contains the bromide ion, e.g. NaBr or KBr.

3.4 Oxoacids of the Halogens

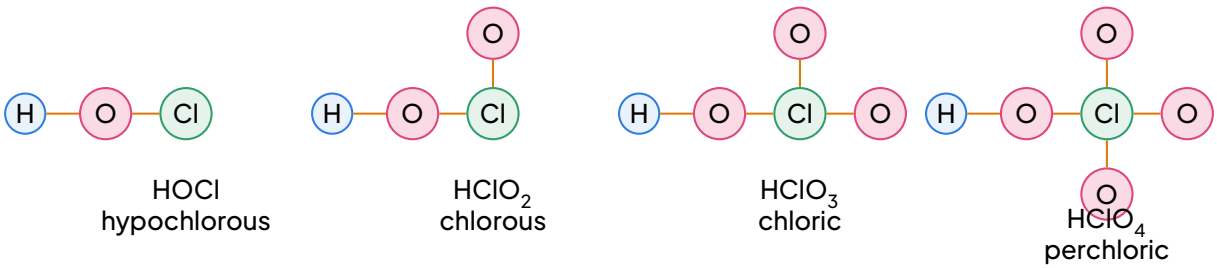
DEFINITION **Oxoacid**

An **oxoacid** is an acid containing hydrogen, oxygen and another element. The halogen oxoacids have the general formula HXO_n with n = 1 to 4, in which the halogen has a positive oxidation state.

For chlorine the four oxoacids are:

Formula	Name	Oxidation state of Cl	Salt name
HOCl (or HClO)	hypochlorous acid	+1	hypochlorite, ClO^-
$HClO_2$	chlorous acid	+3	chlorite, ClO_2^-
$HClO_3$	chloric acid	+5	chlorate, ClO_3^-
$HClO_4$	perchloric acid	+7	perchlorate, ClO_4^-

The naming pattern is systematic: the lowest oxoacid takes the prefix hypo- and suffix -ous; the next is simply -ous; then -ic; and the highest takes per- and -ic. Bromine and iodine form analogous series (e.g. $HBrO_3$ bromic acid, HIO_4 periodic acid), though not all members are stable.



Structures of the four oxoacids of chlorine: the acidic hydrogen is always bonded to an oxygen atom, and the number of oxygen atoms coordinated to chlorine increases from one to four. (Double bonds to oxygen are omitted for clarity; the actual geometry around Cl is tetrahedral.)

WARNING **Where is the acidic hydrogen?**

In all halogen oxoacids the ionisable hydrogen is attached to oxygen (H – O – X), never directly to the halogen. Writing HClO with the H bonded to Cl is a structural error.

3.5 Trends in the Oxoacids: Acid Strength, Oxidising Strength and Thermal Stability

PROPERTY **Trends in the halogen oxoacids**

For oxoacids of the same halogen:

- Acid strength increases with the number of oxygen atoms:

$$HOCl < HClO_2 < HClO_3 < HClO_4$$

  HOCl is a weak acid ($K_a \approx 3.0 \times 10^{-8}$) while $HClO_4$ is one of the strongest acids known. Each additional oxygen atom withdraws electron density from the O – H bond (through the electronegative chlorine), weakening it and stabilising the conjugate anion by delocalising the negative charge over more oxygen atoms.

- Oxidising strength decreases as the number of oxygen atoms increases: HOCl is the most powerful oxidising agent of the series, $HClO_4$ (when cold and dilute) the least. The lower oxoacids contain chlorine in a low positive oxidation state with few stabilising oxygens, so they gain electrons readily.

- Thermal stability increases with the number of oxygen atoms: HOCl decomposes easily ($2\,HOCl \longrightarrow 2\,HCl + O_2$, accelerated by light), whereas $HClO_4$ is the most stable, though it becomes a dangerously powerful oxidant when hot and concentrated.

KEY POINTS **Exam tip: the oxoacid mnemonic**

"More oxygens, stronger acid." Acid strength and thermal stability rise from HOX to HXO_4, while oxidising strength falls. The lower oxoacids (HOCl, $HClO_2$) are the strongest oxidising agents; the higher ones ($HClO_4$) are the strongest acids. Learn the two opposite trends together so you never swap them.

Uses of the oxoacids and their salts

- **Sodium hypochlorite**, NaClO, is the active ingredient of household bleach and of many water-treatment products; it kills germs precisely because ClO^- is a strong oxidising agent. In Cameroon, dilute hypochlorite solutions are routinely used to disinfect drinking water.
- **Chlorates**, e.g. $KClO_3$, are used in match heads and fireworks, where they supply oxygen for rapid combustion.
- **Perchlorates**, e.g. $KClO_4$ and NH_4ClO_4, are used in explosives, pyrotechnics and solid rocket propellants.

DID YOU KNOW? **Bleach and the chemistry of clean water**

The bleaching power of "eau de Javel" sold in every Cameroonian market comes from the hypochlorite ion ClO^-, the conjugate base of hypochlorous acid HOCl — the very species formed when chlorine dissolves in water. The same chemistry that bleaches damp litmus paper in your test for chlorine disinfects wells and hospital surfaces. Never mix bleach with acids: the reaction $ClO^- + Cl^- + 2\,H^+ \longrightarrow Cl_2 + H_2O$ releases toxic chlorine gas!

WORKED EXAMPLE 4 **Oxidation states and acid strength**

(a) Determine the oxidation state of chlorine in HClO, $HClO_2$, $HClO_3$ and $HClO_4$. (b) Arrange the acids in order of increasing acid strength and justify your answer. (c) Which of them is the strongest oxidising agent?

SOLUTION

Solution. (a) Using $x + 1 + n(-2) = 0$ for $HClO_n$:

$$\text{HClO}: \; x = +1 \qquad \text{HClO}_2: \; x = +3$$
$$\text{HClO}_3: \; x = +5 \qquad \text{HClO}_4: \; x = +7$$

(b) Increasing acid strength: $HClO < HClO_2 < HClO_3 < HClO_4$. Justification: each extra oxygen atom withdraws electron density from the O – H bond, weakening it, and stabilises the anion ClO_n^- by spreading the negative charge over more oxygen atoms.

(c) HClO (hypochlorous acid) is the strongest oxidising agent: oxidising strength decreases as the number of oxygen atoms increases.

EXERCISE 3 **Preparation and identification**

(a) Explain, with equations, why pure HI cannot be prepared by the action of concentrated H_2SO_4 on NaI, and describe a suitable alternative preparation.
(b) A solution gives a yellow precipitate with $AgNO_3(aq)$ after acidification with dilute HNO_3; the precipitate is insoluble in concentrated ammonia. Identify the ion present and write the precipitation equation.
(c) Arrange HBrO, $HBrO_3$ and $HBrO_4$ in order of (i) increasing acid strength, (ii) increasing oxidising strength.

SOLUTION **Exercise 1**

(a) I^- is a strong reducing agent; concentrated H_2SO_4 oxidises it: $8\,HI + H_2SO_4 \longrightarrow 4\,I_2 + H_2S + 4\,H_2O$, so the HI is contaminated by I_2, SO_2, S and H_2S. Alternative: use non-oxidising phosphoric(V) acid, $NaI + H_3PO_4 \longrightarrow NaH_2PO_4 + HI$, or hydrolyse PI_3: $PI_3 + 3\,H_2O \longrightarrow H_3PO_3 + 3\,HI$.
(b) The ion is I^-: $Ag^+(aq) + I^-(aq) \longrightarrow AgI(s)$ (yellow, insoluble in NH_3).
(c) (i) Acid strength: $HBrO < HBrO_3 < HBrO_4$. (ii) Oxidising strength: $HBrO_4 < HBrO_3 < HBrO$ (the reverse order).

KEY POINTS **Key points of this section**

- HCl and HF are prepared from metal halides with concentrated H_2SO_4; HBr and HI require H_3PO_4 or hydrolysis of PX_3, because H_2SO_4 oxidises Br^- and I^-.
- Acid strength: HF < HCl < HBr < HI (weakening H – X bond); reducing power increases in the same order.
- Halide test: dilute HNO_3, then $AgNO_3$; AgCl white (soluble in dilute NH_3), AgBr cream (soluble in conc. NH_3), AgI yellow (insoluble). Chlorine bleaches damp litmus paper.
- Oxoacids HXO_n: hypochlorous (+1), chlorous (+3), chloric (+5), perchloric (+7). More oxygens ⇒ stronger acid, more thermally stable, but weaker oxidising agent.
- Uses: NaClO in bleach, chlorates in matches and fireworks, perchlorates in explosives.

4 GROUP IV ELEMENTS: INTRODUCTION, ALLOTROPY AND TRENDS IN PHYSICAL PROPERTIES

In the previous section we completed our study of the halogens by examining the hydrogen halides, the tests for halide ions and the oxoacids of chlorine, bromine and iodine. We now leave Group VII and move two steps to the left of the periodic table, to **Group IV** (also called Group 14). This group is remarkable for one reason above all: it contains **carbon**, the element on which all life and all organic chemistry are built. Yet as we descend the group we meet silicon, the element of computer chips and glass, then germanium, tin and lead — and the character of the elements changes completely from non-metal to metal. Understanding why this happens is the heart of this section.

4.1 The Group IV Elements and Their Electronic Configuration

Group IV occupies the central p-block column of the periodic table, between Group III (B, Al, …) and Group V (N, P, …). Its five elements are carbon (C), silicon (Si), germanium (Ge), tin (Sn) and lead (Pb).

DEFINITION **Group IV elements**

The Group IV elements are the elements of the fourth main group of the periodic table: C, Si, Ge, Sn and Pb. Every Group IV atom has the outer electronic configuration ns^2np^2, i.e. four valence electrons.

Element	Symbol	Outer electronic configuration
Carbon	C	$2s^2 2p^2$
Silicon	Si	$3s^2 3p^2$
Germanium	Ge	$4s^2 4p^2$
Tin	Sn	$5s^2 5p^2$
Lead	Pb	$6s^2 6p^2$

With four valence electrons, each atom can form **four covalent bonds** (as in CH_4 or $SiCl_4$), giving the maximum oxidation state of +4. The alternative oxidation state, +2, appears when only the two *np* electrons are used in bonding — we shall return to this crucial idea when we discuss the **inert pair effect**.

4.2 The Unique Properties of Carbon

Carbon behaves very differently from the rest of its group. Four features explain this uniqueness:

- **Small atomic size.** Carbon is the smallest member of the group. Its atoms can approach each other closely, so its 2*p* orbitals overlap effectively sideways to form strong π bonds. Hence carbon readily forms **multiple bonds**: C = C, C ≡ C, C = O. Silicon and the heavier elements cannot do this efficiently because their larger 3*p*, 4*p* … orbitals overlap poorly side-

ways.

- **Catenation.** Carbon shows **catenation** — the ability to form long chains and rings of identical atoms — to a unique extent. The C – C single bond is very strong (about 348 kJ mol^{-1}), stronger than Si – Si (about 226 kJ mol^{-1}), so carbon chains are stable. This is why millions of organic compounds exist.
- **High electronegativity.** Carbon ($\chi \approx 2.5$ on the Pauling scale) is the most electronegative member of the group, so it forms essentially covalent compounds and never forms a simple C^{4+} ion.
- **Maximum covalency of four.** Carbon has no available *d* orbitals in its valence shell ($n = 2$), so it can never expand its octet: CCl_4 does not hydrolyse, whereas $SiCl_4$ hydrolyses violently because silicon can use its empty 3*d* orbitals to accept a water molecule.

WARNING **Common mistake**

Do not write that carbon forms C^{4+} ions. The energy required to remove four electrons is far too large; carbon is always covalent in its compounds.

4.3 Allotropy of Carbon: Diamond, Graphite and Fullerenes

DEFINITION **Allotropy**

Allotropy is the existence of an element in two or more different structural forms, called allotropes, in the same physical state.

Carbon provides the classic example of allotropy: **diamond** and **graphite** are both pure solid carbon, yet diamond is the hardest natural substance known while graphite is soft enough to write with. The difference lies entirely in the bonding arrangement.

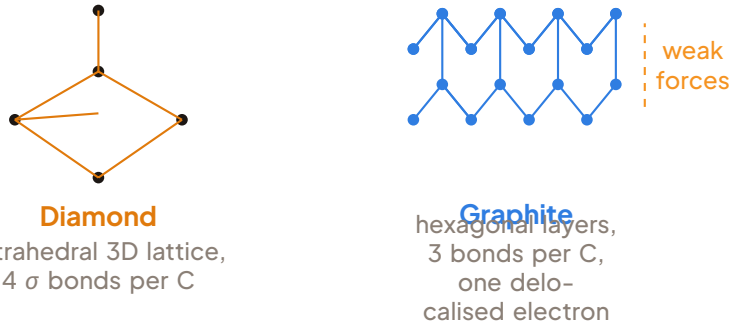
Diamond. Each carbon atom is sp^3 hybridised and bonded tetrahedrally to four neighbours, building a **giant covalent** three-dimensional lattice. All four valence electrons are locked in σ bonds. Consequently diamond is extremely hard, has a very high melting point (sublimes above 3600 °C) and is an electrical **insulator**.

Graphite. Each carbon atom is sp^2 hybridised and bonded to three neighbours in flat hexagonal layers. The fourth electron of each atom occupies a *p* orbital perpendicular to the layer, and these electrons merge into a **delocalised** π system between the layers. The layers are held only by weak van der Waals forces, so they slide over one another: graphite is soft, slippery (a lubricant and pencil "lead") and conducts electricity along the layers.

PROPERTY **Electrical conductivity of graphite**

Graphite conducts electricity because each carbon atom contributes one **delocalised**

electron that moves freely within and between the layers; diamond, in which every valence electron is localised in a σ bond, cannot conduct. (Proof not examinable; the explanation must be quoted.)



Structures of diamond (left) and graphite (right). Diamond: every C forms four σ bonds in a rigid tetrahedral network. Graphite: every C forms three σ bonds in layers; the fourth electron is delocalised, giving conductivity and softness.

DID YOU KNOW? **Fullerenes — an exam-useful extra**

In 1985 a third allotrope family was discovered: the **fullerenes**, molecules such as C_{60} ("buckyballs") in which carbon atoms form closed cages of pentagons and hexagons, like a football. Fullerenes and the related carbon nanotubes are studied for drug delivery and super-strong materials. Mentioning them in an essay on allotropy earns credit for breadth of knowledge.

4.4 Allotropy in Tin and the Metallic Character Trend

Carbon is not the only allotropic member. **Tin** exists as **grey tin** (α-tin, a brittle non-metallic form with a diamond-type structure) stable below 13.2 °C, and **white tin** (β-tin, the familiar metallic form) stable above that temperature. The slow crumbling of tin objects in severe cold ("tin pest") is a real consequence of this transition.

This dual personality of tin illustrates the general trend: **metallic character increases down the group**. Carbon is a non-metal; silicon and germanium are **metalloids** (semiconductors); tin and lead are metals, with positive electrode potentials and basic oxides in their lower oxidation state.

4.5 Trend in Structure: From Giant Covalent to Metallic

The change of character down the group is rooted in a change of **structure**:

Element	Structure of the element	Character
C (diamond)	Giant covalent, tetrahedral	Non-metal
Si, Ge	Giant covalent, diamond-type	Metalloids (semiconductors)
Sn	Giant covalent (grey) or metallic (white)	Metal
Pb	Metallic lattice	Metal

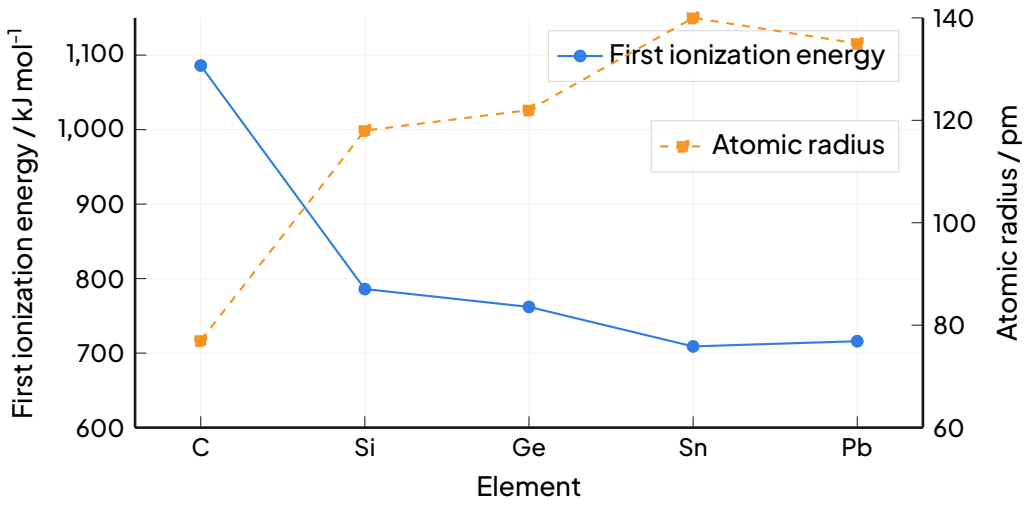
As the atoms grow larger, the E – E covalent bonds become longer and weaker, and the giant covalent lattice loses out to a metallic lattice in which the valence electrons are delocalised over positive ions.

4.6 Trends in Atomic Radius, Ionization Energy, Electronegativity and Melting Point

PROPERTY **Trends down Group IV**

Descending Group IV from C to Pb:

- Atomic radius increases — each element has one more filled electron shell.
- First ionization energy decreases — the outer electrons are further from the nucleus and more shielded.
- Electronegativity decreases (C 2.5 to Pb about 1.8) — bonding electrons are attracted less strongly.
- Melting point decreases overall — but the explanation must be structural (see warning below).



First ionization energy (blue, left axis) falls while atomic radius (orange, right axis) rises from C to Pb. Note the slight irregularity at Pb — a reminder that real trends are smooth only to first approximation.

WARNING **The melting point trap**

Do not write simply that "melting points decrease down Group IV" without explanation. Carbon (as diamond or graphite, subliming above 3600 °C) has an exceptionally high melting point because melting requires breaking strong covalent bonds throughout a giant lattice. Silicon (1410 °C) and germanium (937 °C) are lower because their E – E bonds are longer and weaker. Tin (232 °C) and lead (327 °C) are metallic, and their metallic bonding is weak because the atoms are large. Always explain the trend via the change of structure, from giant covalent to metallic.

4.7 Oxidation States +2 and +4: The Inert Pair Effect

Group IV elements show oxidation states +2 and +4. At the top of the group, +4 dominates: CO_2, CCl_4, SiO_2 are all stable, while CO is easily oxidised. At the bottom, the picture reverses: PbO and $PbCl_2$ are stable, whereas PbO_2 and Pb(IV) compounds are strong **oxidizing agents**, eager to be reduced to Pb(II).

DEFINITION **The inert pair effect**

The **inert pair effect** is the reluctance of the ns^2 pair of valence electrons to participate in bonding. Its importance increases down Groups III, IV and V, so the lower oxidation state (+2 for Group IV) becomes progressively more stable relative to the higher one (+4).

The explanation is energetic: down the group the E – E and E – X bonds become weaker, so the energy released by forming two extra bonds no longer compensates the energy needed to involve the ns^2 pair. Hence:

- Stability of +4 decreases: C > Si > Ge > Sn > Pb.
- Stability of +2 increases: Pb^{2+} is more stable than Pb^{4+}; Sn^{2+} is a mild reducing agent, but Pb(IV) is a strong oxidizing agent.

EXEMPLE **The inert pair effect in action**

Lead(IV) oxide oxidises concentrated hydrochloric acid to chlorine, being itself reduced to lead(II):

$$PbO_2(s) + 4\,HCl(aq) \longrightarrow PbCl_2(aq) + Cl_2(g) + 2\,H_2O(l)$$

This is the reaction used practically to distinguish PbO_2 from PbO: only the Pb(IV) oxide liberates chlorine.

WORKED EXAMPLE 5 **Explaining the oxidizing power of Pb(IV)**

Explain why PbO_2 is a good oxidizing agent whereas CO_2 is not.

SOLUTION

Step 1 — Identify the oxidation states. In PbO_2, lead is in the +4 state; in CO_2, carbon is in the +4 state.
Step 2 — Apply the inert pair effect. For lead, at the bottom of Group IV, the $6s^2$ pair is reluctant to bond (inert pair effect), so the +2 state is more stable than +4. PbO_2 therefore readily gains electrons and is reduced to Pb(II):

$$PbO_2 + 4\,H^+ + 2\,e^- \longrightarrow Pb^{2+} + 2\,H_2O$$

This makes PbO_2 a strong oxidizing agent.
Step 3 — Contrast with carbon. Carbon, at the top of the group, shows no inert pair effect: +4 is its most stable state, so CO_2 has no tendency to be reduced and is not an oxidizing agent under ordinary conditions.
Conclusion. The oxidizing power of Pb(IV) arises from the stability of Pb(II), a direct consequence of the inert pair effect.

METHOD **Exam tip: answering "why is Pb(IV) oxidizing?"**

When asked why Pb(IV) compounds are oxidizing agents:

1. State that the stability of +2 relative to +4 increases down Group IV.

2. Name the cause: the **inert pair effect** — the $6s^2$ electrons of lead are reluctant to participate in bonding.
3. Conclude that Pb(IV) readily accepts electrons to form the more stable Pb(II), hence it oxidizes other species. Quote an example such as PbO_2 with concentrated HCl.

4.8 Uses of the Group IV Elements

The physical trends we have just studied translate directly into everyday applications, several of them familiar in Cameroon:

- **Carbon:** diamond in cutting and drilling tools; graphite in pencil "leads", electrodes and lubricants; coke as a reducing agent in metallurgy.
- **Silicon:** purified silicon is the **semiconductor** at the heart of computer chips and solar panels; silicon(IV) oxide, SiO_2, is the raw material of **glass**.
- **Germanium:** used in semiconductors and in infrared optics.
- **Tin:** an unreactive, non-toxic metal used to **plate** steel food cans ("tin cans") and, alloyed with lead, as **solder** in electronics.
- **Lead:** dense and corrosion-resistant, used in **car batteries** (lead–acid accumulators), in roofing sheets, and as shielding against X-rays; its compounds were formerly used in paints and petrol, uses now abandoned because of lead's toxicity.

KEY POINTS **Key points**

- Group IV elements (C, Si, Ge, Sn, Pb) all have the outer configuration ns^2np^2.
- Carbon is unique: small size, high electronegativity, strong catenation and multiple bonding, maximum covalency of four.
- Carbon shows allotropy: diamond (giant tetrahedral, hard, insulator) and graphite (layered, soft, conductor thanks to delocalised electrons); fullerenes are a modern third form.
- Structure changes from giant covalent (C, Si, Ge) to metallic (Sn, Pb); metallic character increases down the group.
- Atomic radius increases; first ionization energy, electronegativity and melting point decrease — the melting point trend must be explained by the change of structure.
- Oxidation states +2 and +4: the inert pair effect makes +2 increasingly stable down the group, so Pb(IV) compounds are oxidizing agents.

EXERCISE 4 **Checking your understanding**

1. Explain why CCl_4 is not hydrolysed by water whereas $SiCl_4$ is hydrolysed rapidly.
2. Account for the fact that graphite is used as an electrode material while diamond is not.
3. Arrange CO, SnO, PbO_2 and PbO in order of increasing stability and justify your answer using the inert pair effect.
4. Explain why the melting point of lead (327 °C) is far lower than that of carbon, even though lead is much lower in the group.

SOLUTION **Exercise 1**

1. Carbon has no *d* orbitals in its $n = 2$ valence shell, so it cannot expand its octet and cannot accept a lone pair from water; CCl_4 is inert. Silicon has empty 3*d* orbitals, accepts a water molecule as a ligand, and $SiCl_4$ hydrolyses: $SiCl_4(l) + 2\,H_2O(l) \longrightarrow SiO_2(s) + 4\,HCl(aq)$.
2. In graphite each carbon uses only three electrons in σ bonds; the fourth is delocalised between the layers and carries current. In diamond all four electrons are localised in σ bonds, so no charge carriers exist and diamond is an insulator.
3. Increasing stability: $PbO_2 < CO < SnO < PbO$. The inert pair effect makes the +2 state more stable down the group: Pb(II) oxide is the most stable, while Pb(IV) oxide is the least stable (a strong oxidizing agent); for carbon the +4 state is favoured, so CO (C in +2) is readily oxidised.
4. Carbon is a giant covalent structure: melting requires breaking very strong C – C bonds throughout the lattice. Lead has a metallic lattice with large atoms and weak metallic bonding, so far less energy is needed; hence its much lower melting point.

5 REACTIONS AND COMPOUNDS OF GROUP IV: HYDRIDES, OXIDES AND CHLORIDES

In the previous section we surveyed the physical trends across Group IV — atomic radii, ionisation energies, allotropy and the change from non-metallic carbon to metallic lead. We now turn to their chemical behaviour: how the elements react with oxygen, acids and alkalis, and the nature of the hydrides, oxides and chlorides they form. Understanding these reactions is essential for predicting the outcome of qualitative analysis and for explaining the contrasting properties of, for example, gaseous CO_2 versus the refractory solid SiO_2.

5.1 Reactions of the Group IV Elements

EXPERIMENT **Reactions with oxygen**

When the Group IV elements are heated in air or oxygen they form oxides. The conditions and products vary down the group:

- Carbon burns in excess oxygen to give carbon dioxide; in a limited supply it forms carbon monoxide:

$$C(s) + O_2(g) \rightarrow [\equiv] CO_2(g) \qquad 2C(s) + O_2(g) \rightarrow [\equiv] 2CO(g)$$

- Silicon requires strong heating (≈ 1000 °C) to react, yielding silicon dioxide:

$$Si(s) + O_2(g) \rightarrow [\equiv] SiO_2(s)$$

- Germanium behaves similarly to silicon, forming GeO_2.
- Tin and lead are more reactive. Tin gives tin(IV) oxide SnO_2 on strong heating; lead forms a mixture of lead(II) oxide PbO and lead(II,IV) oxide Pb_3O_4 (red lead) depending on temperature and oxygen pressure:

$$2Pb(s) + O_2(g) \rightarrow [\equiv] 2PbO(s) \qquad 3Pb(s) + 2O_2(g) \rightarrow [\equiv] Pb_3O_4(s)$$

EXPERIMENT **Reactions with acids**

- Carbon and silicon are resistant to dilute non-oxidising acids (e.g. HCl, dilute H_2SO_4) because of their strong covalent network structures.
- Tin and lead react with oxidising acids. Concentrated nitric acid oxidises Sn to SnO_2 (or Sn^{4+}) and Pb to PbO_2 (or Pb^{4+}), with evolution of NO_2. Concentrated sulfuric acid (hot) also acts as an oxidising agent:

$$Sn(s) + 4\,HNO_3(conc) \longrightarrow SnO_2(s) + 4\,NO_2(g) + 2\,H_2O(l)$$

$$Pb(s) + 4\,HNO_3(conc) \longrightarrow PbO_2(s) + 4\,NO_2(g) + 2\,H_2O(l)$$

$$Sn(s) + 2H_2SO_4(conc, \equiv) \rightarrow SnO_2(s) + 2SO_2(g) + 2H_2O(l)$$

$$Pb(s) + 2H_2SO_4(conc, \equiv) \rightarrow PbO_2(s) + 2SO_2(g) + 2H_2O(l)$$

- Dilute HCl reacts slowly with Sn and Pb, producing hydrogen and the corresponding M^{2+} chlorides. The reaction with lead is slow due to the protective layer of insoluble $PbCl_2$:

$$Sn(s) + 2\,HCl(aq) \longrightarrow SnCl_2(aq) + H_2(g)$$

$$Pb(s) + 2\,HCl(aq) \longrightarrow PbCl_2(s) + H_2(g) \quad \textbf{(slow)}$$

EXPERIMENT **Reactions with alkalis**

- Silicon dissolves in hot concentrated NaOH to give sodium silicate and hydrogen:

$$Si(s) + 2\,NaOH(aq) + H_2O(l) \longrightarrow Na_2SiO_3(aq) + 2\,H_2(g)$$

- Tin and lead show amphoteric behaviour: they dissolve in both acids and hot concentrated alkalis, forming hexahydroxostannate(IV) $[Sn(OH)_6]^{2-}$ and hexahydroxoplumbate(IV) $[Pb(OH)_6]^{2-}$ respectively (often written as stannates Na_2SnO_3 and plumbates Na_2PbO_3).
- Carbon does not react with alkalis under normal conditions.

PROPERTY **Amphoteric oxides of Sn and Pb**

The oxides SnO, SnO_2, PbO and PbO_2 react with both acids and bases. For example:

$$SnO_2(s) + 2\,H_2SO_4(aq) \longrightarrow Sn(SO_4)_2(aq) + 2\,H_2O(l) \quad \text{(acidic medium)}$$

$$SnO_2(s) + 2\,NaOH(aq) + 2\,H_2O(l) \longrightarrow Na_2[Sn(OH)_6](aq) \quad \text{(basic medium)}$$

This dual reactivity is a hallmark of the metallic side of Group IV.

5.2 Hydrides of Group IV

All Group IV elements form tetrahedral hydrides of general formula XH_4 (X = C, Si, Ge, Sn, Pb). The structures are based on sp^3 hybridisation at the central atom.

DEFINITION **Group IV hydrides**

The binary hydrides are:

- CH_4 (methane) — gas, very stable
- SiH_4 (silane) — gas, pyrophoric
- GeH_4 (germane) — gas, less stable
- SnH_4 (stannane) — unstable, decomposes at room temperature
- PbH_4 (plumbane) — only observed at low temperature, extremely unstable

All adopt a tetrahedral geometry (bond angles ≈ 109.5°).

PROPERTY **Thermal stability and reducing power of Group IV hydrides**

The thermal stability of the hydrides decreases down the group as the X – H bond becomes longer and weaker. Consequently, the reducing power increases down the group: SnH_4 and PbH_4 are strong reducing agents, whereas CH_4 is essentially inert.

KEY POINTS **Key trends in Group IV hydrides**

- Formula: XH_4 (tetrahedral)
- Thermal stability: $CH_4 > SiH_4 > GeH_4 > SnH_4 > PbH_4$
- Reducing power: opposite order
- Catenation: Carbon forms countless C – C chains (hydrocarbons); silicon forms limited silanes (Si_nH_{2n+2} up to $n \approx 8$); heavier analogues show negligible catenation.

EXEMPLE **Comparing CH_4 and SiH_4**

Methane (CH_4) is the main component of natural gas; it burns cleanly to CO_2 and H_2O and does not hydrolyse. Silane (SiH_4) ignites spontaneously in air (pyrophoric) and hydrolyses rapidly:

$$SiH_4(g) + 2\,H_2O(l) \longrightarrow SiO_2(s) + 4\,H_2(g)$$

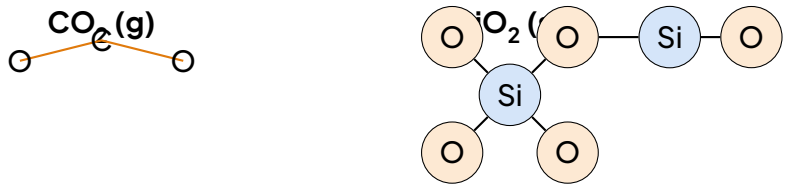
The readiness of SiH_4 to hydrolyse reflects the polarity of the Si – H bond and the accessibility of silicon's empty 3d orbitals.

5.3 Oxides of Group IV

Group IV oxides exist in two oxidation states: +2 (XO) and +4 (XO_2). Their structures and acid–base characters change dramatically down the group.

PROPERTY **Structures of the dioxides**

CO_2 consists of discrete linear O = C = O molecules (gas at room temperature). SiO_2, GeO_2 and SnO_2 adopt giant covalent network structures based on $[XO_4]$ tetrahedra sharing corners. PbO_2 has a layered structure (rutile type) with Pb in octahedral coordination.



Comparison of discrete molecular CO_2 (left) and extended tetrahedral network of SiO_2 (right).

PROPERTY **Acid–base character of Group IV oxides**

- CO_2 and SiO_2 are acidic oxides. They react with bases to form carbonates/silicates.
- CO is a neutral oxide — it does not react with either acids or bases under normal conditions.
- SnO, SnO_2, PbO, PbO_2 are amphoteric.
- GeO_2 is weakly acidic; GeO is amphoteric.

Oxide	Formula	Acid–base character
Carbon dioxide	CO_2	Acidic
Silicon dioxide	SiO_2	Acidic
Germanium dioxide	GeO_2	Acidic (weak)
Tin(II) oxide	SnO	Amphoteric
Tin(IV) oxide	SnO_2	Amphoteric
Lead(II) oxide	PbO	Amphoteric
Lead(IV) oxide	PbO_2	Amphoteric
Carbon monoxide	CO	Neutral

Acid–base classification of Group IV oxides (colour-coded).

WARNING **CO is neutral, not acidic**

A common error is to classify CO as acidic because it is an oxide of carbon. CO does not react with water to form an acid, nor does it react with NaOH or HCl. Only CO_2 is the acidic oxide of carbon.

METHOD **Classifying an oxide as acidic, basic or amphoteric**

1. Write the reaction of the oxide with a strong acid (e.g. HCl). If a salt forms, the oxide shows basic behaviour.
2. Write the reaction with a strong base (e.g. NaOH). If a salt forms, the oxide shows acidic behaviour.
3. If both reactions occur, the oxide is amphoteric. If neither occurs, it is neutral.

EXEMPLE **Amphoteric behaviour of SnO_2**

$$SnO_2(s) + 4\,HCl(aq) \longrightarrow SnCl_4(aq) + 2\,H_2O(l) \quad \text{(basic behaviour)}$$

$$SnO_2(s) + 2\,NaOH(aq) + 2\,H_2O(l) \longrightarrow Na_2[Sn(OH)_6](aq) \quad \text{(acidic behaviour)}$$

Thus SnO_2 is amphoteric.

5.4 Chlorides of Group IV

The chlorides follow the oxidation states +2 and +4: XCl_2 and XCl_4. Their stability and reactivity with water show a clear trend.

PROPERTY **Hydrolysis of tetrachlorides**

CCl_4 is inert to water at room temperature. $SiCl_4$, $GeCl_4$ and $SnCl_4$ hydrolyse violently:

$$SiCl_4(l) + 2\,H_2O(l) \longrightarrow SiO_2(s) + 4\,HCl(aq)$$

The reaction proceeds because silicon (and Ge, Sn) possess low-lying empty d-orbitals (3d for Si) that can accept a lone pair from a water molecule, forming a pentacoordinate intermediate. Carbon lacks accessible d-orbitals, so CCl_4 cannot expand its coordination sphere and does not hydrolyse under normal conditions.

PROPERTY **Stability of the +4 chlorides down the group**

The +4 oxidation state becomes less stable relative to +2 due to the inert pair effect. Consequently:

- CCl_4, $SiCl_4$, $GeCl_4$, $SnCl_4$ are stable liquids (or solids).

- $PbCl_4$ is unstable; it decomposes readily to $PbCl_2$ and chlorine:

$$PbCl_4(s) \rightarrow [\equiv]PbCl_2(s) + Cl_2(g)$$

KEY POINTS **Trends in Group IV chlorides**

- Formulae: XCl_2 and XCl_4 (tetrahedral for XCl_4, bent/angular for XCl_2 due to lone pair).
- Hydrolysis: CCl_4 inert; $SiCl_4$, $GeCl_4$, $SnCl_4$ rapid; $PbCl_4$ hydrolyses but decomposes first.
- Thermal stability of XCl_4: decreases down the group; $PbCl_4$ decomposes to $PbCl_2$ + Cl_2.

EXEMPLE **Contrasting CCl_4 and $SiCl_4$ hydrolysis**

When water is added to CCl_4, two immiscible layers form — no reaction. With $SiCl_4$, white fumes of HCl appear instantly and a white precipitate of SiO_2 forms. The difference is exploited in the laboratory: $SiCl_4$ is used to produce pure silica smoke for smoke screens.

WORKED EXAMPLE 6 **Hydrolysis and acid–base behaviour**

Statement: A student adds a few drops of water to separate test tubes containing CCl_4, $SiCl_4$ and $SnCl_4$. She then tests the resulting mixtures with blue and red litmus. Predict and explain the observations.
Solution:

1. CCl_4: No reaction. Two layers remain. Litmus unchanged (neutral).
2. $SiCl_4$: Violent hydrolysis: $SiCl_4 + 2\,H_2O \longrightarrow SiO_2(s) + 4\,HCl(aq)$. The solution becomes strongly acidic — blue litmus turns red, red litmus stays red.
3. $SnCl_4$: Similarly hydrolyses: $SnCl_4 + 2\,H_2O \longrightarrow SnO_2(s) + 4\,HCl(aq)$. Acidic solution — same litmus result as $SiCl_4$.

Key point: The hydrolysis of $SiCl_4$ and $SnCl_4$ produces HCl, giving an acidic solution, whereas CCl_4 does not hydrolyse because carbon cannot expand its octet.

KEY POINTS **GCE favourite: CO_2 vs SiO_2**

Examiners frequently ask: "Explain why CO_2 is a gas at room temperature while SiO_2 is a high-melting solid, even though both are Group IV dioxides." The answer must mention:

- CO_2: discrete linear molecules, weak van der Waals forces.
- SiO_2: giant covalent network of $[SiO_4]$ tetrahedra, strong Si – O bonds throughout the lattice.

DID YOU KNOW? **Silicon in Cameroon**

Cameroon's volcanic regions (e.g. Mount Cameroon) are rich in silica (SiO_2) deposits. Local industries use quartz sand to produce glass and ceramics. The high melting point of SiO_2 (≈ 1710 °C) makes it ideal for refractory bricks in metallurgical furnaces.

EXERCISE 5 **Practice questions**

1. Write balanced equations for the reaction of (a) Sn with concentrated HNO_3, (b) Si with hot concentrated NaOH.
2. Classify each of the following oxides as acidic, basic, amphoteric or neutral: CO, CO_2, SnO, PbO_2, SiO_2.
3. Explain why $PbCl_4$ decomposes on heating while $SnCl_4$ does not.
4. Describe the trend in thermal stability of the Group IV hydrides and relate it to bond strength.

SOLUTION **Exercise 1**

1. (a) $Sn(s) + 4\,HNO_3(conc) \longrightarrow SnO_2(s) + 4\,NO_2(g) + 2\,H_2O(l)$
   (b) $Si(s) + 2\,NaOH(aq) + H_2O(l) \longrightarrow Na_2SiO_3(aq) + 2\,H_2(g)$
2. CO — neutral; CO_2 — acidic; SnO — amphoteric; PbO_2 — amphoteric; SiO_2 — acidic.
3. $PbCl_4$ decomposes because the +4 oxidation state of lead is destabilised by the inert pair effect (reluctance of the $6s^2$ electrons to participate in bonding). Tin's $5s^2$ pair is less inert, so $SnCl_4$ remains stable.
4. Thermal stability decreases down the group: $CH_4 > SiH_4 > GeH_4 > SnH_4 > PbH_4$. The X – H bond length increases and bond dissociation energy decreases, making the

heavier hydrides easier to decompose.

6 INTRODUCTION TO THE D-BLOCK: ELECTRONIC CONFIGURATIONS AND PHYSICAL TRENDS

Having surveyed the chemistry of Group IV elements — their hydrides, oxides and chlorides — we now turn to the **d-block** of the periodic table. These elements, often called the **transition metals**, are the workhorses of modern industry: iron in construction, copper in electrical wiring, nickel in batteries, and many serve as catalysts in chemical manufacture. In this section we define the d-block, distinguish true transition elements from the borderline cases, master the electronic configurations of atoms and ions, and rationalise the observed trends in atomic radius, ionisation energy, melting point and related properties.

6.1 What are d-Block Elements?

DEFINITION **d-Block Elements**

The d-block comprises the ten groups (3–12) of the periodic table in which the **differentiating electron** enters a **(n–1)d** subshell. For the first transition series this means the 3d subshell is filled after the 4s subshell. The block lies between the s-block (Groups 1–2) and the p-block (Groups 13–18).

The first transition series runs from Sc (Z=21) to Zn (Z=30). All are metals with typical metallic properties: high electrical and thermal conductivity, malleability, ductility and lustre.

6.2 Transition Elements: The Strict Definition

Not every d-block element qualifies as a **transition element** in the strict IUPAC sense.

DEFINITION **Transition Element**

A transition element is a d-block element that forms at least one stable ion with a partially filled d-subshell.

WARNING **Borderline Cases: Scandium and Zinc**

Sc ([Ar] $3d^1\ 4s^2$) forms only Sc^{3+} ([Ar] $3d^0$) — an empty d-subshell. Zn ([Ar] $3d^{10}\ 4s^2$) forms only Zn^{2+} ([Ar] $3d^{10}$) — a completely filled d-subshell. Hence Sc and Zn are d-block elements but are NOT transition elements. This distinction is a favourite GCE examination

point.

6.3 Electronic Configurations of the First Transition Series

6.4 Building Up: The Aufbau Principle and the 4s/3d Order

According to the Aufbau principle, the 4s orbital is lower in energy than 3d for the neutral atoms of the first series, so it fills first. The general configuration is [Ar] $3d^n$ $4s^2$ (n = 1–10). However, two elements adopt anomalous configurations because a half-filled (d^5) or fully-filled (d^{10}) d-subshell confers extra stability through exchange energy and symmetry.

PROPERTY **Anomalous Configurations of Cr and Cu**

- Cr (Z=24): expected [Ar] $3d^4$ $4s^2$ → actual [Ar] $3d^5$ $4s^1$ (half-filled d-subshell).
- Cu (Z=29): expected [Ar] $3d^9$ $4s^2$ → actual [Ar] $3d^{10}$ $4s^1$ (fully-filled d-subshell).

METHOD **Writing the Configuration of a Transition Metal Ion**

When a transition metal forms a cation, electrons are removed from the highest principal quantum number first, i.e. from the 4s orbital before the 3d orbital.

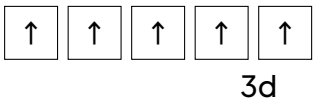
1. Write the ground-state configuration of the neutral atom (including anomalies for Cr and Cu).
2. Remove the required number of electrons from the 4s subshell.
3. If more electrons must be removed, take them from the 3d subshell.

Example: Fe (Z=26) is [Ar] $3d^6$ $4s^2$. Fe^{2+}: remove two 4s electrons → [Ar] $3d^6$. Fe^{3+}: remove two 4s then one 3d → [Ar] $3d^5$.

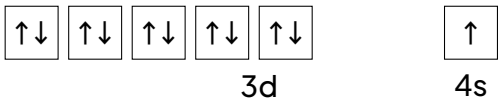
EXEMPLE **Electron-in-Boxes Diagrams**

The orbital (electron-in-boxes) notation makes the pairing and unpaired electrons explicit. Below are the diagrams for Cr, Cu and Fe^{2+}. Each box represents one orbital; arrows represent electrons with opposite spins.

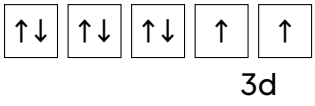
Cr: [Ar] $3d^5$ $4s^1$



Cu: [Ar] $3d^{10}$ $4s^1$



Fe^{2+}: [Ar] $3d^6$



Electron-in-boxes (orbital) diagrams for Cr, Cu and Fe^{2+}. Each box is one orbital; arrows show electron spin. Hund's rule: electrons occupy degenerate orbitals singly before pairing.

WORKED EXAMPLE 7 **Worked Example: Configurations of Co^{2+} and Ni^{3+}**

Statement: Write the full electronic configuration (in spdf notation) and the electron-in-boxes diagram for the 3d subshell of Co^{2+} and Ni^{3+}.
Solution:

1. Co (Z=27): neutral = [Ar] $3d^7$ $4s^2$. Co^{2+}: remove two 4s electrons → [Ar] $3d^7$. 3d boxes (5 orbitals, 7 electrons): three orbitals doubly occupied, two singly occupied. Diagram: ↑↓ ↑↓ ↑↓ ↑ ↑.
2. Ni (Z=28): neutral = [Ar] $3d^8$ $4s^2$. Ni^{3+}: remove two 4s then one 3d → [Ar] $3d^7$. Same d-electron count as Co^{2+}, hence identical 3d diagram.

Key point: Ions with the same d^n configuration have similar magnetic and colour properties.

KEY POINTS **Complete Configurations of the First Transition Series (Neutral Atoms and Common Ions)**

Element (Z)	Neutral Atom	M^{2+}	M^{3+}	Transition Element?
Sc (21)	[Ar] $3d^1$ $4s^2$	[Ar] $3d^1$	[Ar] $3d^0$	No (Sc^{3+} is d^0)
Ti (22)	[Ar] $3d^2$ $4s^2$	[Ar] $3d^2$	[Ar] $3d^1$	Yes
V (23)	[Ar] $3d^3$ $4s^2$	[Ar] $3d^3$	[Ar] $3d^2$	Yes
Cr (24)	[Ar] $3d^5$ $4s^1$	[Ar] $3d^4$	[Ar] $3d^3$	Yes
Mn (25)	[Ar] $3d^5$ $4s^2$	[Ar] $3d^5$	[Ar] $3d^4$	Yes
Fe (26)	[Ar] $3d^6$ $4s^2$	[Ar] $3d^6$	[Ar] $3d^5$	Yes
Co (27)	[Ar] $3d^7$ $4s^2$	[Ar] $3d^7$	[Ar] $3d^6$	Yes
Ni (28)	[Ar] $3d^8$ $4s^2$	[Ar] $3d^8$	[Ar] $3d^7$	Yes
Cu (29)	[Ar] $3d^{10}$ $4s^1$	[Ar] $3d^9$	[Ar] $3d^8$	Yes
Zn (30)	[Ar] $3d^{10}$ $4s^2$	[Ar] $3d^{10}$	—	No (Zn^{2+} is d^{10})

6.5 Trends in Physical Properties Across the First Transition Series

6.6 Atomic Radius

Across the series the atomic radius shows a **small decrease from Sc to Cr**, then remains **nearly constant** from Mn to Cu, with a slight increase at Zn.

KEY POINTS **Why the Radius is Nearly Constant**

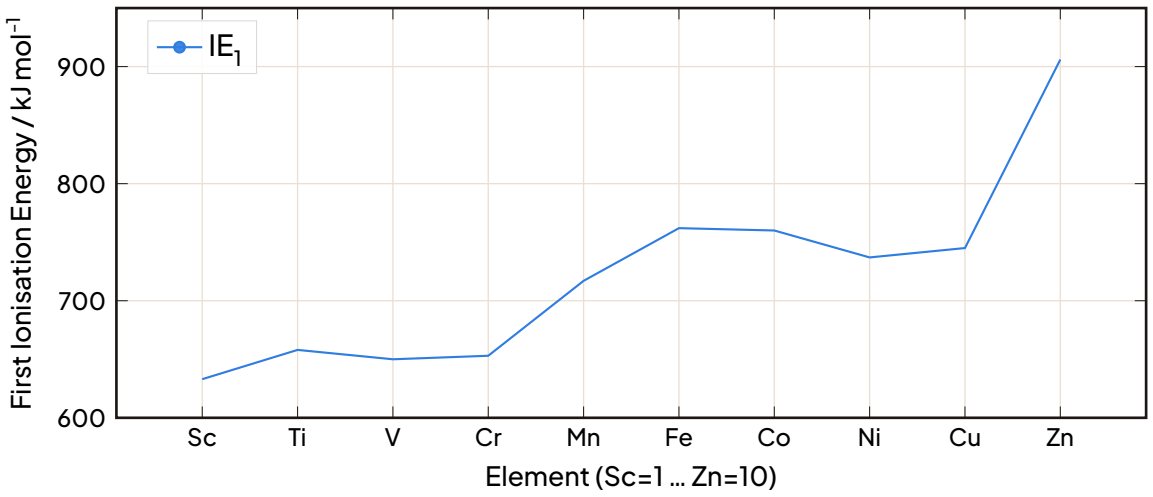
- Nuclear charge increases by +1 each step.
- The added electrons enter the inner 3d subshell, which shields the outer 4s electrons effectively.
- The increased attraction is largely offset by increased shielding → 4s electron cloud experiences similar effective nuclear charge.

WARNING **Exam Tip**

When asked to explain the near-constancy of atomic radii, always mention: "The additional 3d electrons shield the 4s electrons from the increasing nuclear charge, so the effective nuclear charge felt by the 4s electrons changes very little." Do not mention lanthanide contraction — that applies to the 4d/5d series.

6.7 First Ionisation Energy

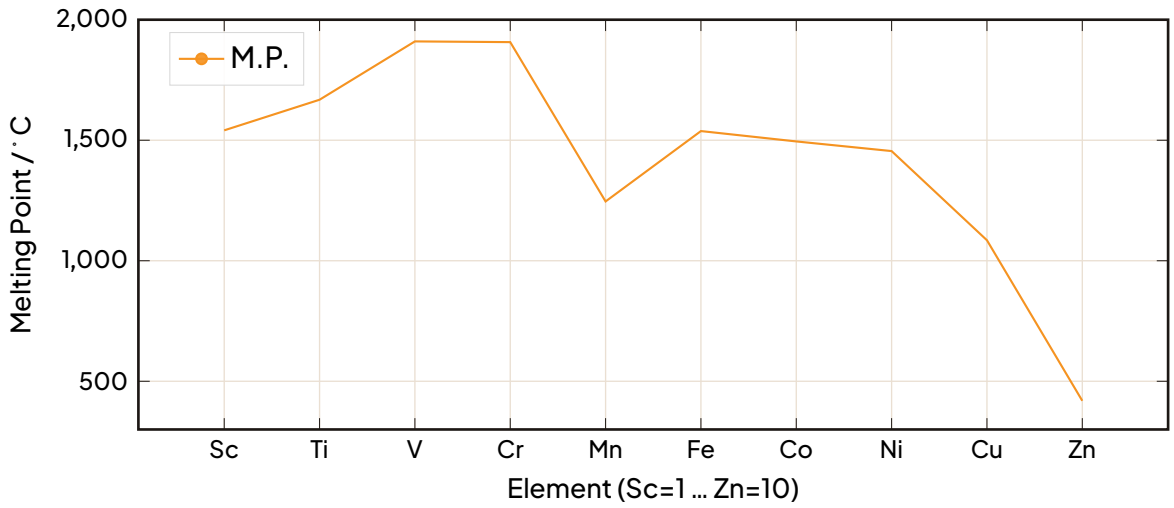
The first ionisation energy (I_1) generally increases across the series but with **irregularities** due to exchange energy and subshell stability. The overall increase reflects the rising nuclear charge. The small dips at Cr and Cu arise because the electron removed comes from a singly occupied 4s orbital, leaving a stable half-filled (d^5) or fully-filled (d^{10}) d-subshell in the cation. The sharp rise at Zn occurs because the electron is removed from a stable $4s^2$ orbital into a stable $3d^{10}$ core.



First ionisation energy across the first transition series. Note the general increase, small irregularities at V, Cr, Ni, Cu, and the sharp rise at Zn.

6.8 Melting and Boiling Points

Transition metals have high melting and boiling points due to strong metallic bonding involving both 4s and 3d electrons. The melting point peaks around the middle of the series (V, Cr) where the number of unpaired d-electrons available for metallic bonding is maximal. Mn has a lower melting point due to its complex crystal structure. Zn, with a full $3d^{10}$ subshell, has a much lower melting point because only the 4s electrons contribute to metallic bonding.



Melting points across the first transition series. The peak at V/Cr reflects maximum unpaired d-electrons; Mn dips due to structure; Zn is anomalously low.

6.9 Density and Hardness Compared with s-Block Metals

Transition metals are **much denser and harder** than s-block metals (e.g. Na, Mg, Ca). This is because:

- Smaller atomic radii → closer packing.
- Greater number of valence electrons (4s + 3d) participating in metallic bonding → stronger electrostatic attraction between cations and electron sea.

For example, Fe density = 7.87 g cm^{-3} vs Mg = 1.74 g cm^{-3}; Fe is hard, Mg is relatively soft.

PROPERTY **Summary of Physical Trends (Sc → Zn)**

Property	Trend	Reason
Atomic radius	Small decrease, then nearly constant	3d electrons shield 4s from increasing nuclear charge
First ionisation energy	General increase with irregularities	Rising nuclear charge; dips at Cr, Cu due to stable d^5/d^{10} in M^+; sharp rise at Zn
Melting point	High, peaks at V/Cr, dips at Mn, drops at Zn	Number of unpaired d-electrons for metallic bonding; Mn structure; Zn only 4s electrons
Density / Hardness	Much higher than s-block	Stronger metallic bonding (more valence electrons, smaller size)

6.10 Common Mistakes and Exam Tips

WARNING **Frequent Errors**

- Writing ion configurations as $3\,d^4\,4\,s^2$ for Fe^{2+} instead of $3\,d^6$. Always remove 4s electrons first.
- Calling Sc and Zn "transition elements". Remember the definition: partially filled d-subshell in at least one common ion.
- Explaining constant atomic radius by "lanthanide contraction" — that applies to the 4d/5d series, not the first series.
- Forgetting that Cr and Cu have $4\,s^1$ configurations, which affects their ionisation energies and chemistry.
- Drawing electron-in-boxes diagrams without applying Hund's rule (maximise unpaired electrons before pairing).

6.11 Did You Know?

DID YOU KNOW? **Transition Metals in Cameroon**

- Iron ore (magnetite, hematite) deposits in the Adamawa and East regions feed the local steel industry.

- Copper is mined in the Nkam area; its excellent conductivity makes it vital for the national electricity grid.
- Nickel and cobalt lateritic ores in the South-East are prospective for battery-grade metals.
- Many homogeneous catalysts (e.g. Fe, Co complexes) are used in the SONARA refinery (Limbe) for hydrodesulfurisation of petroleum fractions.

6.12 Consolidation Exercise

EXERCISE 6 **Practice Problems**

1. Write the ground-state electron configuration (spdf and electron-in-boxes) for Mn and Mn^{2+}.
2. Explain why the first ionisation energy of Zn is significantly higher than that of Cu.
3. Predict which of the following ions will be coloured in aqueous solution: Sc^{3+}, Ti^{3+}, V^{2+}, Zn^{2+}. Justify your answer.
4. The melting point of Cr is 1907 °C while that of Zn is 419 °C. Use metallic bonding theory to account for this difference.

SOLUTION **Exercise 6.1**

1. Mn (Z=25): [Ar] $3d^5\ 4s^2$. Boxes: 3d — five singly occupied (↑ ↑ ↑ ↑ ↑); 4s — paired (↑↓). Mn^{2+}: remove $4s^2$ → [Ar] $3d^5$. Boxes: five singly occupied.
2. Zn has configuration $3\,d^{10}\ 4\,s^2$; the electron removed comes from a stable, fully-filled 4s orbital, and the resulting Zn^+ has a stable $3\,d^{10}$ core. Cu is $3\,d^{10}\ 4\,s^1$; removing the single 4s electron yields stable Cu^+ ($3\,d^{10}$), so less energy is required.
3. Coloured ions have partially filled d-subshells: Ti^{3+} ($3d^1$), V^{2+} ($3d^3$) → coloured. Sc^{3+} ($3d^0$) and Zn^{2+} ($3d^{10}$) → colourless.
4. Cr has six unpaired valence electrons ($3d^5\ 4s^1$) contributing to a strong metallic bond; Zn has only two 4s electrons ($3d^{10}$ core is inert), giving a much weaker metallic bond and lower melting point.

This completes the foundation on d-block electronic structure and physical trends. In the next section we will explore the hallmark chemical behaviours of transition metals: **variable oxidation states** and the fascinating world of **complex formation**.

7 VARIABLE OXIDATION STATES AND COMPLEX FORMATION IN TRANSITION METALS

In the previous section we saw how the d-block elements of the first transition series (Sc to Zn) are built up by filling the 3d and 4s orbitals. We also observed trends in atomic radius, ionisation energy and melting points. Now we turn to the two most striking chemical consequences of that electronic structure: the **variable oxidation states** that make transition metal chemistry so rich in redox reactions, and the **formation of complex ions** through coordinate bonding with ligands. These two features are the hallmark of transition metal chemistry and underlie their importance in catalysis, biological systems and industrial processes.

7.1 Why Transition Metals Exhibit Variable Oxidation States

Look at the electronic configurations of the first-row transition metals and their common ions. For example, manganese:

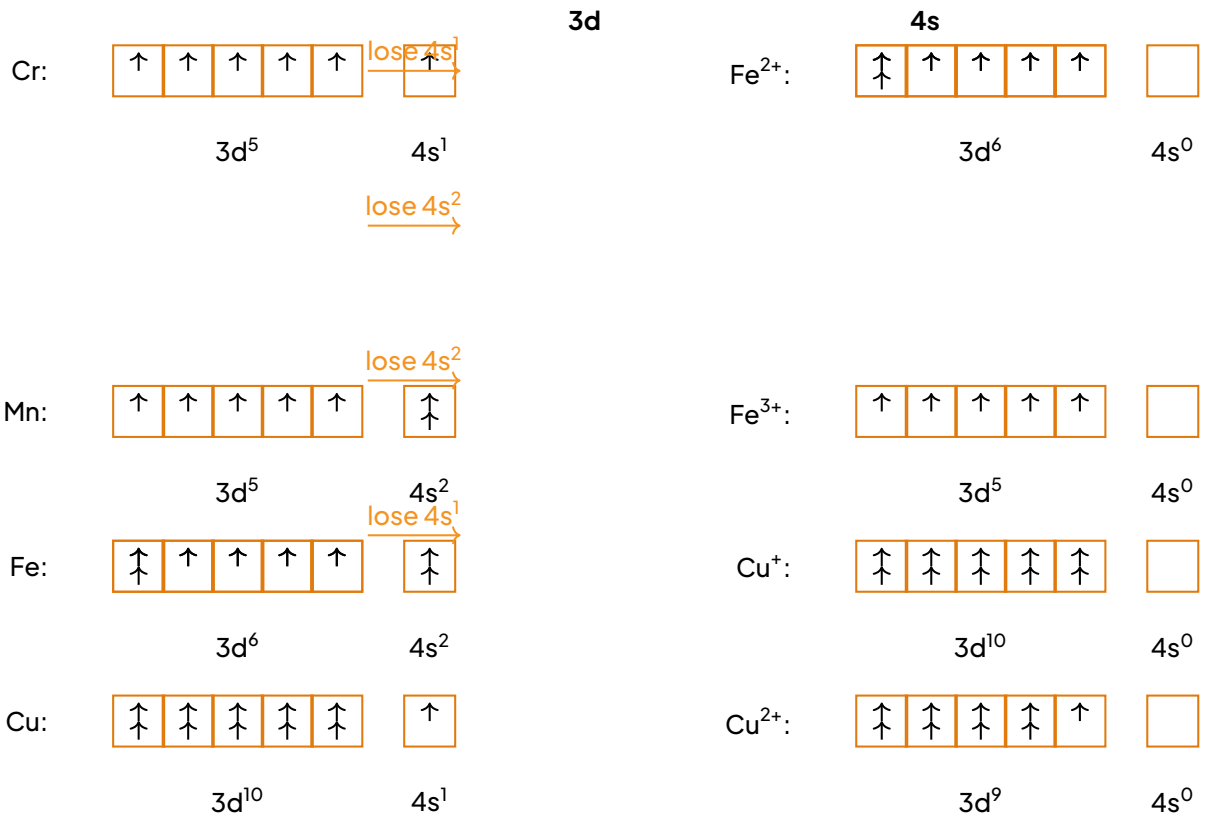
$$\mathrm{Mn}: [\mathrm{Ar}]\,3d^5\,4s^2 \quad \mathrm{Mn}^{2+}: [\mathrm{Ar}]\,3d^5 \quad \mathrm{Mn}^{7+}\,(\text{in } \mathrm{MnO_4^-}): [\mathrm{Ar}]$$

The 4s electrons are removed first, then the 3d electrons. Because the 3d and 4s energy levels are very close, successive ionisation energies do not jump dramatically until a stable configuration (empty, half-filled or fully-filled d subshell) is reached. This allows a metal to lose different numbers of electrons under different conditions, giving a range of oxidation states.

PROPERTY **Origin of Variable Oxidation States**

Transition metals show variable oxidation states because the 3d and 4s energy levels are very close, allowing successive removal of electrons from both subshells without a prohibitively large energy gap. The maximum oxidation state usually equals the number of 4s + 3d electrons (e.g. +7 for Mn, +6 for Cr), while the minimum is typically +2 (loss of the two 4s electrons).

Electron-in-boxes representation (required by the syllabus) helps visualise the filling and ionisation. The 3d subshell has five degenerate orbitals; electrons occupy them singly before pairing (Hund's rule).



Electron-in-boxes diagrams for selected first-row transition metals and their common ions. Arrows indicate electron spin. Note the anomalous configurations of Cr ($3d^5 4s^1$) and Cu ($3d^{10} 4s^1$) due to extra stability of half-filled and fully-filled d subshells.

Element	Common Oxidation States	Most Stable / Typical Compounds
Sc	+3	Sc_2O_3, $ScCl_3$
Ti	+2, +3, **+4**	TiO_2, $TiCl_4$
V	+2, +3, +4, **+5**	VO^{2+}, VO_3^-
Cr	+2, **+3**, +6	Cr^{3+}, $Cr_2O_7^{2-}$
Mn	+2, +3, +4, +6, **+7**	Mn^{2+}, MnO_2, MnO_4^-
Fe	**+2**, **+3**	Fe^{2+}, Fe^{3+}
Co	+2, +3	Co^{2+}, Co^{3+} (in complexes)
Ni	**+2**	Ni^{2+}
Cu	**+1**, **+2**	Cu^+, Cu^{2+}
Zn	+2	Zn^{2+}

Note: Bold oxidation states are the most stable in aqueous solution. Higher oxidation states (+5, +6, +7) are strongly oxidising and are stabilised in oxides or oxoanions (e.g. MnO_4^-, $Cr_2O_7^{2-}$, VO_3^-) where the metal is bonded to highly electronegative oxygen. Lower oxidation states (+2, +3) predominate in simple ionic compounds (halides, sulfates) and aqua ions.

7.2 Redox Interconversions of Key Oxidation States

The variable oxidation states allow a rich redox chemistry. Three systems are fundamental for GCE Advanced Level: iron, chromium and manganese. You must know the half-equations, the colour changes and the conditions (acidic medium).

DEFINITION **Half-Equation**

A half-equation shows either the oxidation or the reduction process separately, with electrons explicitly written. In acidic medium, balance O with H_2O, H with H^+, and charge with e^-.

METHOD **Balancing a Half-Equation in Acidic Medium**

1. Write the species containing the element being reduced/oxidised on the left, the product on the right.
2. Balance the element (other than O and H).
3. Balance O by adding H_2O to the side deficient in O.
4. Balance H by adding H^+ to the side deficient in H.
5. Balance charge by adding electrons (e^-) to the more positive side.

1. Iron: Fe^{3+}/Fe^{2+}

$$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq) \qquad E^\circ = +0.77\,V$$

Colour change: Pale yellow (Fe^{3+}) ⇌ Pale green (Fe^{2+}). This couple is used in redox titrations (e.g. with MnO_4^- or $Cr_2O_7^{2-}$).

2. Chromium: $Cr_2O_7^{2-}/Cr^{3+}$ (acidic medium)

$$Cr_2O_7^{2-}(aq) + 14\,H^+(aq) + 6\,e^- \rightleftharpoons 2\,Cr^{3+}(aq) + 7\,H_2O(l) \qquad E^\circ = +1.33\,V$$

Colour change: Orange ($Cr_2O_7^{2-}$) → Green (Cr^{3+}). $K_2Cr_2O_7$ is a primary standard in redox titrations.

3. Manganese: MnO_4^-/Mn^{2+} (acidic medium)

$$MnO_4^-(aq) + 8\,H^+(aq) + 5\,e^- \rightleftharpoons Mn^{2+}(aq) + 4\,H_2O(l) \qquad E^\circ = +1.51\,V$$

Colour change: Deep purple (MnO_4^-) → Colourless (Mn^{2+}). The reaction is self-indicating; the first permanent pink colour signals the endpoint.

WARNING **Common Mistakes in Half-Equations**

- Forgetting to balance oxygen with H_2O and hydrogen with H^+ in acidic medium.
- Writing the wrong number of electrons: count the change in oxidation state per formula unit (e.g. Cr in $Cr_2O_7^{2-}$ is +6 each, total +12; two Cr^{3+} give +6; change = 6 electrons).
- Using OH^- instead of H^+ when the problem states "acidic medium".
- Confusing the colour of Cr^{3+} (green) with Cr^{2+} (blue) or Fe^{2+} (pale green) with Fe^{3+} (yellow/brown).

WORKED EXAMPLE 8 **Balancing a Redox Reaction: MnO_4^- with Fe^{2+}**

Statement: Write the balanced ionic equation for the reaction between acidified potassium manganate(VII) and iron(II) sulfate.
Detailed solution:

1. Write the two half-equations (reduction of MnO_4^-, oxidation of Fe^{2+}):

$$\text{Reduction: } MnO_4^- + 8\,H^+ + 5\,e^- \longrightarrow Mn^{2+} + 4\,H_2O$$

$$\text{Oxidation: } Fe^{2+} \longrightarrow Fe^{3+} + e^-$$

2. Equalise electrons: multiply the oxidation half-equation by 5.

$$5\,Fe^{2+} \longrightarrow 5\,Fe^{3+} + 5\,e^-$$

3. Add the two half-equations, cancelling electrons:

$$MnO_4^- + 8\,H^+ + 5\,Fe^{2+} \longrightarrow Mn^{2+} + 4\,H_2O + 5\,Fe^{3+}$$

4. Check: atoms and charge balanced. Left: charge = −1 + 8 + 10 = +17; Right: +2 + 15 = +17. ✓

7.3 Complex Ions: Ligands, Coordination and Classification

When a transition metal ion is in solution or reacts with certain molecules/ions, it does not remain as a bare ion. It attracts species that have lone pairs of electrons, forming **coordinate (dative) bonds**. The resulting assembly is a **complex ion**.

DEFINITION **Ligand**

A ligand is a molecule or ion that donates a lone pair of electrons to a central metal ion to form a coordinate bond. The ligand acts as a Lewis base; the metal ion acts as a Lewis acid.

DEFINITION **Coordination Number**

The coordination number is the number of coordinate bonds formed between the ligands and the central metal ion. It is not necessarily the number of ligands (a bidentate ligand forms two bonds).

Ligands are classified by the number of donor atoms (**denticity**):

Type	Examples	Formula / Charge	Denticity
Monodentate (neutral)	Water, Ammonia, Carbon monoxide	H_2O, NH_3, CO (0)	1
Monodentate (anionic)	Chloride, Cyanide, Hydroxide, Thiocyanate	Cl^-, CN^-, OH^-, SCN^- (-1)	1
Bidentate	Ethylenediamine (en), Oxalate	$NH_2CH_2CH_2NH_2$ (0), $C_2O_4^{2-}$ (-2)	2
Polydentate	$EDTA^{4-}$	$[EDTA]^{4-}$	6
Ambidentate	Thiocyanate, Nitrite	SCN^- (S or N), NO_2^- (N or O)	1 (two possible donor atoms)

Common ligands, their charges and denticity.

KEY POINTS **Chelation and the Chelate Effect**

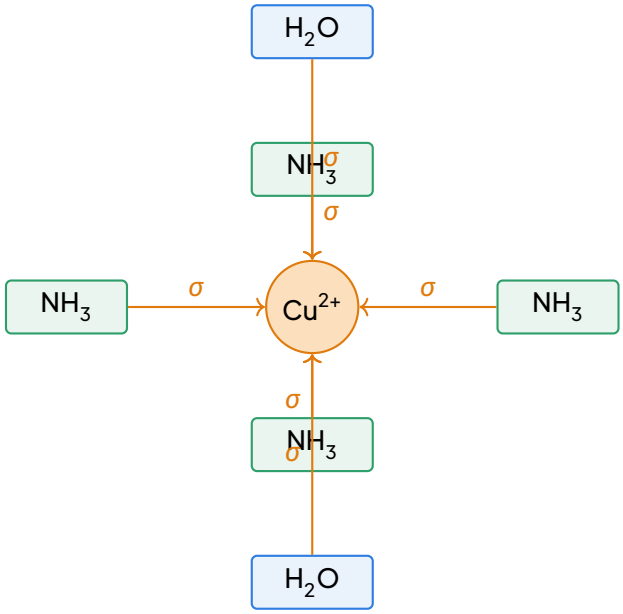
A ligand that forms more than one bond to the same metal ion is a chelating ligand. The resulting ring structure (chelate ring, usually 5- or 6-membered) gives extra thermodynamic stability compared to similar monodentate ligands. This is the chelate effect.

Ethylenediamine (en) and EDTA are classic chelating agents. $EDTA^{4-}$ wraps around the metal ion in an octahedral arrangement, forming five chelate rings.

7.4 Coordination Number and Geometry

The coordination number (CN) determines the shape of the complex. For the first transition series, CN = 6 (octahedral), 4 (tetrahedral or square planar) and 2 (linear) are most common.

CN	Geometry	Examples
2	Linear	$[Ag(NH_3)_2]^+$, $[CuCl_2]^-$ (in conc. HCl)
4	Tetrahedral	$[CoCl_4]^{2-}$, $[CuCl_4]^{2-}$, $[Zn(NH_3)_4]^{2+}$
4	Square planar	$[Ni(CN)_4]^{2-}$, $[Pt(NH_3)_2Cl_2]$ (Pt is 2nd row)
6	Octahedral	$[Fe(H_2O)_6]^{2+}$, $[Co(NH_3)_6]^{3+}$, $[Cu(NH_3)_4(H_2O)_2]^{2+}$



Octahedral $[Cu(NH_3)_4(H_2O)_2]^{2+}$
NH_3 (green) and H_2O (blue) donate lone pairs (arrows) to Cu^{2+}.
Coordination number = 6.
Jahn-Teller distortion: axial bonds longer than equatorial.

Structure of the tetraamminediaquacopper(II) ion showing dative bonds from ligands to the central metal ion.

7.5 Classification of Complexes by Overall Charge

The overall charge of a complex ion is the sum of the oxidation state of the metal and the charges of the ligands.

PROPERTY **Overall Charge of a Complex**

For a complex $[M_xL_y]^z$, the overall charge z is given by:

$$z = \text{Oxidation state of M} + \sum(\text{charges of ligands})$$

Type	Overall Charge	Examples
Cationic	Positive	$[Cu(NH_3)_4(H_2O)_2]^{2+}$, $[Co(NH_3)_6]^{3+}$, $[Fe(H_2O)_6]^{2+}$
Anionic	Negative	$[Fe(CN)_6]^{4-}$, $[CuCl_4]^{2-}$, $[MnO_4]^-$
Neutral	Zero	$[Ni(CO)_4]$, $[Pt(NH_3)_2Cl_2]$, $Fe(acac)_3$

METHOD **Finding the Oxidation State of the Metal in a Complex**

1. Identify the overall charge of the complex ion (from the formula of the salt, e.g. $K_4[Fe(CN)_6]$ → complex is $[Fe(CN)_6]^{4-}$).
2. List the ligands and their charges (e.g. CN^- is −1 each).
3. Let x be the oxidation state of the metal. Write: $x + \sum$(ligand charges) = overall charge.
4. Solve for x.

WORKED EXAMPLE 9 **Oxidation State in a Complex**

Statement: Determine the oxidation state of chromium in $K_3[Cr(C_2O_4)_3]$ and cobalt in $[Co(NH_3)_5Cl]Cl_2$.
Detailed solution:

1. $K_3[Cr(C_2O_4)_3]$: The salt has 3 K^+ ions, so the complex anion is $[Cr(C_2O_4)_3]^{3-}$. Ligand: oxalate $C_2O_4^{2-}$ (charge −2), three of them → total ligand charge = −6. Let x = oxidation state of Cr.

$$x + (-6) = -3 \quad \Rightarrow \quad x = +3.$$

2. $[Co(NH_3)_5Cl]Cl_2$: Two chloride ions outside the bracket are counter-ions, so the complex cation is $[Co(NH_3)_5Cl]^{2+}$. Ligands: 5 NH_3 (neutral, charge 0) + 1 Cl^- (charge −1). Total ligand charge = −1. Let x = oxidation state of Co.

$$x + (-1) = +2 \quad \Rightarrow \quad x = +3.$$

WARNING **Charge of Complex vs. Oxidation State of Metal**

Do not confuse the charge of the complex ion with the oxidation state of the metal. They are equal only when all ligands are neutral. If any ligand carries a charge, you must do the calculation.

7.6 Ligand Exchange Reactions: The Chemistry of Cu^{2+}

Aqua ions like $[Cu(H_2O)_6]^{2+}$ (blue) undergo ligand exchange when a stronger ligand (or a higher concentration of a ligand) is added. Two classic examples are tested in GCE practical and theory papers.

EXPERIMENT **Ligand Exchange with Cu^{2+}**

Reaction 1: Addition of aqueous ammonia (limited then excess).

1. Few drops $NH_3(aq)$: Cu^{2+} acts as an acid; NH_3 acts as a base.

$$[Cu(H_2O)_6]^{2+}(aq) + 2NH_3(aq) \rightarrow Cu(OH)_2(s) + 2NH_4^+(aq) + 4H_2O(l)$$

   Observation: Pale blue precipitate of $Cu(OH)_2$.

2. Excess $NH_3(aq)$: NH_3 acts as a ligand, dissolving the precipitate via ligand exchange.

$$Cu(OH)_2(s) + 4NH_3(aq) + 2H_2O(l) \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+}(aq) + 2OH^-(aq)$$

   Observation: Precipitate dissolves giving a deep blue solution (tetraamminediaquacopper(II) ion).

Reaction 2: Addition of concentrated hydrochloric acid.

$$[Cu(H_2O)_6]^{2+}(aq) + 4Cl^-(aq) \rightleftharpoons [CuCl_4]^{2-}(aq) + 6H_2O(l)$$

Observation: Blue solution turns yellow-green (tetrahedral $[CuCl_4]^{2-}$). The equilibrium shifts right with high Cl^- concentration (conc. HCl) and left on dilution with water.

KEY POINTS **Key Points for Ligand Exchange**

- Ligand exchange is a substitution reaction: one ligand replaces another in the coordination sphere.
- The driving force can be: (a) stronger ligand field (e.g. $NH_3 > H_2O$), (b) chelate effect (e.g. en > NH_3), (c) concentration (Le Chatelier's principle, e.g. conc. HCl).
- Colour change is a direct indicator of the change in d-orbital splitting (Δ) caused by

the new ligand set.

- Cu^{2+} (d9) in octahedral field shows Jahn-Teller distortion: two axial bonds longer than four equatorial. In $[Cu(NH_3)_4(H_2O)_2]^{2+}$ the four NH_3 occupy the equatorial plane.

7.7 Summary and Exam Focus

KEY POINTS **What You Must Master for the GCE**

1. Variable oxidation states: Explain using 3d/4s energy proximity. List common states for Cr, Mn, Fe, Cu. Know that high states are oxidising and stabilised by O; low states are stable in simple salts.
2. Redox half-equations: Write and balance MnO_4^-/Mn^{2+}, $Cr_2O_7^{2-}/Cr^{3+}$, Fe^{3+}/Fe^{2+} in acidic medium. Know the colour changes and $E^\ominus$ values.
3. Complex terminology: Ligand, coordinate bond, coordination number, chelate, denticity (mono-, bi-, poly-). Classify ligands with charge.
4. Shapes: Linear (CN=2), tetrahedral/square planar (CN=4), octahedral (CN=6). Draw/describe $[Cu(NH_3)_4(H_2O)_2]^{2+}$.
5. Oxidation state calculation: Use overall charge = metal oxidation state + sum of ligand charges.
6. Ligand exchange: Write equations for $[Cu(H_2O)_6]^{2+}$ with NH_3 (two stages) and conc. HCl. Explain observations.
7. Electron-in-boxes: Draw configurations for atoms and ions of the first transition series, noting anomalies for Cr and Cu.

DID YOU KNOW? **Transition Metals in Cameroon**

The lateritic soils of the Adamawa and East regions of Cameroon are rich in iron and manganese oxides (Fe_2O_3, MnO_2) — the high oxidation states stabilised by oxygen. These ores are the basis for potential ferroalloy industries. In our rivers, the Fe^{2+}/Fe^{3+} redox couple plays a role in natural water chemistry, while the blue $[Cu(NH_3)_4]^{2+}$ complex is a classic qualitative test for copper(II) in the school laboratory.

EXERCISE 7 Practice Problems

- Write the half-equation for the reduction of $Cr_2O_7^{2-}$ to Cr^{3+} in acidic medium. What is the colour change?
- Determine the oxidation state of the metal in: (a) $[Fe(CN)_6]^{3-}$, (b) $[CoCl_4]^{2-}$, (c) $[Ni(NH_3)_6]Cl_2$.
- A solution of $CuSO_4$ is treated with (i) a few drops of NaOH, (ii) excess NH_3(aq), (iii) conc. HCl. Describe the observations and write equations for each step.
- Explain why Mn shows oxidation states from +2 to +7 while Zn shows only +2.
- The complex $[Cr(en)_3]Cl_3$ is a green crystalline solid. (en = ethylenediamine). State the coordination number, geometry, oxidation state of Cr, and overall charge of the complex ion.
- Draw the electron-in-boxes diagram for Cr atom and Cr^{3+} ion. Explain the stability of Cr^{3+}.

SOLUTION Exercise 7

- $Cr_2O_7^{2-} + 14\,H^+ + 6\,e^- \longrightarrow 2\,Cr^{3+} + 7\,H_2O$. Colour: orange → green.
- (a) CN^- is -1, six of them = -6. Complex charge -3. $x - 6 = -3 \Rightarrow x = +3$. (b) Cl^- is -1, four of them = -4. Complex charge -2. $x - 4 = -2 \Rightarrow x = +2$. (c) Two Cl^- counter-ions → complex is $[Ni(NH_3)_6]^{2+}$. NH_3 neutral. $x = +2$.
- (i) $Cu^{2+} + 2\,OH^- \longrightarrow Cu(OH)_2$(s) (pale blue ppt). (ii) $Cu(OH)_2 + 4NH_3 + 2H_2O \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 2OH^-$ (deep blue solution). (iii) $[Cu(H_2O)_6]^{2+} + 4Cl^- \rightleftharpoons [CuCl_4]^{2-} + 6H_2O$ (yellow-green solution).
- Mn has configuration [Ar] $3\,d^5\,4\,s^2$; all seven electrons can be lost because 3d and 4s are close in energy. Zn has [Ar] $3\,d^{10}\,4\,s^2$; after losing 4s electrons, the 3d subshell is full and stable, so further ionisation requires much higher energy.
- en is bidentate → 3 en ligands give 6 bonds → CN = 6, octahedral. Complex ion is $[Cr(en)_3]^{3+}$ (3 Cl^- outside). en is neutral. Oxidation state of Cr = +3. Overall charge of complex ion = +3.
- Cr: $3d^5\,4s^1$ (five unpaired electrons in 3d, one in 4s). Cr^{3+}: loses $4s^1$ and two 3d electrons → $3d^3$ (three unpaired electrons). The Cr^{3+} ion has a half-filled t_{2g} set ($t_{2g}^3\,e_g^0$) in an octahedral field, giving extra ligand field stabilisation energy (LFSE), making it particularly stable.

8 NAMING, SHAPES AND ISOMERISM OF COMPLEXES

In the previous section we discovered that transition metals form **complex ions** by accepting electron pairs from ligands. We also saw that a single metal ion can exist in several oxidation states and coordinate with different numbers of ligands. This richness gives rise to an enormous variety of compounds. To communicate about them unambiguously, chemists use a systematic nomenclature. Moreover, the spatial arrangement of ligands around the metal centre — the **geometry** of the complex — determines not only its name but also its physical properties, chemical reactivity and even its biological activity. In this section we master the IUPAC naming rules, explore the common shapes of complexes, and study the fascinating phenomenon of **isomerism** where compounds with the same formula behave differently because their atoms are arranged differently in space.

8.1 IUPAC Nomenclature of Coordination Compounds

Naming a coordination compound follows a precise sequence. The guiding principle is: **name the ligands first (in alphabetical order), then the metal centre with its oxidation state**. If the complex is an anion, the metal name takes an **–ate** ending.

METHOD **Naming a Coordination Compound — Step by Step**

1. Identify the complex ion and the counter-ions. Write the formula with the coordination sphere in square brackets, e.g. $[Cu(NH_3)_4]SO_4$.
2. Name the ligands in alphabetical order (ignoring multiplicative prefixes di-, tri-, tetra-, etc.). Use the standard ligand names:
   - Neutral ligands: H_2O = aqua, NH_3 = ammine, CO = carbonyl, NO = nitrosyl.
   - Anionic ligands: Cl^- = chloro, Br^- = bromo, I^- = iodo, F^- = fluoro, CN^- = cyano, OH^- = hydroxo, SO_4^{2-} = sulfato, NO_2^- = nitro (N-bonded) or nitrito (O-bonded), SCN^- = thiocyanato (S-bonded) or isothiocyanato (N-bonded).
   - For organic ligands use the common name: en = ethylenediamine, ox = oxalate, bipy = 2,2'-bipyridine, phen = 1,10-phenanthroline.
3. Indicate the number of each ligand with prefixes: di- (2), tri- (3), tetra- (4), penta- (5), hexa- (6). If the ligand name itself contains a prefix (e.g. ethylenediamine), use bis-, tris-, tetrakis- and enclose the ligand name in parentheses.
4. Name the metal. For a cationic or neutral complex, use the element name (e.g. iron, copper). For an anionic complex, use the Latin root + -ate: **ferrate** (Fe), **cuprate** (Cu), **chromate** (Cr), **argentate** (Ag), **stannate** (Sn), **plumbate** (Pb).
5. Give the oxidation state of the metal in Roman numerals in parentheses immediately after the metal name (no space): iron(III), copper(II), ferrate(II).
6. Name the counter-ions (if any) after the complex cation or before the complex anion,

as for simple ionic compounds.

WARNING **Common Naming Traps**

- Alphabetical order ignores prefixes. ammine (a) comes before chloro (c) even if you have triammine and dichloro.
- Oxidation state ≠ charge on the complex ion. The oxidation state is the charge the metal would have if all ligands were removed with their donor electrons. Calculate it from the overall charge of the complex sphere.
- The '-ate' ending is only for anionic complexes. $K_4[Fe(CN)_6]$ → potassium hexacyanoferrate(II); but $[Fe(CN)_6]^{3-}$ → hexacyanoferrate(III) ion.
- Ligand names: NH_3 is ammine (two m's), not amine; H_2O is aqua, not aquo.
- Ambidentate ligands: Specify the donor atom: NO_2^- = nitro (N) vs nitrito (O); SCN^- = thiocyanato-S vs isothiocyanato-N; CN^- = cyano (C) vs isocyano (N).

WORKED EXAMPLE 10 **Worked Naming Examples**

Name the following compounds:

1. $[Cu(H_2O)_6]SO_4$
2. $K_4[Fe(CN)_6]$
3. $[Co(NH_3)_5Cl]Cl_2$
4. $[Cr(NH_3)_2Cl_2(en)]Cl$ (en = ethylenediamine)

SOLUTION

Solution:

1. Complex cation: $[Cu(H_2O)_6]^{2+}$. Ligands: six aqua → hexaaqua. Metal: copper(II) (overall charge +2, ligands neutral). Counter-ion: sulfate. Name: **hexaaquacopper(II) sulfate**.
2. Complex anion: $[Fe(CN)_6]^{4-}$. Ligands: six cyano → hexacyano. Metal: ferrate(II) (anionic complex, Fe oxidation state +2 because $6\times(-1) + x = -4 \Rightarrow x = +2$). Counter-ion: potassium. Name: **potassium hexacyanoferrate(II)**.
3. Complex cation: $[Co(NH_3)_5Cl]^{2+}$. Ligands: five ammine, one chloro → alphabetical: ammine (a) before chloro (c) → pentaamminechloro. Metal: cobalt(III) (overall +2, $5\times0 + (-1) + x = +2 \Rightarrow x = +3$). Counter-ions: two chloride. Name: **pentaamminechloro-**

cobalt(III) chloride.

4. Complex cation: $[Cr(NH_3)_2Cl_2(en)]^+$. Ligands: two ammine, two chloro, one ethylenediamine → alphabetical: ammine (a), chloro (c), ethylenediamine (e) → diammine dichloro ethylenediamine. Use bis for ethylenediamine? No, ethylenediamine doesn't contain a multiplicative prefix, so di- is fine. But wait: ethylenediamine is a bidentate ligand, its name is "ethylenediamine" which contains "di", so we use bis(ethylenediamine). Actually, the rule is: if the ligand name is complex or contains a multiplicative prefix, use bis, tris, etc. Ethylenediamine contains "di", so we use bis(ethylenediamine). Metal: chromium(III) (overall +1, 2×0 + 2×(−1) + 0 + x = +1 ⇒ x = +3). Counter-ion: chloride. Name: **diamminedichlorido-bis(ethylenediamine)chromium(III) chloride**.

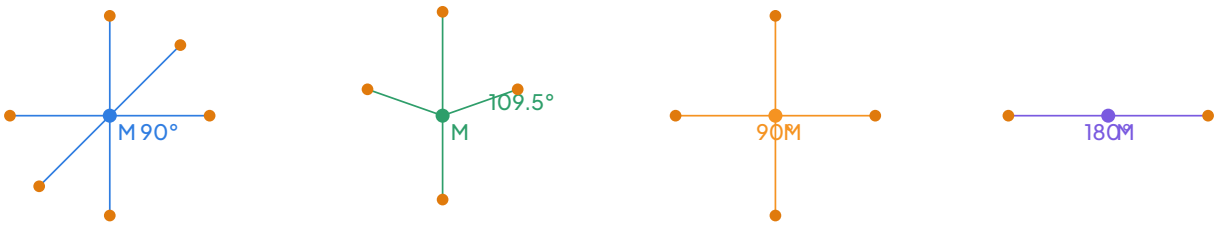
EXERCISE 8 **Naming Practice**

Name the following coordination compounds:

1. $[Ni(NH_3)_6]Cl_2$
2. $Na_2[NiCl_4]$
3. $[Cr(H_2O)_4Cl_2]Cl \cdot_2 H_2O$
4. $[Pt(NH_3)_2Cl_2]$ (neutral complex)
5. $K_3[Cr(C_2O_4)_3]$
6. $[Co(en)_2(NO_2)_2]NO_3$ (en = ethylenediamine)

8.2 Shapes of Complex Ions — Geometry and Coordination Number

The **coordination number** (CN) is the number of donor atoms attached to the metal centre. The geometry adopted minimises ligand–ligand repulsion (VSEPR-like arguments) and is influenced by the metal's d-electron count and ligand field effects. The four most common geometries in the first transition series are:



Octahedral (CN=6)
e.g. $[Fe(H_2O)_6]^{2+}$

Tetrahedral (CN=4)
e.g. $[CoCl_4]^{2-}$

Square Planar (CN=4)
e.g. $[Pt(NH_3)_2Cl_2]$

Linear (CN=2)
e.g. $[Ag(NH_3)_2]^{+}$

Common geometries of transition metal complexes. Bond angles are indicated.

KEY POINTS **Key Geometries and Examples**

- Octahedral (CN=6): Six ligands at the vertices of an octahedron. All bond angles 90° (or 180° for trans pairs). Very common for d^6 (low spin), d^3, d^8 (e.g. $[Co(NH_3)_6]^{3+}$, $[Cr(H_2O)_6]^{3+}$, $[Ni(H_2O)_6]^{2+}$).
- Tetrahedral (CN=4): Four ligands at the vertices of a regular tetrahedron. Bond angles 109.5°. Favoured by d^0, d^5 (high spin), d^{10} metals and bulky ligands (e.g. $[CoCl_4]^{2-}$, $[FeCl_4]^-$, $[Zn(NH_3)_4]^{2+}$).
- Square Planar (CN=4): Four ligands in a plane at the corners of a square. Bond angles 90°. Characteristic of d^8 metals with strong-field ligands: Pt^{2+}, Pd^{2+}, Au^{3+}, Ni^{2+} (e.g. $[Pt(NH_3)_2Cl_2]$, $[Ni(CN)_4]^{2-}$).
- Linear (CN=2): Two ligands opposite each other. Bond angle 180°. Typical for d^{10} ions: Ag^+, Cu^+, Au^+ (e.g. $[Ag(NH_3)_2]^+$, $[CuCl_2]^-$).

8.3 Isomerism in Coordination Compounds

DEFINITION **Isomers**

Isomers are compounds that have the same molecular formula but different arrangements of atoms in space or in the bonding network. In coordination chemistry, isomerism is classified into two broad categories:

1. Structural (constitutional) isomerism — different connectivity (which atoms are bonded to which).
2. Stereoisomerism — same connectivity but different spatial arrangement.

8.4 Structural Isomerism

Three important types occur in coordination compounds:

PROPERTY **Ionization Isomerism**

Arises when a ligand that can also act as a counter-ion exchanges places with the counter-ion outside the coordination sphere. The composition of the coordination sphere changes, so the ions precipitated from solution differ.
Classic example: $[Co(NH_3)_5Br]SO_4$ (red-violet) and $[Co(NH_3)_5SO_4]Br$ (red).

- In $[Co(NH_3)_5Br]SO_4$, the **free ion** in solution is SO_4^{2-} → gives white $BaSO_4$ precipitate with $BaCl_2$; no precipitate with $AgNO_3$ (Br– is bound).
- In $[Co(NH_3)_5SO_4]Br$, the free ion is Br^- → gives pale yellow AgBr precipitate with $AgNO_3$; no precipitate with $BaCl_2$.

PROPERTY **Hydrate (Solvate) Isomerism**

A special case of ionization isomerism where water molecules act as ligands inside the coordination sphere or as water of crystallization outside. The three forms of $CrCl_3 \cdot 6\,H_2O$ are the textbook example:

Formula (modern)	Traditional name	Free Cl^- per formula	Colour
$[Cr(H_2O)_6]Cl_3$	hexaaquachromium(III) chloride	3 (all ionic)	Violet
$[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$	pentaaquachloridochromium(III) chloride monohydrate	2	Light green
$[Cr(H_2O)_4Cl_2]Cl \cdot 2\,H_2O$	tetraaquadichloridochromium(III) chloride dihydrate	1	Dark green

Adding $AgNO_3$ to aqueous solutions of each form precipitates 3, 2, or 1 mole of AgCl per mole of compound respectively.

PROPERTY **Coordination (Linkage) Isomerism**

Arises from **ambidentate ligands** that can bind through two different donor atoms. The connectivity differs: the ligand is attached to the metal via a different atom.
Examples:

- NO_2^-: nitro (N-bonded, M – NO_2) vs nitrito (O-bonded, M – ONO)
- SCN^-: thiocyanato (S-bonded, M – SCN) vs isothiocyanato (N-bonded, M – NCS)
- CN^-: cyano (C-bonded) vs isocyano (N-bonded)

Example pair: $[Co(NH_3)_5NO_2]Cl_2$ (nitropentaamminecobalt(III) chloride) and $[Co(NH_3)_5ONO]Cl_2$ (nitritopentaamminecobalt(III) chloride).

WARNING **Identifying Structural Isomers — Exam Tip**

To distinguish ionization/hydrate isomers, test the free anions in aqueous solution:

- Add $AgNO_3(aq)$: precipitate → free Cl^-, Br^-, I^- present.
- Add $BaCl_2(aq)$: white precipitate → free SO_4^{2-} present.

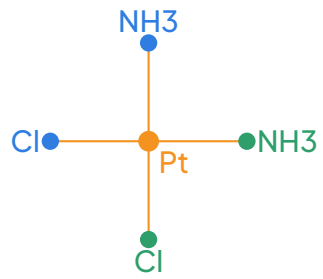
The isomer that gives a precipitate with $AgNO_3$ has the halide as counter-ion; the one that gives a precipitate with $BaCl_2$ has sulfate as counter-ion. For linkage isomers: IR spectroscopy distinguishes M–N vs M–O or M–S vs M–N bonds.

EXPERIMENT **Distinguishing the Three Hydrate Isomers of $CrCl_3 \cdot 6\,H_2O$**

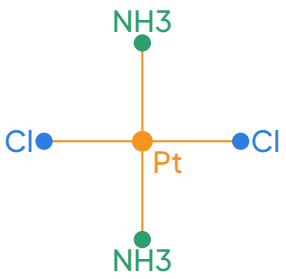
Procedure: Dissolve equal molar amounts of each isomer in water. To each solution add excess $AgNO_3(aq)$. Filter, dry and weigh the AgCl precipitate. Observation: The violet isomer gives 3 mol AgCl, the light green gives 2 mol, the dark green gives 1 mol. Conclusion: The number of free chloride ions (outside the coordination sphere) differs, proving hydration isomerism.

8.5 Stereoisomerism — Geometric (Cis–Trans) Isomerism

Geometric isomerism occurs when ligands can be arranged in different relative positions around the metal centre. It is possible in **square planar** (CN=4) and **octahedral** (CN=6) complexes, but **not** in tetrahedral or linear geometries (all positions equivalent).



cis-$[Pt(NH_3)_2Cl_2]$
Cl–Pt–Cl = 90°
Dipole moment ≠ 0



trans-$[Pt(NH_3)_2Cl_2]$
Cl–Pt–Cl = 180°
Dipole moment = 0

Geometric isomers of square planar $[Pt(NH_3)_2Cl_2]$. cis-platin (left) and trans-platin (right).

DID YOU KNOW? **Cis-Platin: Geometry Saves Lives**

Cis-platin (cis-diamminedichloridoplatinum(II)) is one of the most successful **anti-cancer drugs**, used to treat testicular, ovarian, bladder and lung cancers. It works by binding to DNA, forming cross-links that block replication. The trans isomer is inactive — its geometry prevents the necessary bifunctional binding to adjacent guanine bases. This is a spectacular example of how **stereochemistry controls biological activity**. The discovery of cisplatin's activity (Rosenberg, 1965) revolutionised chemotherapy and earned a place in every inorganic chemistry syllabus.

PROPERTY **Geometric Isomerism in Octahedral Complexes**

For MA_4B_2 type (e.g. $[Co(NH_3)_4Cl_2]^+$):

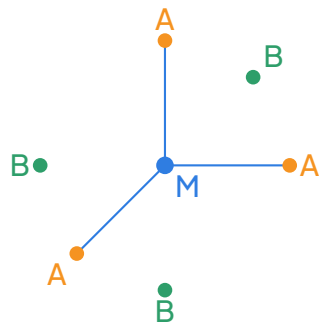
- cis: two B ligands adjacent (90°).
- trans: two B ligands opposite (180°).

For MA_3B_3 type (e.g. $[Cr(NH_3)_3Cl_3]$):

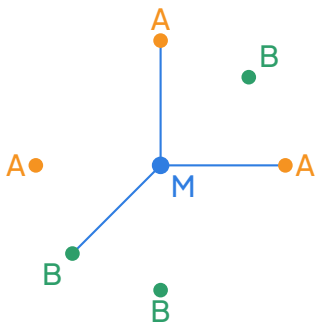
- fac (facial): three identical ligands occupy one face of the octahedron (all mutually cis).
- mer (meridional): three identical ligands lie in a plane passing through the metal (one trans pair).

For $M(AA)_2B_2$ type (AA = bidentate ligand, e.g. $[Co(en)_2Cl_2]^+$):

- cis: two B adjacent. Chiral (exists as enantiomers).
- trans: two B opposite. Achiral (centre of symmetry).



fac-[MA_3B_3]
A's on one face



mer-[MA_3B_3]
A's in a meridian

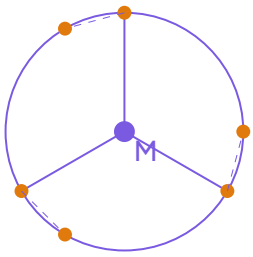
Facial (fac) and meridional (mer) isomers of octahedral MA_3B_3 complexes.

8.6 Stereoisomerism — Optical Isomerism (Enantiomerism)

Optical isomers (**enantiomers**) are non-superimposable mirror images of each other, like left and right hands. They rotate plane-polarised light in opposite directions. A complex is **chiral** (optically active) if it lacks:

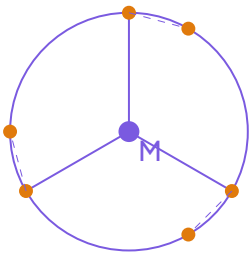
- a plane of symmetry (σ),
- a centre of inversion (i),
- an improper rotation axis (S_n).

Octahedral complexes with **bidentate ligands** are the classic sources of optical isomerism. The tris-chelate [$M(AA)_3$] (AA = bidentate ligand like ethylenediamine, en) exists as a pair of enantiomers (Λ and Δ).



Λ-**[M(en)3]^{3}+**
Left-handed propeller

Mirror



Δ-**[M(en)3]^{3}+**
Right-handed propeller

Optical isomers (enantiomers) of $[M(en)_3]^{3+}$ (Λ and Δ). They are non-superimposable mirror images.

KEY POINTS **Optical Isomerism Checklist**

- Requires: Octahedral geometry + at least two bidentate ligands (or three) OR specific unsymmetrical tetradentate ligands.
- Common chiral formulas: $[M(AA)_3]$, $[M(AA)_2B_2]$ (cis only), $[M(AA)_2BC]$ (cis), [M(ABCD)EF] (rare).
- Never chiral: trans isomers (have a centre of symmetry), tetrahedral complexes with monodentate ligands (rapid racemisation), square planar complexes.
- Separation: Enantiomers have identical physical properties (except optical rotation) and chemical properties towards achiral reagents. They can be separated by chromatography on a chiral stationary phase or by forming diastereomeric salts with a chiral resolving agent.

WORKED EXAMPLE 11 **Identifying Isomer Types**

For each pair, state the type of isomerism and suggest a simple chemical test to distinguish them.

1. $[Co(NH_3)_5Cl]SO_4$ and $[Co(NH_3)_5SO_4]Cl$
2. $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$
3. cis- and trans-$[Pt(NH_3)_2Cl_2]$
4. Λ-$[Cr(en)_3]^{3+}$ and Δ-$[Cr(en)_3]^{3+}$
5. $[Co(NH_3)_5NO_2]Cl_2$ and $[Co(NH_3)_5ONO]Cl_2$

SOLUTION

Solution:

1. Ionization isomerism. Test: Add $AgNO_3$(aq) → first gives no precipitate (Cl- bound), second gives AgCl(s). Add $BaCl_2$(aq) → first gives $BaSO_4$(s), second gives no precipitate.
2. Hydrate isomerism. Test: Add $AgNO_3$(aq) to equal molar solutions. First gives 3 mol AgCl, second gives 2 mol AgCl per mol of compound.
3. Geometric (cis–trans) isomerism. Test: Measure dipole moment (cis has $\mu > 0$, trans has $\mu = 0$). Or test biological activity: cis-platin binds DNA, trans does not.
4. Optical isomerism. Test: Polarimetry — they rotate plane-polarised light equally but in opposite directions. Chemical properties identical towards achiral reagents.

5. Linkage (coordination) isomerism. Test: IR spectroscopy — M – NO_2 (nitro) shows N–O stretches 1300–1400 cm^{-1}; M – ONO (nitrito) shows asymmetric O–N–O stretch 1000–1100 cm^{-1}.

8.7 Summary of Isomerism in Coordination Compounds

Type	Key Feature	Examples
Ionization	Counter-ion exchanges with ligand in coordination sphere	$[Co(NH_3)_5Br]SO_4$ vs $[Co(NH_3)_5SO_4]Br$
Hydrate	Water as ligand vs water of crystallization	$[Cr(H_2O)_6]Cl_3$, $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$, $[Cr(H_2O)_4Cl_2]Cl \cdot 2\,H_2O$
Coordination (Linkage)	Ambidentate ligand binds through different donor atoms	$[Co(NH_3)_5NO_2]^{2+}$ (nitro) vs $[Co(NH_3)_5ONO]^{2+}$ (nitrito); SCN^- (thiocyanato-S vs isothiocyanato-N)
Geometric (cis–trans)	Different relative positions (square planar, octahedral)	cis-/trans-$[Pt(NH_3)_2Cl_2]$, fac-/mer-$[Cr(NH_3)_3Cl_3]$
Optical	Non-superimposable mirror images (chiral)	Λ- and Δ-$[Cr(en)_3]^{3+}$, cis-$[Co(en)_2Cl_2]^+$

WARNING **Exam Checklist for Isomerism Questions**

- Draw structures clearly — use wedge/dash or proper 3D sketches for octahedral complexes.
- State the type of isomerism precisely (e.g. "geometric isomerism", not just "isomerism").
- For ionization/hydrate isomers: specify which ions are free in solution and give the observation with $AgNO_3$ and $BaCl_2$.
- For linkage isomers: name the ligand correctly (nitro vs nitrito, thiocyanato vs isothiocyanato) and mention IR spectroscopy.
- For optical isomers: mention "non-superimposable mirror images", "chiral", "rotate plane-polarised light in opposite directions".
- Don't confuse geometric and optical: cis-$[M(AA)_2B_2]$ is chiral (optical isomers exist); trans-$[M(AA)_2B_2]$ is achiral.
- Count total isomers: e.g. $[M(AA)_2B_2]$ has 3 stereoisomers (1 trans + 2 cis enantiomers).

EXERCISE 9 **Comprehensive Isomerism Exercise**

1. Draw all possible isomers (geometric and optical) for $[Co(en)_2Cl_2]^+$. Label each as cis or trans, and indicate which are chiral.
2. A compound $CrCl_3 \cdot 6\,H_2O$ (isomer A) gives 3 mol AgCl per mole with $AgNO_3$. Isomer B gives 1 mol AgCl. Write the modern formulas for A and B and name them.
3. $[Pt(NH_3)_2Cl_2]$ exists as two geometric isomers. Which one has a net dipole moment? Explain why the other has zero dipole moment.
4. The complex $[Ni(CN)_4]^{2-}$ is square planar while $[NiCl_4]^{2-}$ is tetrahedral. Explain this difference in terms of the ligand field strength and the d-electron configuration of Ni^{2+}.
5. How many stereoisomers exist for $[Cr(ox)_3]^{3-}$ (ox = oxalate, bidentate)? Draw them and label their absolute configurations (Λ/Δ).

SOLUTION **Exercise 1**

1. $[Co(en)_2Cl_2]^+$ (octahedral, en = bidentate):

- trans isomer: two Cl opposite. Has a centre of symmetry → achiral. One structure.
- cis isomer: two Cl adjacent (90°). No plane of symmetry → chiral. Exists as a pair of enantiomers (Λ and Δ).
- Total: 3 stereoisomers (1 trans + 2 cis enantiomers).

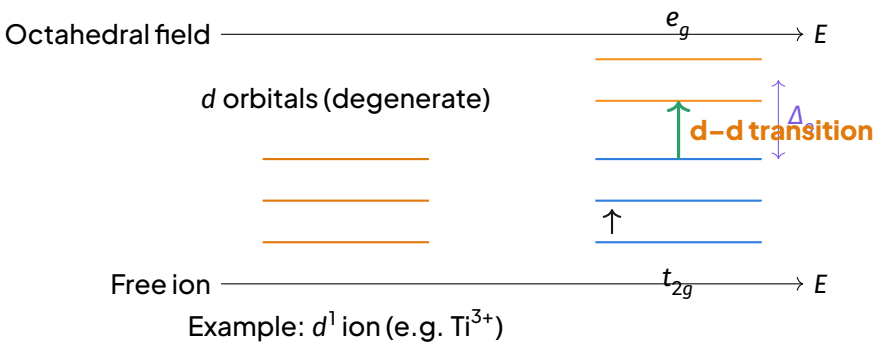
2. Isomer A: $[Cr(H_2O)_6]Cl_3$ — hexaaquachromium(III) chloride. Isomer B: $[Cr(H_2O)_4Cl_2]Cl \cdot 2\,H_2O$ — tetraaquadichloridochromium(III) chloride dihydrate.
3. cis-$[Pt(NH_3)_2Cl_2]$ has a net dipole moment because the two Pt – Cl bond dipoles and two Pt – NH_3 bond dipoles do not cancel (vector sum ≠ 0). trans-$[Pt(NH_3)_2Cl_2]$ has zero dipole moment because the two Pt – Cl dipoles are opposite and equal (cancel), and the two Pt – NH_3 dipoles are opposite and equal (cancel).
4. Ni^{2+} is d^8. CN^- is a strong-field ligand → large Δ → low-spin configuration favours square planar (all electrons paired in lower orbitals, $d_{x^2-y^2}$ empty). Cl^- is a weak-field ligand → small Δ → high-spin → tetrahedral geometry avoids pairing energy penalty.
5. $[Cr(ox)_3]^{3-}$ is of type $[M(AA)_3]$. It exists as a pair of enantiomers: Λ (left-handed propeller) and Δ (right-handed propeller). Total = 2 optical isomers. No geometric isomers possible.

With naming, shapes and isomerism mastered, we have the language and the spatial intuition to discuss the remaining hallmark properties of transition metal complexes: **colour**, **catalysis**, **magnetism** and their **technological applications** — the subject of our final section.

9 COLOUR, CATALYSIS, MAGNETISM AND USES OF D-BLOCK ELEMENTS

In the previous section we mastered the nomenclature, geometries and isomerism of coordination complexes. We now turn to the consequences of the partially filled d-subshell that defines a transition element. It is this incomplete d-shell that gives rise to the vivid colours, remarkable catalytic power, magnetic behaviour and the myriad industrial uses of the d-block metals and their compounds. Understanding these properties not only explains everyday observations — from the blue of copper(II) sulfate to the platinum in a car's exhaust — but also provides the conceptual toolkit that GCE examiners love to test.

9.1 Origin of Colour in Transition Metal Compounds



d-Orbital splitting in an octahedral ligand field. The five degenerate d-orbitals split into a lower-energy t_{2g} triplet (d_{xy}, d_{yz}, d_{zx}) and a higher-energy e_g doublet ($d_{z^2}, d_{x^2-y^2}$). A d-electron can absorb a photon of visible light and jump from t_{2g} to e_g (d–d transition). The energy gap Δ_o corresponds to the wavelength absorbed; the complementary colour is observed.

When ligands approach a transition metal ion, the electrostatic field they create lifts the degeneracy of the five d-orbitals. In an octahedral complex the orbitals split into a lower-energy set (t_{2g}: d_{xy}, d_{yz}, d_{zx}) and a higher-energy set (e_g: $d_{z^2}, d_{x^2-y^2}$). The energy gap is denoted Δ_o (crystal field splitting parameter). If the metal ion has a partially filled d-subshell (d^1–d^9), an electron in a lower orbital can absorb a photon of visible light and be promoted to a higher orbital — a **d–d transition**. The wavelength absorbed depends on Δ_o; the colour we see is the complementary colour of the absorbed light.

PROPERTY **Origin of Colour**

Transition metal compounds are coloured because d-electrons absorb specific wavelengths of visible light when jumping between split d-orbitals (d–d transitions). The colour observed is the complementary colour of the absorbed light.

If the d-subshell is empty (d^0) or completely full (d^{10}), no d–d transition is possible and the com-

pound is typically colourless (or white as a solid). This explains why Sc^{3+} ($3d^0$) and Zn^{2+} ($3d^{10}$) form colourless solutions and white salts. Note that Sc^{3+} and Zn^{2+} are d-block elements but not transition metals by the IUPAC definition (no partially filled d-subshell in any stable oxidation state).

EXEMPLE **Why $[Cu(H_2O)_6]^{2+}$ is blue**

The Cu^{2+} ion has a d^9 configuration. In an octahedral aqua complex the single hole in the e_g level allows a d–d transition from t_{2g} to e_g. The absorption maximum lies in the orange-red region (≈ 600–800 nm), so the transmitted light appears blue.

9.2 Effect of Ligand on Colour

The magnitude of Δ_o depends strongly on the nature of the ligand (spectrochemical series). Strong-field ligands (e.g. CN^-, CO, NH_3) cause a larger splitting than weak-field ligands (e.g. H_2O, F^-, Cl^-). A larger Δ_o means higher energy (shorter wavelength) light is absorbed, shifting the observed colour.

PROPERTY **Spectrochemical Series (common ligands, increasing field strength)**

$I^- < Br^- < SCN^-(S) < Cl^- < F^- < OH^- < H_2O < NH_3 < en < CN^- < CO$

Complex	Ligand field strength	Δ_o	Observed colour
$[Cu(H_2O)_6]^{2+}$	Weak field (H_2O)	Moderate	Blue
$[Cu(NH_3)_4(H_2O)_2]^{2+}$	Stronger field (NH_3)	Larger	Deep blue (indigo)
$[CuCl_4]^{2-}$	Weak field (Cl^-), tetrahedral	Smaller ($\Delta_t \approx \frac{4}{9}\Delta_o$)	Yellow-green

The tetrahedral $[CuCl_4]^{2-}$ ion has a much smaller splitting, so it absorbs lower-energy (red) light and appears yellow-green. This ligand-dependent colour change is a classic qualitative analysis observation.

WARNING **Charge-Transfer Colour**

Some intensely coloured complexes (e.g. MnO_4^-, $Cr_2O_7^{2-}$, $[Fe(SCN)]^{2+}$) owe their colour to ligand-to-metal charge transfer (LMCT) or metal-to-ligand charge transfer (MLCT), not d–d transitions. These occur when the metal has a high oxidation state (d^0 or d^{10}) or low-lying empty ligand orbitals. They are not forbidden by selection rules and give very intense colours.

9.3 Catalytic Activity of Transition Metals

Transition metals and their compounds are outstanding catalysts because they can:

1. **Adopt multiple oxidation states** — providing alternative reaction pathways with lower activation energies via redox cycles.
2. **Adsorb reactants on their surface** — using vacant d-orbitals to form temporary bonds, weakening reactant bonds and increasing collision efficiency (heterogeneous catalysis).

EXPERIMENT **Industrial Catalysts — Key Examples for GCE**

- Haber process (ammonia synthesis): Finely divided **Fe** (with K_2O/Al_2O_3 promoters) catalyses $N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$. Fe cycles between oxidation states and provides a surface for N ≡ N bond cleavage.
- Contact process (sulfuric acid): **V2O5** catalyses $2\,SO_2(g) + O_2(g) \rightleftharpoons 2\,SO_3(g)$. The vanadium cycles between +5 and +4 oxidation states: $V_2O_5 + SO_2 \longrightarrow V_2O_4 + SO_3$, $V_2O_4 + \frac{1}{2}\,O_2 \longrightarrow V_2O_5$.
- Hydrogenation of oils: **Ni** (or Pt/Pd) catalyses addition of H_2 to C=C bonds in unsaturated fats. H_2 dissociates on the metal surface; H atoms add to the adsorbed alkene.
- Catalytic converters: **Pt**/**Rh** (on a ceramic honeycomb) oxidise CO and hydrocarbons to CO_2 and H_2O, and reduce NO_x to N_2 and O_2. The metals cycle between oxidation states (e.g. Pt^0/Pt^{2+}).

9.4 Magnetic Properties: Paramagnetism and Diamagnetism

Substances respond differently to an applied magnetic field. The behaviour is dictated by the presence or absence of unpaired electrons.

DEFINITION **Paramagnetism**

Paramagnetism is the weak attraction of a substance to a magnetic field due to the presence of unpaired electrons. Each unpaired electron acts as a tiny magnet (spin magnetic moment). The magnetic moment μ_s (spin-only formula) is $\mu_s = \sqrt{n(n+2)}\,\mu_B$, where n = number of unpaired electrons and μ_B = Bohr magneton.

DEFINITION **Diamagnetism**

Diamagnetism is the weak repulsion of a substance from a magnetic field. It occurs in all matter but is only observable when there are no unpaired electrons (all electrons paired). A diamagnetic substance is not "non-magnetic" — it is repelled.

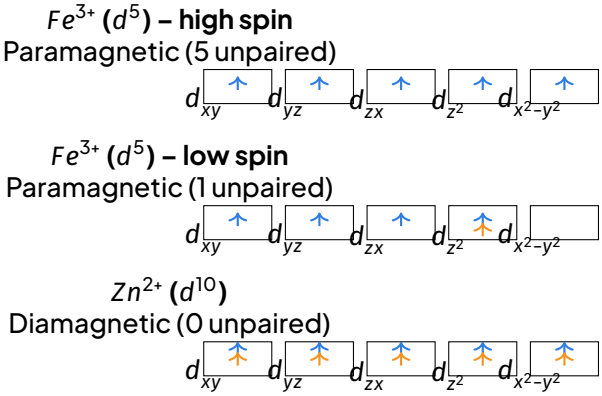
WARNING **Diamagnetic ≠ Non-magnetic**

A substance with all electrons paired is **diamagnetic** — it is weakly repelled by a magnetic field. Do not call it "non-magnetic"; that term is ambiguous and incorrect in a physics/chemistry context.

To predict magnetic behaviour we write the d-electron configuration in **electron-in-boxes** notation (Hund's rule: maximise unpaired electrons before pairing). For octahedral complexes we must consider whether the ligand field is strong (low-spin) or weak (high-spin).

METHOD **Predicting Magnetic Behaviour of Transition Metal Ions/Complexes**

1. Determine the oxidation state of the metal ion.
2. Write its d-electron count (d^n).
3. Draw the five d-orbitals as boxes. For an octahedral complex, split into t_{2g} (3 boxes) and e_g (2 boxes).
4. Decide high-spin or low-spin: weak-field ligands → high-spin (Hund's rule across all five boxes); strong-field ligands → low-spin (fill t_{2g} completely before e_g).
5. Fill the boxes: one electron per box before pairing.
6. Count unpaired electrons.
7. Unpaired > 0 → paramagnetic; Unpaired = 0 → diamagnetic.



Electron-in-boxes diagrams. Fe^{3+} (d^5, high-spin) has five unpaired electrons → strongly paramagnetic. Fe^{3+} (d^5, low-spin) has one unpaired electron → paramagnetic. Zn^{2+} (d^{10}) has all electrons paired → diamagnetic.

WORKED EXAMPLE 12 **Predicting magnetic behaviour**

Predict the magnetic behaviour (paramagnetic/diamagnetic) and number of unpaired electrons for: (a) Cr^{3+}, (b) Cu^{2+}, (c) Zn^{2+}, (d) $[Fe(CN)_6]^{4-}$, (e) $[Fe(H_2O)_6]^{2+}$.

SOLUTION

Solution:

1. Cr^{3+}: Cr is [Ar] $3d^5 4s^1$. Cr^{3+} loses $4s^1$ and two $3d$ electrons → d^3. High-spin (weak field): three electrons occupy t_{2g} singly → 3 unpaired → paramagnetic. $\mu_s = \sqrt{15}\,\mu_B \approx 3.87\,\mu_B$.
2. Cu^{2+}: Cu is [Ar] $3d^{10} 4s^1$. Cu^{2+} loses $4s^1$ and one $3d$ electron → d^9. In octahedral field: $t_{2g}^6 e_g^3$ → 1 unpaired → paramagnetic. $\mu_s = \sqrt{3}\,\mu_B \approx 1.73\,\mu_B$.
3. Zn^{2+}: Zn is [Ar] $3d^{10} 4s^2$. Zn^{2+} loses $4s^2$ → d^{10}. All orbitals full → 0 unpaired → diamagnetic.
4. [Fe(CN)6]$^{4-}$: Fe oxidation state +2 (d^6). CN^- is strong field → low-spin. $t_{2g}^6 e_g^0$ → 0 unpaired → diamagnetic.
5. [Fe(H2O)6]$^{2+}$: Fe oxidation state +2 (d^6). H_2O is weak field → high-spin. $t_{2g}^4 e_g^2$ → 4 unpaired → paramagnetic. $\mu_s = \sqrt{24}\,\mu_B \approx 4.90\,\mu_B$.

9.5 Uses of d-Block Elements and Compounds

The unique combination of mechanical strength, electrical conductivity, variable oxidation states, catalytic activity and colour makes d-block elements indispensable.

Element / Compound	Key Uses
Fe (steel)	Construction (reinforced concrete, girders), vehicle bodies, machinery. Alloyed with C, Cr, Ni, Mn for tailored properties.
Cu	Electrical wiring (high conductivity, ductility), plumbing, alloys (brass Cu – Zn, bronze Cu – Sn).
Ti	Lightweight high-strength alloys (aircraft, implants); TiO_2 as brilliant white pigment in paints, plastics, sunscreen.
Cr	Electroplating (decorative, corrosion-resistant); stainless steel (Fe – Cr – Ni); $K_2Cr_2O_7$ as oxidising agent.
Ag	Photography (AgBr decomposition), jewellery, electrical contacts, antimicrobial coatings.
KMnO4	Powerful oxidising agent in redox titrations, water treatment, disinfectant, qualitative analysis.
Ni	Hydrogenation catalyst (margarine), rechargeable batteries (Ni-Cd, Ni-MH), stainless steel.
Pt/Rh	Catalytic converters, laboratory crucibles, chemotherapy drugs (cisplatin).

9.6 Summary Table: Characteristic Properties and the Partially Filled d-Subshell

All the hallmark properties of transition metals stem from the same root: an incomplete d-subshell in at least one stable oxidation state.

Property	Origin in partially filled d-subshell	Typical GCE question
Variable oxidation states	Small energy gap between $(n - 1)d$ and ns orbitals; electrons can be lost from both.	"Explain why Mn shows oxidation states from +2 to +7."
Formation of coloured compounds	d-d transitions between split d-orbitals absorb visible light.	"Why is $ZnSO_4$ white while $CuSO_4$ is blue?"
Catalytic activity	Ability to change oxidation state (redox cycles) and to adsorb reactants on surface via vacant d-orbitals.	"Describe the role of V_2O_5 in the Contact process."
Paramagnetism	Unpaired d-electrons → net spin magnetic moment.	"Predict the magnetic behaviour of $[Fe(CN)_6]^{4-}$."
Complex formation	Vacant low-energy d-orbitals accept lone pairs from ligands; crystal field stabilisation.	"Draw the shape of $[Ni(CN)_4]^{2-}$ and state its magnetic property."

KEY POINTS **Key Points for Revision**

- Colour arises from d–d transitions in split d-orbitals; d^0 and d^{10} ions are colourless (white solids).
- Ligand field strength alters Δ_o, changing the colour (spectrochemical series: $I^- < Br^- < Cl^- < F^- < H_2O < NH_3 < CN^- < CO$).
- Tetrahedral splitting $\Delta_t \approx \frac{4}{9}\Delta_o$; tetrahedral complexes often have different colours.
- Charge-transfer (LMCT/MLCT) gives intense colours in some d^0/d^{10} species (e.g. MnO_4^-).
- Catalysis exploits variable oxidation states (homogeneous) and surface adsorption (heterogeneous).
- Know the major industrial catalysts: Fe (Haber), V_2O_5 (Contact), Ni (hydrogenation), Pt/Rh (converters).
- Paramagnetism = unpaired electrons (attracted); diamagnetism = all electrons paired (repelled).
- Use electron-in-boxes to count unpaired electrons; remember high-spin vs low-spin for octahedral complexes.
- Spin-only magnetic moment $\mu_s = \sqrt{n(n+2)}\,\mu_B$.
- Common uses: Fe/steel, Cu/wiring, Ti/TiO_2, Cr/plating, Ag/photography, $KMnO_4$/oxidant.

KEY POINTS **GCE Favourite**

A classic question: "Explain why zinc(II) compounds are white." The expected answer: "Zn^{2+} has a $3d^{10}$ configuration — the d-subshell is completely full, so no d–d transitions are possible; no visible light is absorbed." Always mention the d^{10} configuration explicitly. Another favourite: "State the magnetic behaviour of $[Fe(CN)_6]^{4-}$ and explain." Answer: "Diamagnetic; Fe(II) is d^6, CN^- is strong field → low-spin $t_{2g}^6 e_g^0$, no unpaired electrons."

With this section we complete the descriptive inorganic chemistry of the halogens, Group IV and the d-block. You now have a coherent framework linking electronic structure to the rich chemical behaviour that makes these elements central to both the laboratory and the modern world. Keep practising the electron-in-boxes diagrams and the qualitative analysis tables — they are the keys to high marks in Paper 2 and Paper 3.

10 INTEGRATION ACTIVITY

11 INTEGRATED PRACTICAL ACTIVITY: QUALITATIVE ANALYSIS OF HALOGENS, GROUP IV AND D-BLOCK ELEMENTS

11.1 Aims of the Activity

- Consolidate and integrate knowledge from the whole chapter: halogen chemistry (halide tests, conc. H_2SO_4 reactions), Group IV chemistry (lead(II) chloride behaviour), and d-block chemistry (transition metal identification, complex formation, naming, shapes, isomerism, colour, magnetism).
- Develop practical planning skills: design a logical flowchart for qualitative analysis using limited reagents.
- Practice writing balanced ionic equations with state symbols and drawing mechanisms where appropriate.
- Apply IUPAC nomenclature for coordination compounds (oxidation states, ligand names, shapes, stereochemistry).
- Evaluate safety hazards associated with chlorine gas, lead compounds, and cyanide complexes in a Cameroonian school and domestic context.

11.2 Situation

Context: The chemistry laboratory of Lycée Bilingue d'Etoug-Ebe in Yaoundé is preparing for the practical session of the GCE Advanced Level mock examinations. During the inventory, the laboratory technician discovers five unlabelled containers left on the preparation bench after the previous term's practicals.

The Unknowns:

- Four white crystalline solids labelled **Solid A**, **Solid B**, **Solid C**, and **Solid D**.
- One pale green aqueous solution labelled **Solution E**.

The Suspects (known to be in the school's chemical store):

1. Sodium chloride (NaCl)
2. Sodium bromide (NaBr)
3. Lead(II) chloride ($PbCl_2$)
4. Potassium hexacyanoferrate(II) ($K_4[Fe(CN)_6]$)

5. Either Iron(II) sulfate ($FeSO_4$) or Chromium(III) sulfate ($Cr_2(SO_4)_3$) — the label on the stock bottle is torn.

Available Reagents (standard school bench reagents): Dilute nitric acid ($HNO_3(aq)$), Silver nitrate solution ($AgNO_3(aq)$), Ammonia solution ($NH_3(aq)$) — both dilute and concentrated, Concentrated sulfuric acid ($H_2SO_4(l)$), Sodium hydroxide solution ($NaOH(aq)$), Bunsen burners, Nichrome wires, Test tubes, Safety goggles, Fume cupboard (functional).

Your Mission: As a team of Upper Sixth chemists, you are tasked with identifying all five unknowns conclusively. You must design a systematic experimental flowchart, predict every observation, write all relevant ionic equations, name any coordination complexes formed (giving their shapes and types of isomerism where applicable), and produce a final identification report. The activity concludes with a safety briefing on the handling of chlorine-based bleach (common in Cameroonian households as eau de Javel) and lead compounds.

11.3 Tasks

1. **Flowchart Design (Solids):** Draw a detailed flowchart showing the sequence of tests to distinguish Solids A–D. Start with solubility in water, then use $AgNO_3/NH_3$ for halide identification, concentrated H_2SO_4 test for halide vs non-halide behaviour, and a specific test for the ferrocyanide ion. Indicate expected observations at each decision node.

2. **Flowchart Design (Solution E):** Draw a flowchart to distinguish between $Fe^{2+}(aq)$ and $Cr^{3+}(aq)$. Use $NaOH(aq)$ (dropwise then excess), $NH_3(aq)$ (dropwise then excess), and a confirmatory test for Fe^{2+} (e.g., oxidation to Fe^{3+} followed by $K_4[Fe(CN)_6]$ or $K_3[Fe(CN)_6]$).

3. **Prediction Table:** For each solid (A–D) and Solution E, tabulate the expected observations for every test in your flowcharts.

4. **Ionic Equations:** Write balanced ionic equations (with state symbols) for every positive reaction observed in Tasks 1 and 2. Include the formation of precipitates, dissolution of precipitates in excess reagent (complex formation), and redox reactions with conc. H_2SO_4.

5. **Complex Analysis:** For every coordination complex formed in your equations (e.g., $[Ag(NH_3)_2]^+$, $[PbCl_4]^{2-}$, $[Fe(CN)_6]^{4-}$, $[Cr(NH_3)_6]^{3+}$, $[Cr(OH)_4]^-$, $[Fe(CN)_6]^{3-}$), provide:
   - The systematic IUPAC name.
   - The oxidation state of the central metal.
   - The coordination number and predicted shape (geometry).
   - The type(s) of isomerism possible for that complex formula (geometric, optical, ionization, hydrate).

6. **Identification Report:** Based on the hypothetical observations below, identify Solids A–D and Solution E. Justify each identification by referencing the specific observation and the underlying chemical principle.

   Hypothetical Observations:

   - **Solid A:** Dissolves in water. Gives white ppt with $AgNO_3/HNO_3$, soluble in dilute NH_3. With conc. H_2SO_4, evolves colourless acidic gas (fumes in air), no colour change.

- **Solid B:** Dissolves in water. Gives cream ppt with $AgNO_3/HNO_3$, soluble in conc. NH_3 but not dilute. With conc. H_2SO_4, evolves brown/orange fumes and a gas that bleaches damp litmus.
- **Solid C:** Sparingly soluble in cold water, soluble in hot water. Gives white ppt with $AgNO_3/HNO_3$, insoluble in NH_3. With conc. H_2SO_4, no gas evolution, dissolves on heating.
- **Solid D:** Dissolves in water. No ppt with $AgNO_3/HNO_3$. With conc. H_2SO_4, no visible reaction. Adding Fe^{3+} solution gives a deep blue precipitate (Prussian blue).
- **Solution E:** Pale green. With NaOH (dropwise): dirty green ppt, insoluble in excess, turns brown on standing. With NH_3 (dropwise): same dirty green ppt, insoluble in excess. Acidified solution + $K_3[Fe(CN)_6]$ gives a deep blue ppt.

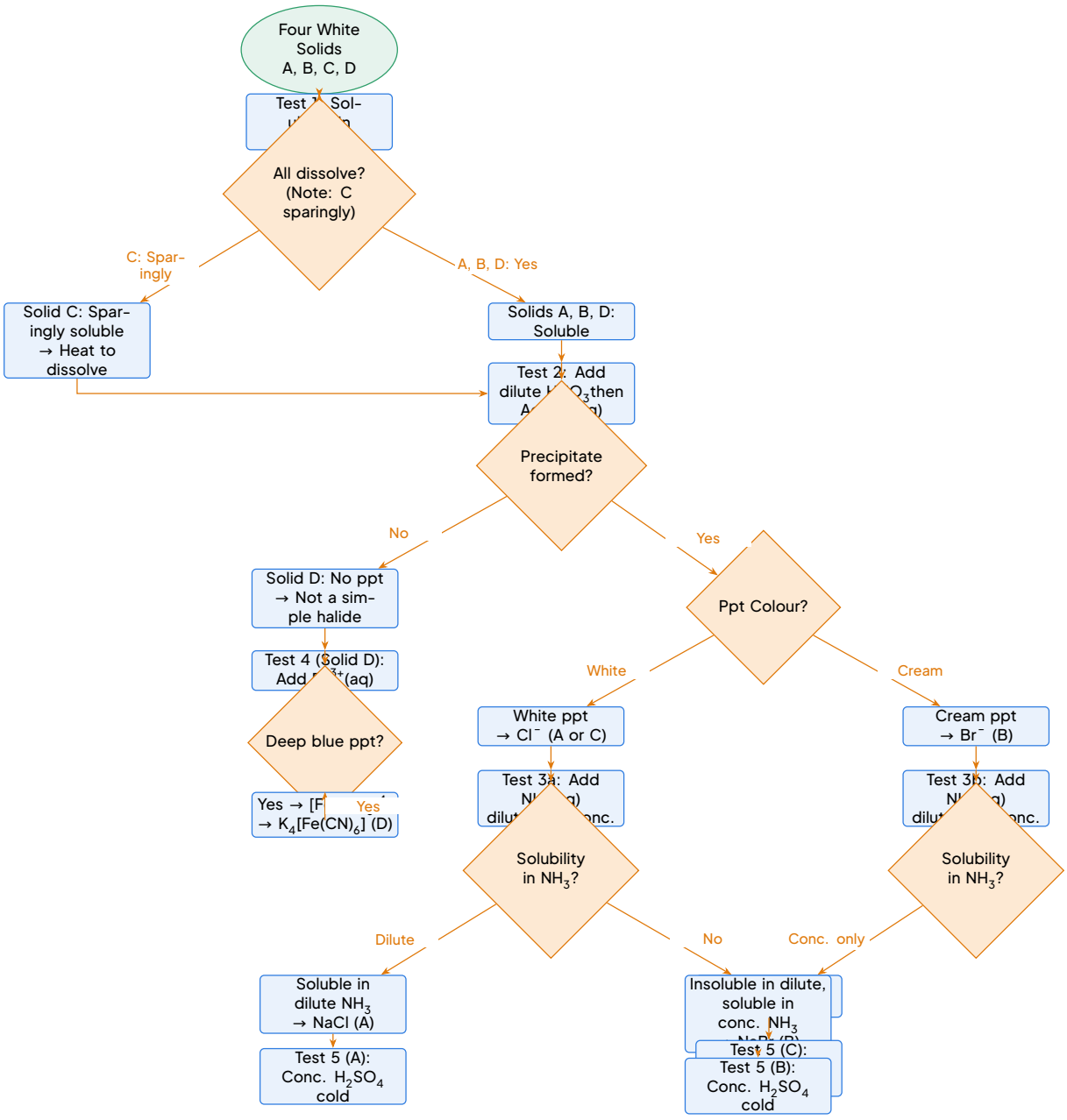
7. **Safety Discussion:** Write a short briefing (150–200 words) for a school assembly in Yaoundé on the dangers of:

   - Mixing household bleach (eau de Javel, NaOCl) with acidic toilet cleaners (HCl-based) or ammonia.
   - Using lead-based pigments in traditional paints or pottery glazes.
   - Improper disposal of potassium ferrocyanide waste.

   Include the relevant chemical equations for the hazardous reactions.

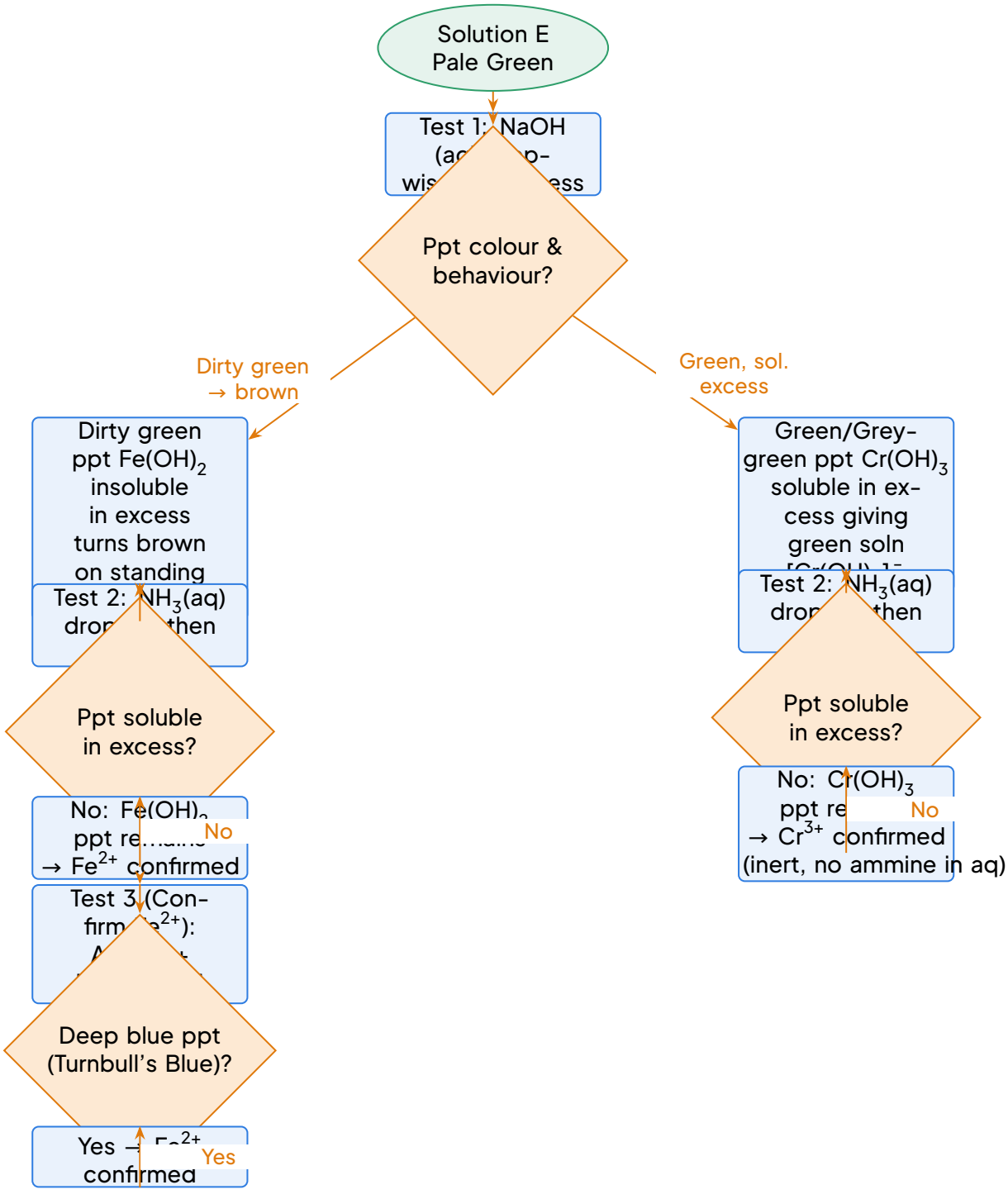
11.4 Model Answer

11.5 Task 1: Flowchart for Solids A–D



Flowchart for identification of Solids A–D.

11.6 Task 2: Flowchart for Solution E



Flowchart for identification of Solution E (Fe^{2+} vs Cr^{3+}).

11.7 Task 3: Prediction Table of Observations

Test	Solid A (NaCl)	Solid B (NaBr)	Solid C ($PbCl_2$)	Solid D ($K_4[Fe(CN)_6]$)	Solution E (Fe^{2+})
Solubility (cold)	Soluble, colourless	Soluble, colourless	Sparingly soluble	Soluble, colourless	— (already solution)
$AgNO_3/HNO_3$	White ppt (AgCl)	Cream ppt (AgBr)	White ppt (AgCl)	No ppt	—
NH_3 (dilute)	AgCl dissolves	AgBr insoluble	AgCl insoluble	—	—
NH_3 (conc.)	—	AgBr dissolves	AgCl insoluble	—	—
Conc. H_2SO_4 (cold) no colour change SO_2 (colourless), bleaches litmus	HCl gas (white fumes), Br_2 vapour (brown), No reaction (cold)	No visible reaction	—		
Conc. H_2SO_4 (heat) toxic)	— —	—	Dissolves on heating	Decomposes (HCN gas,	
Fe^{3+}(aq) (Prussian Blue)	— —	—	—	Deep blue ppt	
NaOH (dropwise) $Fe(OH)_2$	—	—	—	—	Dirty green ppt
NaOH (excess) $(Fe(OH)_3)$	—	—	—	—	Insoluble, turns brown
NH_3 (dropwise) $Fe(OH)_2$	—	—	—	—	Dirty green ppt
NH_3 (excess)	—	—	—	—	Insoluble
Acidified + $K_3[Fe(CN)_6]$ (Turnbull's Blue)	—	—	—	—	Deep blue ppt

11.8 Task 4: Ionic Equations

Solid A (NaCl):

$$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s) \quad \text{(white ppt)}$$
$$AgCl(s) + 2\,NH_3(aq) \longrightarrow [Ag(NH_3)_2]^+(aq) + Cl^-(aq)$$
$$NaCl(s) + H_2SO_4(l) \rightarrow NaHSO_4(s) + HCl(g)\uparrow \quad \text{(white fumes with } NH_3\text{)}$$

Solid B (NaBr):

$$Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s) \quad \text{(cream ppt)}$$
$$AgBr(s) + 2\,NH_3(aq) \longrightarrow [Ag(NH_3)_2]^+(aq) + Br^-(aq) \quad \text{(requires conc. } NH_3\text{)}$$
$$2\,NaBr(s) + 3\,H_2SO_4(l) \longrightarrow 2\,NaHSO_4(s) + Br_2(g) + SO_2(g) + 2\,H_2O(l)$$
$$Br_2(g) + H_2O(l) \rightleftharpoons HBr(aq) + HOBr(aq) \quad \text{(bleaching action)}$$

Solid C ($PbCl_2$):

$$PbCl_2(s) \rightleftharpoons Pb^{2+}(aq) + 2\,Cl^-(aq) \quad \text{(sparingly soluble, shifts right o}$$
$$Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s) \quad \text{(from dissolved } PbCl_2\text{)}$$
$$PbCl_2(s) + 2Cl^-(aq) \rightarrow [PbCl_4]^{2-}(aq) \quad \text{(in conc. } Cl^-\text{ excess)}$$

With conc. H_2SO_4 (heat): $PbCl_2(s) + H_2SO_4(l) \longrightarrow PbSO_4(s) + 2\,HCl(g)$ or dissolution in hot acid

Solid D ($K_4[Fe(CN)_6]$):

$$2Fe^{3+}(aq) + [Fe(CN)_6]^{4-}(aq) \rightarrow Fe_2[Fe(CN)_6](s) \quad \text{(Prussian Blue)}$$

With conc. H_2SO_4 (heat): $[Fe(CN)_6]^{4-}(aq) + 6H^+(aq) + 6H_2O(l) \rightarrow Fe^{2+}(aq) + 6HCN(g) + 6H_2O(l)$ (Highly to

Solution E (Fe^{2+}):

$$Fe^{2+}(aq) + 2OH^-(aq) \rightarrow Fe(OH)_2(s) \quad \text{(dirty green)}$$
$$4Fe(OH)_2(s) + O_2(g) + 2H_2O(l) \rightarrow 4Fe(OH)_3(s) \quad \text{(brown)}$$
$$Fe^{2+}(aq) + 2\,NH_3(aq) + 2\,H_2O(l) \longrightarrow Fe(OH)_2(s) + 2\,NH_4^+(aq) \quad \text{(no dissolution in excess)}$$
$$Fe^{2+}(aq) + [Fe(CN)_6]^{3-}(aq) \rightarrow Fe[Fe(CN)_6](s) \quad \text{(Turnbull's Blue)}$$

Note: $3[Fe(CN)_6]^{4-}(aq) + 4Fe^{3+}(aq) \rightarrow Fe_4[Fe(CN)_6]_3(s)$ (Prussian Blue)

$$Fe^{2+}(aq) + [Fe(CN)_6]^{3-}(aq) \rightarrow Fe^{3+}(aq) + [Fe(CN)_6]^{4-}(aq) \quad \text{(electron transfer in solid lattice)}$$

11.9 Task 5: Complex Analysis

Complex Formula	Ox. State	Coord. No.	Shape	IUPAC Name	Possible Isomerism
$[Ag(NH_3)_2]^+$	+1	2	Linear	Diamminesilver(I) ion	None (CN=2)
$[PbCl_4]^{2-}$	+2	4	Tetrahedral	Tetrachloridoplumbate(II) ion	None (tetrahedral, identical ligands)
$[Fe(CN)_6]^{4-}$	+2	6	Octahedral	Hexacyanoferrate(II) ion	None (Ma_6 type)
$[Cr(NH_3)_6]^{3+}$	+3	6	Octahedral	Hexaamminechromium(III) ion	None (Ma_6 type)
$[Cr(OH)_4]^-$	+3	4	Tetrahedral	Tetrahydroxidochromate(III) ion	None (tetrahedral, identical ligands)
$[Fe(CN)_6]^{3-}$	+3	6	Octahedral	Hexacyanoferrate(III) ion	None (Ma_6 type)

Notes on Isomerism:

- $[Ag(NH_3)_2]^+$ (Linear, CN=2): No geometric or optical isomerism possible.
- $[PbCl_4]^{2-}$, $[Cr(OH)_4]^-$ (Tetrahedral, CN=4, Ma_4 type): No geometric isomerism (all vertices equivalent). Optical isomerism only possible with four different ligands (chiral tetrahedral centre), not applicable here.
- $[Fe(CN)_6]^{4-}$, $[Fe(CN)_6]^{3-}$, $[Cr(NH_3)_6]^{3+}$ (Octahedral, CN=6, Ma_6 type): No geometric or optical isomerism.
- **Ionization Isomerism:** Possible for salts where a ligand can exchange with a counter-ion, e.g., $[Co(NH_3)_5Br]SO_4$ (red) vs $[Co(NH_3)_5SO_4]Br$ (violet). Not exhibited by the simple complex ions above.
- **Hydrate Isomerism:** Arises from different numbers of water molecules as ligands vs. water of crystallization, e.g., $[Cr(H_2O)_6]Cl_3$ (violet), $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$ (green), $[Cr(H_2O)_4Cl_2]Cl \cdot 2\,H_2O$ (dark green). Relevant for chromium(III) salts crystallised from aqueous solution.
- **Geometric (cis-trans) Isomerism:** Possible for Ma_4b_2 or Ma_3b_3 octahedral complexes (e.g., $[Co(NH_3)_4Cl_2]^+$). Not applicable to the homoleptic complexes listed above.
- **Optical Isomerism:** Possible for octahedral complexes with bidentate ligands ($[M(AA)_3]$, $[M(AA)_2b_2]$) or certain $Ma_2b_2c_2$ types. Not applicable to the complexes listed above.

11.10 Task 6: Identification Report

KEY POINTS **Final Identification**

- Solid A is NaCl. Justification: Soluble in water; gives white AgCl precipitate with

$AgNO_3/HNO_3$ soluble in dilute NH_3 (high solubility product Ksp(AgCl) ≈ 1.8×10^{-10}, complex formation constant Kf($[Ag(NH_3)_2]^+$) ≈ 1.6×10^{7} shifts equilibrium); conc. H_2SO_4 gives HCl gas (white fumes with NH_3, no redox reaction as Cl^- is a weak reducing agent).

- Solid B is NaBr. Justification: Soluble in water; gives cream AgBr precipitate (Ksp ≈ 5.0×10^{-13}), insoluble in dilute NH_3 but soluble in conc. NH_3 (higher $[NH_3]$ drives complex formation $AgBr + 2\,NH_3 \rightleftharpoons [Ag(NH_3)_2]^+ + Br^-$); conc. H_2SO_4 oxidises Br^- to Br_2 (brown fumes) and is reduced to SO_2 (Br^- is a stronger reducing agent than Cl^-, $E^{\ominus}(Br_2/Br^-) = +1.07$ V allows oxidation by H_2SO_4/SO_2 couple).
- Solid C is $PbCl_2$. Justification: Sparingly soluble in cold water (Ksp ≈ 1.7×10^{-5} at 25°C), soluble in hot water (characteristic property); gives white AgCl precipitate insoluble in NH_3 (the Pb^{2+} ions present precipitate as $Pb(OH)_2$ in NH_3, physically occluding the AgCl or preventing complexation under test conditions — a standard distinguishing observation for $PbCl_2$ in this syllabus context); conc. H_2SO_4 causes no gas evolution (non-reducing Cl^-, Pb^{2+} not oxidised), solid dissolves on heating (formation of soluble species or melting in acid medium).
- Solid D is $K_4[Fe(CN)_6]$. Justification: Soluble in water; no precipitate with $AgNO_3/HNO_3$ (Ag^+ forms soluble $[Ag(CN)_2]^-$ complex instantly due to high Kf ≈ 10^{21} and excess CN^- from the salt); positive Prussian Blue test with Fe^{3+} confirms ferrocyanide ion $[Fe(CN)_6]^{4-}$.
- Solution E contains Fe^{2+}. Justification: Pale green colour characteristic of $[Fe(H_2O)_6]^{2+}$. NaOH/NH_3 give dirty green $Fe(OH)_2$ precipitate, insoluble in excess ($Fe(OH)_2$ is basic, not amphoteric). Oxidation by atmospheric oxygen to brown $Fe(OH)_3$ confirms Fe(II). Turnbull's Blue precipitate with $[Fe(CN)_6]^{3-}$ confirms Fe^{2+}. Cr^{3+} would give green $Cr(OH)_3$ soluble in excess NaOH (amphoteric, forming $[Cr(OH)_4]^-$) and insoluble in excess NH_3 (kinetically inert).

11.11 Task 7: Safety Briefing for School Assembly

EXPERIMENT **Safety Briefing: Hidden Dangers in Our Homes and Labs**

Theme: "Chemistry for Safety: Protecting Our Families in Yaoundé"
1. Bleach (Eau de Javel) + Acid = Chlorine Gas (Deadly) Many households in Yaoundé use eau de Javel (sodium hypochlorite, NaOCl) for disinfecting floors and toilets. Some toilet cleaners contain hydrochloric acid (HCl). Never mix them.

$$NaOCl(aq) + 2\,HCl(aq) \longrightarrow Cl_2(g) + NaCl(aq) + H_2O(l)$$

Chlorine gas (Cl_2) is a pulmonary agent used in WWI. Inhalation causes coughing, pulmonary oedema, and death. Action: Use one product, rinse thoroughly with water, then

use the other. Ventilate always.
2. Bleach + Ammonia = Chloramine / Hydrazine (Toxic) Some cleaning products contain ammonia (NH_3). Mixing with bleach produces chloramines (NH_2Cl, $NHCl_2$, NCl_3) and potentially explosive hydrazine (N_2H_4).

$$NaOCl(aq) + NH_3(aq) \longrightarrow NH_2Cl(aq) + NaOH(aq)$$
$$3\,NH_2Cl(aq) + 2\,NH_3(aq) \longrightarrow N_2(g) + 3\,NH_4Cl(aq) \quad \text{(or } N_2H_4 \text{ formation)}$$

These gases cause severe respiratory distress and eye damage. Action: Read labels. Never mix cleaning agents.
3. Lead Compounds: The Silent Poison Traditional pottery glazes or old paints may contain lead(II) oxide (PbO) or lead chromate ($PbCrO_4$). Lead dust or fumes (from burning waste) are ingested/inhaled. Children are most vulnerable: lead inhibits haem synthesis and damages the developing brain (lower IQ, behavioural issues).

$$2\,PbO(s) + C(s) \longrightarrow 2\,Pb(l) + CO_2(g) \quad \text{(improvised smelting - DANGEROUS fumes)}$$

Pb^{2+} + biological ligands→ Enzyme inhibition

Action: In school labs, always wear gloves handling $PbCl_2$, $Pb(NO_3)_2$. Wash hands immediately. Dispose of lead waste in designated heavy-metal containers, never down the sink.
4. Potassium Ferrocyanide: Safe if Cold, Deadly if Hot/Acid $K_4[Fe(CN)_6]$ is low toxicity (LD_{50} high) because CN^- is tightly bound to Fe^{2+}. BUT heating with conc. H_2SO_4 or strong acid releases Hydrogen Cyanide gas (HCN) — "the smell of bitter almonds" (genetic inability to smell in 20–40% of people!).

$$[Fe(CN)_6]^{4-}(aq) + 6H^+(aq) + 6H_2O(l) \rightarrow Fe^{2+}(aq) + 6HCN(g) + 6H_2O(l)$$

Action: Never heat ferro/ferricyanides with conc. acids. Store away from acids. In case of acid spill on cyanide salts: EVACUATE, ventilate, neutralise with NaOH (forming NaCN, still toxic but non-volatile) before cleanup by trained personnel.
Conclusion: Chemistry gives us comfort (clean water, medicine) but demands respect. "La chimie est une science de vie, pas un jeu." (Chemistry is a life science, not a game). Know your reagents. Read the MSDS. Protect your neighbour.

11.12 Examiner's Comments (for Teacher Reference)

WARNING **Marking Points Summary**

- Flowcharts (Tasks 1&2): Logical sequence (4 marks), correct reagents (2 marks), decision nodes with observations (4 marks). [Total 10]
- Table (Task 3): Accuracy of colours, solubility, gas evolution (1 mark per cell ≈ 15

marks).

- Equations (Task 4): Balancing, state symbols, correct redox products for H_2SO_4/Br^- (2 marks each ≈ 12 marks).
- Complexes (Task 5): Name (1), Ox state (1), Shape (1), Isomerism analysis (1) per complex (24 marks).
- Report (Task 6): Correct ID (1 each), Justification linking obs to theory (2 each) [15 marks].
- Safety (Task 7): Equations (3), Hazards identified (3), Practical advice (2), Tone/Context (2) [10 marks].
- Total: ≈ 86 marks (scalable to 20–30 for a single session).

12 METHODS AND WORKED EXAMPLES

12.1 Method 1: Predicting and explaining trends in the halogen group

METHOD **Explaining physical trends down Group VII**

To answer any question on trends down the halogens ($F_2 \rightarrow I_2$), follow these reflexes:

1. Identify the type of bonding. The halogens are **diatomic molecules** X_2 held together in the solid/liquid by van der Waals (London dispersion) forces only.
2. Link the force to the number of electrons. Down the group the number of electrons per molecule increases, so the electron cloud is larger and more easily polarised and the van der Waals forces become stronger.
3. Deduce the physical state and melting/boiling point. Stronger intermolecular forces ⇒ higher melting and boiling points: F_2 (pale yellow gas), Cl_2 (greenish-yellow gas), Br_2 (red-brown liquid), I_2 (grey-black solid subliming to a violet vapour).
4. Colour deepens down the group because the energy gap for electronic absorption decreases, so absorption moves from the ultraviolet into the visible region.
5. Reactivity decreases down the group (opposite trend!): the atom gains an electron less easily because the incoming electron enters a shell further from the nucleus and is more shielded from the nuclear charge. Oxidising power: $F_2 > Cl_2 > Br_2 > I_2$.

Key idea to remember: physical trends (m.p., b.p., state) are governed by intermolecular forces; chemical trends (oxidising power, reactivity) are governed by the ease of gaining an electron. Never confuse the two in an explanation.

WORKED EXAMPLE 13 **Trends in the halogens**

Statement (GCE style). (a) State the colour and physical state of each of the halogens F_2, Cl_2, Br_2 and I_2 at room temperature. (b) Explain why the boiling points of the halogens increase down the group. (c) Explain why fluorine is the most reactive halogen although it has the weakest intermolecular forces.

SOLUTION

Solution.
(a) Colours and states at room temperature:

Halogen	Colour	Physical state
F_2	Pale yellow	Gas
Cl_2	Greenish-yellow	Gas
Br_2	Red-brown	Volatile liquid (orange vapour)
I_2	Grey-black	Solid; sublimes to a violet vapour

(b) The halogen molecules X_2 are non-polar; the only forces between them are van der Waals (London dispersion) forces. Down the group the number of electrons in each molecule increases, the electron cloud becomes larger and more easily polarised, so the instantaneous dipole–induced dipole attractions become stronger. More energy is therefore needed to separate the molecules, and the boiling point increases: $F_2 < Cl_2 < Br_2 < I_2$.
(c) Reactivity of a halogen means its power as an oxidising agent, i.e. the ease with which the atom gains one electron to form X^-. Fluorine is the smallest halogen atom; the incoming electron enters the $n = 2$ shell, very close to the nucleus, with little shielding, so it is strongly attracted: fluorine is the strongest oxidising agent of the group. The weakness of the van der Waals forces between F_2 molecules concerns only the physical state (a gas of low boiling point) and has nothing to do with the chemical electron-gaining process. Hence physical and chemical trends need not run in parallel.

12.2 Method 2: Displacement reactions and the oxidising power ladder

METHOD **Predicting halogen–halide displacement reactions**

1. Rank the oxidising agents: $F_2 > Cl_2 > Br_2 > I_2$. A halogen oxidises (displaces) the halide of any halogen below it in the group.
2. Write the half-equations. Oxidation: $2\,X^- \longrightarrow X_2 + 2\,e^-$; Reduction: $X_2' + 2e^- \rightarrow 2X'^-$.
3. Combine and cancel the electrons to obtain the ionic equation, e.g. $Cl_2(aq) + 2\,Br^-(aq) \longrightarrow 2\,Cl^-(aq) + Br_2(aq)$.
4. State the observation: appearance of the colour of the displaced halogen (orange/brown for Br_2; brown solution, or a dark precipitate if concentrated, for I_2). Confirm with an organic solvent such as cyclohexane: an orange organic layer indicates Br_2, a violet layer indicates I_2.
5. No reaction occurs if the added halogen is below the halide's halogen in the group (e.g. $Br_2 + Cl^-$: no reaction).

Reflex: the same order applies to the reducing power of the halide ions, but reversed: $I^- > Br^- > Cl^- > F^-$.

WORKED EXAMPLE 14 **Displacement reactions of the halogens**

Statement. Chlorine water is added to aqueous potassium bromide; bromine water is added to aqueous potassium iodide; iodine is added to aqueous potassium chloride. For each experiment, state what is observed and write an ionic equation where a reaction occurs. Explain the results in terms of oxidising power.

SOLUTION

Solution.
Experiment 1: Cl_2 + KBr. Chlorine is above bromine, hence a stronger oxidising agent; it oxidises Br^-:

$$Cl_2(aq) + 2\,Br^-(aq) \longrightarrow 2\,Cl^-(aq) + Br_2(aq)$$

Observation: the solution turns orange-brown (bromine liberated); on shaking with cyclohexane an orange upper (organic) layer forms.
Experiment 2: Br_2 + KI. Bromine is above iodine:

$$Br_2(aq) + 2\,I^-(aq) \longrightarrow 2\,Br^-(aq) + I_2(aq)$$

Observation: a brown solution of iodine appears (a dark solid may deposit if concentrated); with cyclohexane a violet upper layer forms.
Experiment 3: I_2 + KCl. Iodine is below chlorine; it is a weaker oxidising agent than Cl_2 and cannot oxidise Cl^-. No reaction, no observable change.
Explanation. A halogen can only remove electrons from the halide ion of a halogen below it, because the outer electrons are held less tightly in the larger ion and are lost more easily. The experiments confirm the oxidising power order $Cl_2 > Br_2 > I_2$ and, equivalently, the reducing power order $I^- > Br^- > Cl^-$.

12.3 Method 3: Testing for halide ions

METHOD **Identifying Cl^-, Br^-, I^- in solution**

1. Acidify a portion of the solution with dilute HNO_3 (this destroys interfering ions such as CO_3^{2-} that would also precipitate with Ag^+).
2. Add $AgNO_3(aq)$. A precipitate of silver halide forms: $Ag^+(aq) + X^-(aq) \longrightarrow AgX(s)$.
3. Read the colour: AgCl white, AgBr cream, AgI pale yellow.
4. Confirm with ammonia solution: AgCl dissolves in dilute $NH_3(aq)$; AgBr dissolves only in concentrated $NH_3(aq)$; AgI is insoluble even in concentrated ammonia.
5. Dissolution is due to complex formation: $AgCl(s) + 2\,NH_3(aq) \longrightarrow [Ag(NH_3)_2]^+(aq) + Cl^-(aq)$.

Reflex: never use HCl to acidify — you would be adding chloride ions yourself and would obtain a white precipitate whatever the unknown!

WORKED EXAMPLE 15 **Identifying an unknown halide**

Statement. A student is given a solution Q containing one halide ion. On adding dilute nitric acid followed by silver nitrate solution, a cream precipitate forms. The precipitate does not dissolve in dilute ammonia but dissolves in concentrated ammonia. (a) Identify the halide in Q, giving reasons. (b) Write ionic equations for the reactions observed. (c) Explain why nitric acid is added before the silver nitrate.

SOLUTION

Solution.
(a) The cream colour of the silver halide precipitate suggests AgBr. Confirmation: AgCl (white) dissolves in dilute ammonia, whereas AgI (pale yellow) does not dissolve even in concentrated ammonia. A precipitate that resists dilute ammonia but dissolves in concentrated ammonia must be AgBr. Hence Q contains the bromide ion Br^-.
(b) Precipitation:

$$Ag^+(aq) + Br^-(aq) \longrightarrow AgBr(s)$$

Dissolution in concentrated ammonia (formation of the soluble diamminesilver(I) complex):

$$AgBr(s) + 2\,NH_3(aq) \longrightarrow [Ag(NH_3)_2]^+(aq) + Br^-(aq)$$

(c) Dilute HNO_3 is added first to remove interfering anions, in particular carbonate ions CO_3^{2-}, which would otherwise give a precipitate of Ag_2CO_3 with Ag^+ and falsify the test:

$$2\,H^+(aq) + CO_3^{2-}(aq) \longrightarrow H_2O(l) + CO_2(g)$$

Nitric acid is chosen because it introduces no halide ion and because its own silver salt, $AgNO_3$, is soluble, so it cannot produce a precipitate.

12.4 Method 4: Comparing the oxoacids of the halogens

METHOD **Acid strength, oxidising strength and thermal stability of oxoacids**

For the chlorine oxoacids HOCl, $HClO_2$, $HClO_3$, $HClO_4$ (oxidation states of Cl: +1, +3, +5, +7):

1. Acid strength increases with the oxidation state of the halogen: $HOCl < HClO_2 < HClO_3 < HClO_4$. Reason: each additional oxygen withdraws electron density from the O – H bond, weakening it, and stabilises the conjugate anion

ClO_n^- by delocalisation of the negative charge over more oxygen atoms.

2. Oxidising strength decreases in the same direction: HOCl is the strongest oxidising agent (chlorine in the low +1 state is easily reduced); dilute $HClO_4$ is a comparatively weak oxidising agent.
3. Thermal stability increases with oxidation state: HOCl decomposes readily ($2\,HOCl \longrightarrow 2\,HCl + O_2$, accelerated by light); $HClO_4$ is the most stable, although the pure concentrated acid can decompose explosively.
4. Names (Stock notation): HOCl chloric(I) acid (hypochlorous acid), $HClO_2$ chloric(III) acid (chlorous acid), $HClO_3$ chloric(V) acid (chloric acid), $HClO_4$ chloric(VII) acid (perchloric acid). The Roman numeral is the oxidation number of Cl.

Key formula: for $HClO_n$, the oxidation number of Cl is given by $1 + x - 2n = 0$, i.e. $x = 2n - 1$.

WORKED EXAMPLE 16 **Oxoacids of chlorine**

Statement. (a) Give the systematic (Stock) name and the oxidation number of chlorine in each of the acids HOCl, $HClO_3$ and $HClO_4$. (b) Arrange the four acids HOCl, $HClO_2$, $HClO_3$, $HClO_4$ in order of increasing acid strength and justify the order. (c) Which of these acids is the strongest oxidising agent? Explain. (d) Write an equation for the decomposition of HOCl in sunlight and state one use of this acid.

SOLUTION

Solution.
(a) Using $1 + x - 2n = 0$:

Formula	Oxidation number of Cl	Stock name
HOCl	+1	Chloric(I) acid
$HClO_3$	+5	Chloric(V) acid
$HClO_4$	+7	Chloric(VII) acid

(b) Increasing acid strength:

$$HOCl < HClO_2 < HClO_3 < HClO_4$$

As the number of oxygen atoms increases, the highly electronegative oxygens pull electron density away from the O – H bond, weakening it and making loss of H^+ easier. In addition, the negative charge of the conjugate base ClO_n^- is delocalised over more oxygen atoms, so the anion is increasingly stabilised and the acid increasingly strong. $HClO_4$ is a very strong acid; HOCl is a weak acid.
(c) HOCl is the strongest oxidising agent: chlorine in the +1 state is easily reduced to Cl^- (oxidation state −1), and the O – Cl bond is weak. Oxidising power therefore decreases as the oxidation state of chlorine rises — the opposite order to acid strength.

(d) In sunlight:

$$2\,HOCl(aq) \longrightarrow 2\,HCl(aq) + O_2(g)$$

Use: HOCl (and the OCl^- ion present in bleach) is used as a bleaching agent and disinfectant, e.g. in the treatment of drinking water, because of its strong oxidising (germ-killing) action.

12.5 Method 5: Group IV — structures, allotropes and the inert pair effect

METHOD **Handling Group IV (C to Pb) questions**

1. Electronic configuration ns^2np^2 for all the elements; oxidation states +4 and +2. Down the group the +2 state becomes more stable (the **inert pair effect**: the ns^2 pair is increasingly reluctant to take part in bonding). Hence PbO is more stable than PbO_2; Pb(IV) compounds are strong oxidising agents, while Sn(II) compounds behave as reducing agents.
2. Allotropy: carbon exists as diamond (giant covalent, tetrahedral sp^3, very hard, non-conductor), graphite (layers of sp^2 hexagons, delocalised electrons within the layers ⇒ conducts electricity, soft, lubricant) and fullerenes. Tin also shows allotropy (grey tin, non-metallic, and white tin, metallic).
3. Metallic character increases down the group: C (non-metal), Si and Ge (metalloids), Sn and Pb (metals). Atomic radius increases and first ionisation energy decreases down the group.
4. Oxides: CO_2 is a simple molecular gas (acidic); SiO_2 is a giant covalent solid of very high melting point (acidic); SnO_2 and PbO_2 are amphoteric with ionic character. Chlorides: CCl_4 and $SiCl_4$ are covalent liquids; $SiCl_4$ is hydrolysed by water ($SiCl_4 + 2\,H_2O \longrightarrow SiO_2 + 4\,HCl$) but CCl_4 is not (carbon has no accessible orbital to accept a lone pair from water).
5. Hydrides: CH_4, SiH_4, GeH_4, SnH_4, PbH_4 are tetrahedral; thermal stability decreases down the group as the E – H bond becomes longer and weaker.

WORKED EXAMPLE 17 **Group IV: structure and stability**

Statement. (a) Describe the structure and bonding in diamond and in graphite, and explain why graphite conducts electricity while diamond does not. (b) Explain why $SiCl_4$ is hydrolysed by water whereas CCl_4 is not. Write the equation for the hydrolysis of $SiCl_4$. (c) Explain why PbO_2 oxidises concentrated hydrochloric acid to chlorine while CO_2 shows no oxidising action.

SOLUTION

Solution.
(a) In diamond each carbon atom is sp^3 hybridised and covalently bonded tetrahedrally to four other carbon atoms, giving a giant three-dimensional covalent lattice. All four outer electrons of each atom are localised in bonds, so there are no mobile charge carriers: diamond is a non-conductor (and is extremely hard). In graphite each carbon is sp^2 hybridised, bonded to three neighbours in flat hexagonal layers; the fourth electron of each atom is delocalised over the layer. These mobile electrons carry current, so graphite conducts electricity. The layers are held together by weak van der Waals forces and slide over one another, which makes graphite soft and useful as a lubricant and in pencil "leads".
(b) Silicon, in period 3, is a larger atom than carbon and has accessible empty $3d$ orbitals: a water molecule can donate a lone pair to silicon to form an intermediate, and the polar Si – Cl bond is then readily attacked. Carbon has no available low-energy orbital (there is no $2d$ subshell) to accept the lone pair from water, so CCl_4 is kinetically inert to hydrolysis even though the reaction would be energetically feasible.

$$SiCl_4(l) + 2\,H_2O(l) \longrightarrow SiO_2(s) + 4\,HCl(aq)$$

(White fumes of HCl are observed.)
(c) This illustrates the inert pair effect. For lead, the $6s^2$ pair is held strongly and Pb(IV) is unstable relative to Pb(II); PbO_2 therefore readily accepts electrons (it is reduced to Pb^{2+}) and acts as a strong oxidising agent:

$$PbO_2(s) + 4\,HCl(aq) \longrightarrow PbCl_2(s) + Cl_2(g) + 2\,H_2O(l)$$

For carbon, at the top of the group, the +4 state is very stable (the inert pair effect is negligible), so CO_2 has no tendency to be reduced and shows no oxidising action under these conditions.

12.6 Method 6: Writing electronic configurations of d-block atoms and ions

METHOD **Configurations of the first transition series (Sc to Zn)**

1. Atoms: fill $4s$ before $3d$ (Aufbau order). General pattern: $[Ar]\,4s^2 3d^n$. Two exceptions to memorise: Cr = $[Ar]\,4s^1 3d^5$ and Cu = $[Ar]\,4s^1 3d^{10}$ (extra stability of the half-filled and fully filled d subshells).
2. Ions: remove electrons from the $4s$ orbital first, then from $3d$. Example: $Fe^{2+} = [Ar]\,3d^6$ (not $4s^2 3d^4$!); $Fe^{3+} = [Ar]\,3d^5$.
3. Electron-in-boxes notation: draw five boxes for $3d$ and one for $4s$; fill with single parallel spins first (Hund's rule) before pairing.

4. Definitions: a **d-block element** has its highest-energy (differentiating) electron in a *d* subshell; a **transition element** forms at least one ion with a partially filled *d* subshell. Hence Sc (Sc^{3+} = $3d^0$) and Zn (Zn^{2+} = $3d^{10}$) are d-block elements but not transition elements.

5. Quick check: the total number of electrons must equal Z minus the positive charge.

WORKED EXAMPLE 18 **Electronic configurations of iron and copper species**

Statement. (a) Write the full electronic configuration (spdf notation) of Cr (Z = 24) and Cu (Z = 29), explaining any anomaly. (b) Write the configurations of Fe^{2+} and Fe^{3+} (Z = 26) and explain why Fe^{3+} is the more stable ion in aqueous solution. (c) Using electron-in-boxes notation for the 3*d* and 4*s* orbitals, represent Fe^{2+}. (d) Explain why zinc is not regarded as a transition element.

SOLUTION

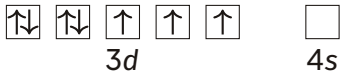
Solution.
(a) Simple filling order would predict Cr = $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^4$ and Cu = $[Ar]\,4s^2 3d^9$. The observed configurations are:

$$Cr = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5, \qquad Cu = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$$

A half-filled ($3d^5$) or completely filled ($3d^{10}$) *d* subshell confers extra stability (symmetrical distribution of electron density and maximum exchange energy), so one 4*s* electron is effectively promoted into 3*d*.
(b) Fe = $[Ar]\,4s^2 3d^6$. Transition metals lose the 4*s* electrons first because, once the 3*d* subshell is occupied, the 4*s* electrons are on average further from the nucleus and are removed more easily:

$$Fe^{2+} = [Ar]\,3d^6, \qquad Fe^{3+} = [Ar]\,3d^5$$

Fe^{3+} possesses the particularly stable half-filled $3d^5$ subshell, which is why iron(III) is favoured over iron(II) in aqueous (especially aerated) solution.
(c) Fe^{2+}: the 4*s* box is empty; the six 3*d* electrons fill the five boxes singly first, then one pair forms:



(d) A transition element must form at least one ion with a partially filled *d* subshell. Zinc's only common ion is Zn^{2+} = $[Ar]\,3d^{10}$, with a completely filled *d* subshell; the Zn atom itself is $[Ar]\,3d^{10}4s^2$. Since neither the atom nor its ion has an incomplete *d* subshell, zinc is a d-block element but not a transition element. This is consistent with its properties: Zn^{2+} compounds are white (colourless ions) and zinc shows only one oxidation state, +2.

12.7 Method 7: Naming, shaping and finding isomers of complexes

METHOD **Naming complexes and predicting shapes and isomerism**

1. Name the ligands first (in alphabetical order), with prefixes di-, tri-, tetra- (or bis-, tris- for composite ligand names). Anionic ligands end in -o (Cl^- chloro, CN^- cyano, OH^- hydroxo); neutral ligands keep their name except H_2O = aqua and NH_3 = ammine.
2. Then name the metal with its oxidation state in Roman numerals; if the complex is an anion, the metal name ends in -ate (e.g. ferrate, cuprate).
3. Oxidation state: (oxidation state of metal) + (sum of ligand charges) = overall charge of the complex ion.
4. Shape from coordination number: C.N. 6 ⇒ octahedral; C.N. 4 ⇒ tetrahedral (or square planar, e.g. $[PtCl_4]^{2-}$); C.N. 2 ⇒ linear.
5. Structural isomerism: ionisation isomerism (a ligand and an external ion exchange places, e.g. $[Co(NH_3)_5Br]SO_4$ and $[Co(NH_3)_5SO_4]Br$); hydrate isomerism (water inside vs outside the coordination sphere, e.g. $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$).
6. Stereoisomerism: geometric (cis/trans) isomerism in square planar MA_2B_2 and octahedral MA_4B_2 complexes; optical isomerism (non-superimposable mirror images), e.g. cis-$[Co(en)_2Cl_2]^+$ or $[M(AA)_3]$ with bidentate ligands.

WORKED EXAMPLE 19 **Complexes: names, shapes, isomers**

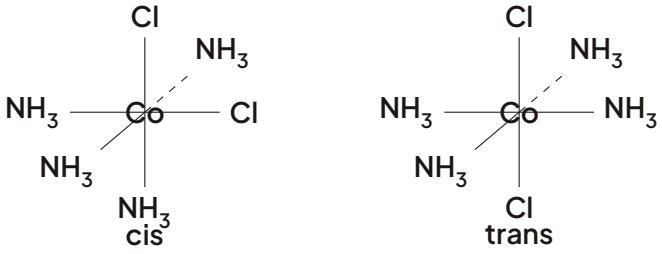
Statement. (a) Name the following complexes and give the oxidation state of the metal: (i) $[Cu(NH_3)_4]SO_4$; (ii) $K_4[Fe(CN)_6]$; (iii) $[CoCl_2(NH_3)_4]Cl$. (b) Draw and name the geometric isomers of the complex ion in (iii). (c) A compound of formula $CrCl_3 \cdot 6\,H_2O$ exists in two forms: one gives three moles of AgCl per mole with $AgNO_3$, the other gives only two. Identify the type of isomerism and write the formula of each form.

SOLUTION

Solution.
(a) (i) The complex cation is $[Cu(NH_3)_4]^{2+}$: $x + 4(0) = +2 \Rightarrow x = +2$. Name: tetraamminecopper(II) sulphate. Shape: coordination number 4 (approximately tetrahedral/square planar).
(ii) The complex anion is $[Fe(CN)_6]^{4-}$: $x + 6(-1) = -4 \Rightarrow x = +2$. Name: potassium hexacyanoferrate(II). Shape: octahedral.
(iii) The complex cation is $[CoCl_2(NH_3)_4]^+$: $x + 2(-1) + 4(0) = +1 \Rightarrow x = +3$. Name: tetraamminedichlorocobalt(III) chloride (ligands cited alphabetically: ammine before chloro).

Shape: octahedral (C.N. 6).
(b) In the octahedral ion $[CoCl_2(NH_3)_4]^+$ the two Cl^- ligands can be:

- cis: adjacent (90° apart) — cis-tetraamminedichlorocobalt(III) ion;
- trans: opposite (180° apart) — trans-tetraamminedichlorocobalt(III) ion.



(c) This is hydrate isomerism (a form of structural isomerism): water molecules and chloride ions exchange places between the coordination sphere and the outside of the complex. Only the Cl^- ions outside the sphere are free and are precipitated by Ag^+.

- Form giving 3 mol AgCl: all three Cl^- are outside the sphere: $[Cr(H_2O)_6]Cl_3$.
- Form giving 2 mol AgCl: one Cl^- is bonded inside the sphere (not precipitated): $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$.

(A third isomer, $[Cr(H_2O)_4Cl_2]Cl \cdot 2\,H_2O$, would give only one mole of AgCl.)

12.8 Method 8: Colour, magnetism and catalysis in d-block chemistry

METHOD **Explaining colour and predicting magnetic behaviour**

1. Colour: in a complex, the ligands split the five d orbitals into two sets of different energy (separation Δ). A d electron absorbs visible light of energy $E = h\nu = \Delta$ and is promoted to the higher set; the complementary colour is transmitted and seen. Ions with d^0 (Sc^{3+}) or d^{10} (Zn^{2+}, Cu^+) configurations have no possible d–d transition and are colourless.
2. Changing the ligand changes Δ, and hence the colour: e.g. $[Cu(H_2O)_6]^{2+}$ is pale blue; adding excess NH_3 gives the deep blue $[Cu(NH_3)_4(H_2O)_2]^{2+}$.
3. Magnetism: count the unpaired electrons from the 3d box diagram. Unpaired electrons ⇒ paramagnetic (attracted into a magnetic field); all electrons paired ⇒ diamagnetic. The spin-only magnetic moment is $\mu = \sqrt{n(n+2)}$ B.M., where n is the number of unpaired electrons.
4. Catalysis: transition metals catalyse by (i) providing a surface on which reactants adsorb (heterogeneous catalysis: Fe in the Haber process, Ni in the hydrogenation of

oils, Pt/Rh in car exhaust converters, V_2O_5 in the Contact process) and (ii) using their variable oxidation states to provide an alternative reaction path of lower activation energy (e.g. Fe^{2+}/Fe^{3+} catalysing the reaction between $S_2O_8^{2-}$ and I^-).

WORKED EXAMPLE 20 **Colour, magnetism and catalysis**

Statement. (a) Explain why aqueous solutions of Cu^{2+} salts are blue whereas solutions of Zn^{2+} salts are colourless. (b) Calculate the spin-only magnetic moment of Fe^{3+} ($3d^5$) and of Ni^{2+} ($3d^8$) and state whether each ion is paramagnetic or diamagnetic. (c) State the catalyst used in (i) the Haber process, (ii) the Contact process, and explain in one sentence why transition metals make good catalysts.

SOLUTION

Solution.
(a) Cu^{2+} has the configuration [Ar] $3d^9$: the d subshell is partially filled. In the octahedral aqua complex $[Cu(H_2O)_6]^{2+}$ the ligands split the d orbitals into two energy levels separated by Δ. Absorption of visible light (in the orange–red region) promotes a d electron across Δ; the complementary colour, blue, is observed. Zn^{2+} is $3d^{10}$: the subshell is full, no d–d promotion is possible, no visible light is absorbed, and the ion is colourless.
(b) Fe^{3+} ($3d^5$): by Hund's rule the five electrons occupy the five d orbitals singly, so $n = 5$ unpaired electrons:

$$\mu = \sqrt{n(n+2)} = \sqrt{5 \times 7} = \sqrt{35} \approx 5.92\ \text{B.M.} \quad \Rightarrow \text{paramagnetic.}$$

Ni^{2+} ($3d^8$): the eight electrons fill the five orbitals as three pairs plus two single electrons, so $n = 2$:

$$\mu = \sqrt{2 \times 4} = \sqrt{8} \approx 2.83\ \text{B.M.} \quad \Rightarrow \text{paramagnetic.}$$

(c) (i) Haber process ($N_2 + 3\,H_2 \rightleftharpoons 2\,NH_3$): iron (with promoters). (ii) Contact process ($2\,SO_2 + O_2 \rightleftharpoons 2\,SO_3$): vanadium(V) oxide, V_2O_5. Transition metals make good catalysts because they can change oxidation state easily — offering an alternative pathway of lower activation energy (e.g. V_2O_5 cycling between V(V) and V(IV)) — and because their surfaces adsorb reactant molecules, weakening their bonds and holding them close together in favourable orientations.

13 EXERCISES

13.1 I check my knowledge

EXERCISE 10 **MCQ: Halogen Trends**

Which of the following statements concerning the halogens (Group 17) is incorrect?

A. The oxidizing power decreases down the group.

B. The bond dissociation energy of the X–X bond decreases steadily down the group.

C. The colour intensity increases down the group.

D. The first ionization energy decreases down the group.

EXERCISE 11 **True/False: Group IV Chemistry**

State whether each statement is True or False. Justify your answer with a brief explanation or a balanced chemical equation where appropriate.

1. Carbon tetrachloride (CCl_4) hydrolyses readily in water at room temperature to form CO_2 and HCl.
2. The stability of the +2 oxidation state in Group IV increases down the group relative to the +4 state.
3. Silicon dioxide (SiO_2) reacts with hot concentrated sodium hydroxide solution to form sodium silicate.
4. Lead(IV) oxide (PbO_2) is a stronger oxidizing agent than tin(IV) oxide (SnO_2).

EXERCISE 12 **Definitions and Configurations: d-Block**

1. Define the term transition element according to the IUPAC definition.
2. Write the full electronic configuration (spdf notation) and the electron-in-boxes diagram for the 3 d orbitals of:
   - A chromium atom (Cr, Z=24).
   - A copper(II) ion (Cu^{2+}, Z=29).
3. Explain why the first ionization energy of zinc (Zn) is higher than that of copper (Cu).

EXERCISE 13 **Direct Calculations: Redox Titrations**

25.0 cm^3 of a solution of iron(II) sulfate ($FeSO_4$) was acidified with dilute sulfuric acid and titrated against 0.0200 mol dm^{-3} potassium permanganate ($KMnO_4$). The volume of permanganate required was 22.5 cm^3.

1. Write the balanced ionic half-equations for the oxidation of Fe^{2+} and the reduction of MnO_4^- in acidic medium.
2. Deduce the overall balanced ionic equation for the reaction.
3. Calculate the concentration, in mol dm^{-3}, of the iron(II) sulfate solution.
4. Calculate the mass concentration of $FeSO_4 \cdot 7H_2O$ in g dm^{-3} (Molar mass = 278 g mol^{-1}).

13.2 I practise

EXERCISE 14 **Halogen Displacement and Oxoacids**

1. A student adds chlorine water to an aqueous solution of potassium bromide, then adds an equal volume of tetrachloromethane (CCl_4) and shakes the mixture. Describe the observations in the aqueous layer and the organic layer. Write the balanced ionic equation for the reaction occurring.
2. Write the formulas and systematic names for the oxoacids of chlorine corresponding to oxidation states +1, +3, +5, and +7.
3. Arrange the oxoacids in (ii) in order of increasing acid strength and increasing thermal stability. Explain the trend in acid strength in terms of the structure of the conjugate bases.

EXERCISE 15 **Halide Ion Identification**

Three test tubes labelled A, B, and C each contain an aqueous solution of a sodium halide (NaF, NaCl, NaBr, or NaI). The following tests were performed:

- Test 1: Addition of $AgNO_3$(aq) followed by dilute NH_3(aq).
- Test 2: Addition of $AgNO_3$(aq) followed by concentrated NH_3(aq).
- Test 3: Addition of concentrated H_2SO_4 and warming.

The observations are recorded below:

Test	Tube A	Tube B	Tube C
1 ($AgNO_3$ + dil. NH_3)	Cream ppt, insoluble	White ppt, soluble	Yellow ppt, insoluble
2 ($AgNO_3$ + conc. NH_3)	Cream ppt, insoluble	White ppt, soluble	Yellow ppt, insoluble
3 (conc. H_2SO_4, heat)	Brown fumes, purple vapour	White fumes (dense)	White fumes (dense)

1. Identify the halide ion in each tube (A, B, C).
2. Write the ionic equations for the reactions with $AgNO_3$ for each identified halide.
3. Explain the observations in Test 3 for Tube A, identifying the brown fumes and purple vapour. Why does concentrated H_2SO_4 not produce hydrogen bromide or hydrogen iodide gas in a pure state?

EXERCISE 16 Group IV: Oxides and Chlorides

1. Compare the structures and bonding of CO_2 and SiO_2. Use diagrams to illustrate your answer. Explain why CO_2 is a gas at room temperature while SiO_2 is a high-melting-point solid.
2. Write balanced equations for the hydrolysis of $SiCl_4$ and CCl_4 with water. Explain why $SiCl_4$ hydrolyses violently at room temperature whereas CCl_4 is inert to hydrolysis under the same conditions.
3. Describe the amphoteric nature of SnO_2 and PbO_2. Write equations for the reaction of SnO_2 with hot concentrated HCl and with hot concentrated NaOH.

EXERCISE 17 Complexes: Naming, Shapes and Isomerism

1. Name the following complexes using IUPAC nomenclature and state the oxidation state of the central metal ion, the coordination number, and the likely shape:
   - $[Co(NH_3)_5Cl]Cl_2$
   - $K_3[Fe(CN)_6]$
   - $[CuCl_4]^{2-}$
2. Draw the structures (using wedge/dash notation where necessary) and name the two geometric isomers of $[Pt(NH_3)_2Cl_2]$. State which isomer is used as an anti-cancer drug (cisplatin).
3. The complex $[Cr(H_2O)_5Cl]SO_4 \cdot H_2O$ and $[Cr(H_2O)_5SO_4]Cl \cdot H_2O$ are isomers. Identify the type of isomerism. Describe a simple chemical test (with observations) to distin-

guish between these two isomers in aqueous solution.

13.3 I prepare for the GCE

EXERCISE 18 **GCE Structured Question: Halogen Chemistry (20 marks)**

1. (a) Describe and explain the trend in the oxidizing power of the halogens (F_2, Cl_2, Br_2, I_2) down Group 17. Your answer should refer to electrode potentials, bond dissociation energies, and electron affinities. [6 marks]

2. (b) A student carries out displacement reactions by adding chlorine water to separate aqueous solutions of potassium bromide and potassium iodide, followed by shaking with an organic solvent (CCl_4 or $CHCl_3$). For each reaction:

   - State the colour changes observed in the aqueous and organic layers.
   - Write the balanced ionic equation.
   - Explain why chlorine displaces bromine and iodine but fluorine is not used for such displacements in the laboratory. [6 marks]

3. (c) Concentrated sulfuric acid is added to solid sodium iodide and the mixture is heated gently.

   - List all the observation(s) made.
   - Identify all the sulfur-containing and iodine-containing products formed.
   - Write balanced equations for the redox reactions occurring.
   - Explain why this reaction is not a suitable method for preparing pure hydrogen iodide. [8 marks]

EXERCISE 19 **GCE Essay Question: Group IV and Transition Metals (20 marks)**

1. (a) The elements of Group IV show a transition from non-metallic to metallic character down the group.

   - Discuss the variation in the stability of the +2 and +4 oxidation states from carbon to lead. [4 marks]
   - Compare the structures, bonding, and acid/base nature of the dioxides CO_2, SiO_2, GeO_2, SnO_2, and PbO_2. [6 marks]

2. (b) Transition metals exhibit variable oxidation states, form coloured compounds, and act as catalysts.

   - Using the first transition series (Sc to Zn), explain the origin of variable oxidation states in terms of electronic configuration. Why do the maximum oxidation states occur around the middle of the series (e.g., Mn)? [4 marks]
   - Explain why most transition metal complexes are coloured while Sc^{3+} (d^0) and Zn^{2+} (d^{10}) complexes are colourless. Use the concept of d-orbital splitting in an octahedral field. [4 marks]
   - Give one example of a homogeneous catalyst and one example of a heterogeneous catalyst involving transition metals or their compounds. State the reaction catalysed in each case. [2 marks]

EXERCISE 20 **GCE Data Analysis and Application: Complexes and Spectrochemical Series (20 marks)**

1. (a) The complex ion $[Cu(H_2O)_6]^{2+}$ is pale blue. When excess concentrated ammonia is added, the solution turns deep blue due to the formation of $[Cu(NH_3)_4(H_2O)_2]^{2+}$.

   - Draw the shape of $[Cu(NH_3)_4(H_2O)_2]^{2+}$ and name its geometry. Is it a regular octahedron? Explain. [3 marks]
   - Using Crystal Field Theory, explain the colour change in terms of the spectrochemical series and the magnitude of Δ_o (crystal field splitting parameter). [4 marks]

2. (b) Consider the complex $[Co(en)_3]^{3+}$ (en = ethylenediamine, $NH_2CH_2CH_2NH_2$).

   - Draw the two optical isomers (enantiomers) of this complex. [3 marks]
   - Explain why this complex does not have geometric isomers. [2 marks]
   - Describe how a polarimeter could be used to distinguish between the two enantiomers. [2 marks]

3. (c) A solution contains a mixture of $[Co(NH_3)_5SO_4]Br$ and $[Co(NH_3)_5Br]SO_4$ (ionization isomers).

   - Describe a chemical test, with expected observations, to distinguish between these two solid isomers. [3 marks]
   - Write the ionic equations for the reactions used in your test. [3 marks]

14 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — MCQ: Halogen Trends**

Answer: B.
Statement B is incorrect. The bond dissociation energy of the X–X bond does not decrease steadily down the group. The actual order (increasing bond energy) is:

$$I_2 < F_2 < Br_2 < Cl_2$$

Fluorine has an anomalously low bond dissociation energy ($\approx 158\ kJ\ mol^{-1}$) compared with chlorine ($\approx 242\ kJ\ mol^{-1}$) and bromine ($\approx 193\ kJ\ mol^{-1}$). This is because the fluorine atom is very small; the lone pairs on the two adjacent F atoms repel each other strongly, weakening the F–F bond. From Cl_2 to I_2, the bond energy decreases steadily as the atoms become larger and the orbital overlap poorer (Cl_2 242, Br_2 193, I_2 151 $kJ\ mol^{-1}$).
The other statements are correct:

- A: Oxidizing power decreases down the group (F_2 is the strongest oxidizing agent, I_2 the weakest).
- C: Colour intensity increases down the group: F_2 pale yellow, Cl_2 greenish-yellow, Br_2 red-brown, I_2 grey-black solid (violet vapour).
- D: First ionization energy decreases down the group as the atomic radius increases and the outer electron is further from the nucleus and more shielded.

SOLUTION **Exercise 2 — True/False: Group IV Chemistry**

1. False. CCl_4 is inert to hydrolysis at room temperature. Although the hydrolysis is thermodynamically favourable, the carbon atom has no available low-energy d-orbitals (it is in period 2) and is sterically shielded by the four chlorine atoms, so water cannot attack it. The activation energy is too high.
2. True. This is the **inert pair effect**. Down the group, the ns^2 electrons become increasingly reluctant to participate in bonding, so the stability of the lower (+2) oxidation state increases relative to +4. Thus Pb^{2+} compounds are stable while PbO_2 is a strong oxidizing agent; conversely CO_2 and SiO_2 (+4) are very stable.
3. True. SiO_2 is acidic and reacts with hot concentrated alkali:

$$SiO_2(s) + 2\,NaOH(aq) \longrightarrow Na_2SiO_3(aq) + H_2O(l)$$

4. True. Because of the inert pair effect, Pb(IV) is unstable relative to Pb(II), so PbO_2

readily accepts electrons (it is reduced to Pb^{2+}) and is a strong oxidizing agent, e.g.:

$$PbO_2(s) + 4\,HCl(aq) \longrightarrow PbCl_2(aq) + Cl_2(g) + 2\,H_2O(l)$$

Sn(IV) is much more stable, so SnO_2 is a weaker oxidizing agent.

SOLUTION **Exercise 3 — Definitions and Configurations: d-Block**

1. A **transition element** is an element which forms at least one ion with a partially filled d-subshell. (Note: by this definition Sc and Zn are d-block elements but not transition elements, since Sc^{3+} is d^0 and Zn^{2+} is d^{10}.)

2. Chromium, Cr ($Z = 24$):

$$Cr: \quad 1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^5\,4s^1$$

The $3d^5\,4s^1$ arrangement (rather than $3d^4\,4s^2$) gives extra stability from the half-filled $3d$ subshell. Electron-in-boxes for the $3d$ orbitals:

↑	↑	↑	↑	↑

$3d^5$ (all five orbitals singly occupied, Hund's rule)

Copper(II) ion, Cu^{2+} ($Z = 29$): Cu atom is $[Ar]\,3d^{10}4s^1$; the $4s$ electron and one $3d$ electron are removed:

$$Cu^{2+}: \quad 1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^9$$

↑↓	↑↓	↑↓	↑↓	↑

$3d^9$ (one unpaired electron)

3. Zinc has configuration $[Ar]\,3d^{10}4s^2$ and copper $[Ar]\,3d^{10}4s^1$. Zinc has one more proton in its nucleus, so its effective nuclear charge on the $4s$ electrons is greater; in addition, the filled $4s^2$ subshell in Zn is relatively stable. The $4s$ electron of Cu is therefore removed more easily than a $4s$ electron of Zn, so the first ionization energy of Zn is higher than that of Cu.

SOLUTION **Exercise 4 — Direct Calculations: Redox Titrations**

1. Half-equations:

$$\text{Oxidation: } Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq) + e^-$$

$$\text{Reduction: } MnO_4^-(aq) + 8\,H^+(aq) + 5\,e^- \longrightarrow Mn^{2+}(aq) + 4\,H_2O(l)$$

2. Multiplying the iron half-equation by 5 and adding:

$$MnO_4^-(aq) + 8\,H^+(aq) + 5\,Fe^{2+}(aq) \longrightarrow Mn^{2+}(aq) + 5\,Fe^{3+}(aq) + 4\,H_2O(l)$$

3. Moles of MnO_4^-:

$$n(MnO_4^-) = 0.0200\ \text{mol dm}^{-3} \times \frac{22.5}{1000}\ \text{dm}^3 = 4.50 \times 10^{-4}\ \text{mol}$$

From the equation, $n(Fe^{2+}) = 5 \times n(MnO_4^-) = 2.25 \times 10^{-3}$ mol in 25.0 cm^3.

$$[FeSO_4] = \frac{2.25 \times 10^{-3}}{0.0250} = \boxed{0.0900\ \text{mol dm}^{-3}}$$

4. Mass concentration:

$$c_m = 0.0900 \times 278 = \boxed{25.0\ \text{g dm}^{-3}} \quad \text{of } FeSO_4 \cdot 7\,H_2O$$

SOLUTION **Exercise 5 — Halogen Displacement and Oxoacids**

1. Chlorine oxidizes bromide ions to bromine:

$$Cl_2(aq) + 2\,Br^-(aq) \longrightarrow 2\,Cl^-(aq) + Br_2(aq)$$

Observations: the aqueous layer turns from colourless to orange/yellow (due to Br_2); after shaking with CCl_4, the denser organic layer (lower layer) becomes orange/red-brown, while the aqueous layer becomes paler or colourless, because Br_2 is much more soluble in CCl_4.

2. Oxoacids of chlorine:

Oxidation state	Formula	Name
+1	HOCl (HClO)	Chloric(I) acid (hypochlorous acid)
+3	$HClO_2$	Chloric(III) acid (chlorous acid)
+5	$HClO_3$	Chloric(V) acid (chloric acid)
+7	$HClO_4$	Chloric(VII) acid (perchloric acid)

3. Increasing acid strength: $HOCl < HClO_2 < HClO_3 < HClO_4$.

Increasing thermal stability: $HOCl < HClO_2 < HClO_3 < HClO_4$ (same order).

Explanation of acid strength: as the number of oxygen atoms bonded to chlorine increases, the conjugate base (ClO^-, ClO_2^-, ClO_3^-, ClO_4^-) has its negative charge delocalized over more oxygen atoms. The anion is therefore increasingly stabilized, so the acid dissociates more readily and is stronger. The electron-withdrawing oxygens also weaken the O–H bond, facilitating loss of H^+.

SOLUTION **Exercise 6 — Halide Ion Identification**

1. Tube A: bromide, Br^- — cream precipitate of AgBr, insoluble in dilute NH_3 (only sparingly soluble even in concentrated), and Test 3 gives brown fumes (Br_2) with violet/purple vapour traces; the recorded brown fumes confirm bromide. Tube B: chloride, Cl^- — white precipitate of AgCl, soluble in dilute NH_3; Test 3 gives dense white fumes of HCl only (chloride cannot reduce H_2SO_4). Tube C: iodide, I^- — yellow precipitate of AgI, insoluble even in concentrated NH_3; Test 3 gives white fumes (HI, SO_2) plus violet vapour (I_2).

2. Precipitation equations:

$$Ag^+(aq) + Br^-(aq) \longrightarrow AgBr(s) \quad \text{(cream)}$$

$$Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s) \quad \text{(white)}$$

$$Ag^+(aq) + I^-(aq) \longrightarrow AgI(s) \quad \text{(yellow)}$$

3. Tube A (NaBr) with conc. H_2SO_4: initially HBr forms (misty white fumes):

$$NaBr(s) + H_2SO_4(l) \longrightarrow NaHSO_4(s) + HBr(g)$$

But HBr is a reducing agent and reduces the sulfuric acid, so the brown fumes are bromine, Br_2, and SO_2 is also produced:

$$2\,HBr(g) + H_2SO_4(l) \longrightarrow Br_2(g) + SO_2(g) + 2\,H_2O(l)$$

(Any violet colouration would come from traces of I_2 impurity; the characteristic observation for bromide is the brown Br_2 fumes.)

Concentrated H_2SO_4 is both an acid and an oxidizing agent. HBr and HI are strong reducing agents, so they are oxidized by the hot acid to Br_2 and I_2 respectively (with SO_2, and for iodide also S and H_2S, as reduction products). Hence pure HBr or HI gas cannot be obtained this way — phosphoric(V) acid, H_3PO_4, which is non-oxidizing, is used instead.

SOLUTION **Exercise 7 — Group IV: Oxides and Chlorides**

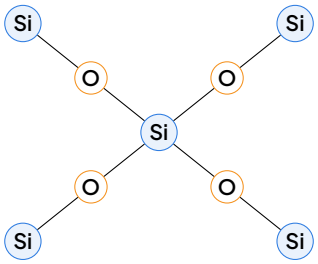
1. CO_2 is a simple molecular compound: carbon forms two double bonds, O = C = O, giving discrete linear molecules held together only by weak van der Waals forces. Little energy is needed to separate the molecules, so CO_2 is a gas at room temperature.

   SiO_2 is a giant covalent (macromolecular) structure: each Si is bonded tetrahedrally to four O atoms, and each O bridges two Si atoms, forming an extended 3-D network of strong Si–O single bonds (silicon does not form stable $p\pi$–$p\pi$ double bonds with oxygen because its 3p orbitals overlap poorly with oxygen's 2p orbitals). Melting re-

quires breaking strong covalent bonds, so SiO_2 is a solid of very high melting point ($\approx 1700\,^{\circ}C$).

O = C = O

discrete linear molecule



giant covalent network

CO_2: simple molecular; SiO_2: giant covalent lattice.

2. $SiCl_4$ hydrolyses violently:

$$SiCl_4(l) + 4\,H_2O(l) \longrightarrow Si(OH)_4(s) + 4\,HCl(aq) \quad (\text{or } SiO_2 \cdot 2\,H_2O)$$

CCl_4: no reaction with water at room temperature (inert).

Explanation: silicon (period 3) has vacant, energetically accessible $3d$ orbitals and a larger, more exposed atom, so a water molecule can donate a lone pair to Si to form an intermediate, allowing nucleophilic attack. Carbon (period 2) has no available d-orbitals, its maximum covalency is 4, and the small C atom is sterically shielded by the four Cl atoms; the activation energy for hydrolysis is therefore prohibitively high.

3. SnO_2 and PbO_2 are **amphoteric** (predominantly basic in character down the group): they react with both acids and alkalis.

With hot concentrated HCl:

$$SnO_2(s) + 4\,HCl(aq) \longrightarrow SnCl_4(aq) + 2\,H_2O(l)$$

With hot concentrated NaOH:

$$SnO_2(s) + 2\,NaOH(aq) + 2\,H_2O(l) \longrightarrow Na_2[Sn(OH)_6](aq)$$

(sodium hexahydroxostannate(IV), a stannate). The reaction with acid shows basic character; the reaction with alkali shows acidic character.

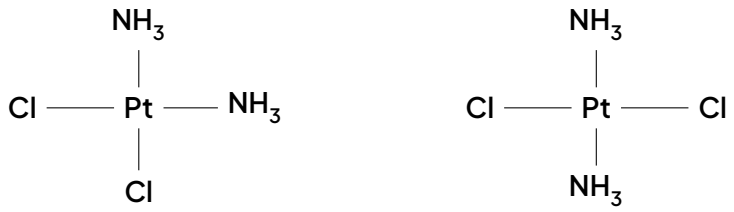
SOLUTION **Exercise 8 — Complexes: Naming, Shapes and Isomerism**

1. • $[Co(NH_3)_5Cl]Cl_2$: pentaamminechloridocobalt(III) chloride. Oxidation state of

Co: +3; coordination number: 6; shape: octahedral.

- $K_3[Fe(CN)_6]$: potassium hexacyanidoferrate(III). Oxidation state of Fe: +3; coordination number: 6; shape: octahedral.
- $[CuCl_4]^{2-}$: tetrachloridocuprate(II) ion. Oxidation state of Cu: +2; coordination number: 4; shape: tetrahedral (distorted).

2. $[Pt(NH_3)_2Cl_2]$ is square planar and exists as two geometric isomers:



cis isomer (cisplatin) trans isomer

cis- and trans- $[Pt(NH_3)_2Cl_2]$ (square planar).

In the cis isomer the two identical ligands are adjacent (90° apart); in the trans isomer they are opposite (180°). The cis isomer is cisplatin, the anti-cancer drug.

3. These are ionization isomers: they differ in which ion is inside the coordination sphere and which is the counter-ion. $[Cr(H_2O)_5Cl]SO_4 \cdot H_2O$ releases SO_4^{2-} in solution; $[Cr(H_2O)_5SO_4]Cl \cdot H_2O$ releases Cl^-.

   Test: dissolve each compound in water and divide into two portions.

   - Add $BaCl_2(aq)$ (or $Ba(NO_3)_2$): the first isomer gives a white precipitate of $BaSO_4$ ($Ba^{2+} + SO_4^{2-} \longrightarrow BaSO_4(s)$); the second gives no precipitate.
   - Add $AgNO_3(aq)$: the second isomer gives a white precipitate of AgCl ($Ag^+ + Cl^- \longrightarrow AgCl(s)$); the first gives no precipitate.

SOLUTION **Exercise 9 — GCE Structured Question: Halogen Chemistry (20 marks)**

(a) Trend in oxidizing power [6 marks]

Oxidizing power decreases down the group: $F_2 > Cl_2 > Br_2 > I_2$. [1]

Oxidizing ability corresponds to the ease of the process $X_2 + 2\,e^- \longrightarrow 2\,X^-$, measured by the standard electrode potential E°, which becomes less positive down the group (e.g. $F_2/F^- \approx +2.87$ V, $Cl_2/Cl^- \approx +1.36$ V, $Br_2/Br^- \approx +1.07$ V, $I_2/I^- \approx +0.54$ V). [2]

Down the group the atomic radius increases, so the incoming electron is added further from the nucleus and is more shielded; the attraction for the extra electron (effective

electron affinity in solution, including hydration) therefore decreases. [2]
Although the X–X bond dissociation energy also generally decreases down the group (which would favour reaction), the dominant factor is the decreasing tendency to attract and accommodate the extra electron, so oxidizing power decreases. [1]
(b) Displacement reactions [6 marks]
Chlorine + potassium bromide: aqueous layer turns orange/yellow; organic layer (lower layer with CCl_4) turns orange/red-brown. [1]

$$Cl_2(aq) + 2\,Br^-(aq) \longrightarrow 2\,Cl^-(aq) + Br_2(aq) \quad (1)$$

Chlorine + potassium iodide: aqueous layer turns brown (dark precipitate of I_2 if concentrated); organic layer turns violet/purple. [1]

$$Cl_2(aq) + 2\,I^-(aq) \longrightarrow 2\,Cl^-(aq) + I_2(aq) \quad (1)$$

Chlorine displaces Br_2 and I_2 because it is a stronger oxidizing agent (higher $E^\ominus$) and accepts electrons from Br^- and I^-. [1]
Fluorine is not used because it is too reactive: it oxidizes water itself, $2\,F_2 + 2\,H_2O \longrightarrow 4\,HF + O_2$, so it cannot exist in aqueous solution, and it is extremely hazardous to handle. [2]
(c) Concentrated H_2SO_4 + solid NaI, heated [8 marks]
Observations: misty white fumes (HI), violet/purple vapour (I_2) and a grey/black solid, smell of rotten eggs (H_2S), yellow solid (S), effervescence of a colourless gas that turns acidified dichromate paper green (SO_2). [2]
Products: sulfur-containing: SO_2, S, H_2S; iodine-containing: HI (little), I_2. [2]
Equations:

$$NaI(s) + H_2SO_4(l) \longrightarrow NaHSO_4(s) + HI(g) \quad (1)$$

$$2\,HI + H_2SO_4 \longrightarrow I_2 + SO_2 + 2\,H_2O$$

$$6\,HI + H_2SO_4 \longrightarrow 3\,I_2 + S + 4\,H_2O$$

$$8\,HI + H_2SO_4 \longrightarrow 4\,I_2 + H_2S + 4\,H_2O \quad (2)$$

Why unsuitable: HI is a strong reducing agent and is oxidized by the oxidizing acid H_2SO_4 to I_2; the gas is therefore contaminated and the yield of HI is very low. Pure HI is prepared using non-oxidizing phosphoric(V) acid, H_3PO_4. [2]

SOLUTION **Exercise 10 — GCE Essay Question: Group IV and Transition Metals (20 marks)**

(a)(i) Stability of +2 and +4 oxidation states [4 marks]
At the top of the group (C, Si) the +4 state is by far the more stable (e.g. CO_2, SiO_2 very stable; CO less so). [1]
Going down the group the +2 state becomes progressively more stable relative to +4: for

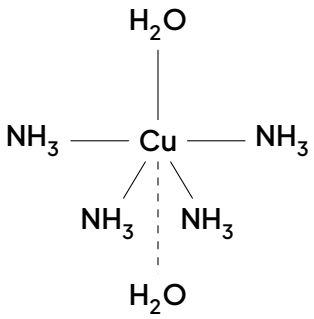
Ge the two are comparable; for Sn, Sn^{2+} is a mild reducing agent; for Pb, +2 is the dominant stable state and Pb(IV) compounds (e.g. PbO_2) are strong oxidizing agents. [2]
This is the **inert pair effect**: the ns^2 electron pair becomes increasingly reluctant to participate in bonding down the group, so only the two np electrons are used, giving the +2 state. [1]
(a)(ii) The dioxides [6 marks]

Oxide	Structure	Bonding	Acid/base nature
CO_2	Simple linear molecules	Covalent (O = C = O)	Acidic
SiO_2	Giant covalent 3-D lattice	Covalent (Si–O)	Acidic (weakly)
GeO_2	Giant/network, some ionic	Intermediate	Amphoteric (mainly acidic)
SnO_2	Giant ionic lattice	Mainly ionic	Amphoteric
PbO_2	Giant ionic lattice	Mainly ionic	Amphoteric (mainly basic)

Award: structure trend molecular → giant covalent → ionic [2]; bonding trend covalent → ionic as metallic character increases [2]; acid/base trend acidic → amphoteric with increasingly basic character down the group [2].
(b)(i) Variable oxidation states [4 marks]
In the first transition series the 3d and 4s orbitals are very close in energy, so both the 4s electrons and varying numbers of 3d electrons can be used in bonding/ion formation with little extra energy cost. [2]
The number of oxidation states increases from Sc to Mn, reaching a maximum at Mn (+2 to +7), because up to Mn all five 3d electrons plus the two 4s electrons can be removed/involved. Beyond Mn, the 3d electrons are held more tightly (increasing effective nuclear charge, paired d-electrons), so fewer are available and the maximum oxidation state decreases (Fe +6 max, commonly +2 / + 3; Zn only +2). [2]
(b)(ii) Colour [4 marks]
In an octahedral ligand field the five 3d orbitals split into a lower set (t_{2g}) and an upper set (e_g) separated by Δ_o. [1]
In ions with a partially filled d-subshell (d^1 to d^9), an electron can absorb visible light of energy $h\nu = \Delta_o$ and be promoted from t_{2g} to e_g (a d–d transition); the transmitted/reflected light is the complementary colour, so the complex appears coloured. [2]
Sc^{3+} (d^0) has no d-electron to excite, and Zn^{2+} (d^{10}) has a full subshell with no vacancy in e_g; no d–d transition is possible, so their complexes are colourless. [1]
(b)(iii) Catalysts [2 marks]
Homogeneous: Fe^{2+}/Fe^{3+} (or Mn^{2+}) catalysing the reaction of persulfate with iodide, $S_2O_8^{2-} + 2\,I^- \longrightarrow 2\,SO_4^{2-} + I_2$ — catalyst and reactants in the same (aqueous) phase. [1]
Heterogeneous: iron in the Haber process ($N_2 + 3\,H_2 \rightleftharpoons 2\,NH_3$), or V_2O_5 in the Contact process, or Ni in the hydrogenation of alkenes — catalyst in a different phase from the reactants. [1]

SOLUTION **Exercise 11 — GCE Data Analysis: Complexes and Spectrochemical Series (20 marks)**

(a)(i) Shape of $[Cu(NH_3)_4(H_2O)_2]^{2+}$ [3 marks]



Distorted octahedron: four NH_3 in a plane, two H_2O axial (elongated).

The geometry is octahedral but distorted (tetragonal distortion): it is not a regular octahedron because of the **Jahn–Teller effect** for the d^9 configuration of Cu^{2+} — the two axial bonds (to H_2O) are longer than the four equatorial bonds. [3]

(a)(ii) Colour change and CFT [4 marks]

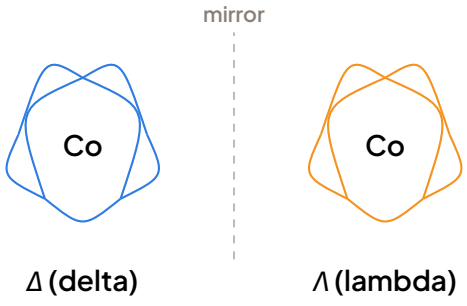
In the spectrochemical series $I^- < Br^- < Cl^- < H_2O < NH_3 < en < CN^-$, NH_3 is a stronger-field ligand than H_2O. [1]

Replacing H_2O by NH_3 therefore increases the splitting Δ_o. [1]

A larger Δ_o means the d–d transition absorbs light of higher energy (shorter wavelength) — absorption shifts from the orange/red region towards the yellow/green. [1]

The colour observed is the complement of the light absorbed, so the solution changes from pale blue to deep (intense) blue/violet-blue. [1]

(b)(i) Optical isomers of $[Co(en)_3]^{3+}$ [3 marks]



The two non-superimposable mirror images (enantiomers) of $[Co(en)_3]^{3+}$; each curved line represents one en ligand.

The two forms are non-superimposable mirror images, labelled Δ and Λ. [3]

(b)(ii) No geometric isomers [2 marks]

All three ligands are identical bidentate ligands, and each en must span two adjacent (cis) positions because its chain is too short to bridge trans positions. With all six coordination sites occupied by three identical symmetrical ligands, there is only one possible connectivity/arrangement; hence no cis/trans (geometric) isomerism is possible — only optical isomerism. [2]

(b)(iii) Polarimetry [2 marks]

A solution of each enantiomer is placed in turn in a polarimeter tube and plane-polarized light is passed through. One enantiomer rotates the plane of polarization clockwise (dextrorotatory, +) and the other by an equal angle anticlockwise (laevorotatory, -). Equal concentrations give equal magnitudes of rotation in opposite directions. [2]

(c)(i) Test to distinguish the ionization isomers [3 marks]

Dissolve each solid separately in distilled water and test the solution:

- Add $BaCl_2(aq)$ (acidified with dilute HCl): $[Co(NH_3)_5Br]SO_4$ gives a white precipitate of $BaSO_4$ (free sulfate ions in solution); $[Co(NH_3)_5SO_4]Br$ gives no precipitate. [1.5]
- Add $AgNO_3(aq)$: $[Co(NH_3)_5SO_4]Br$ gives a cream precipitate of AgBr (free bromide ions); $[Co(NH_3)_5Br]SO_4$ gives no precipitate. [1.5]

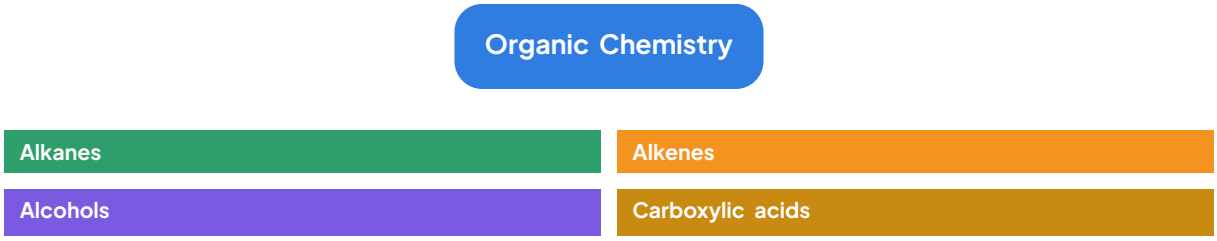
(The ligand inside the coordination sphere is not released into solution, so it does not react.)

(c)(ii) Ionic equations [3 marks]

$$Ba^{2+}(aq) + SO_4^{2-}(aq) \longrightarrow BaSO_4(s) \quad \textbf{(white)} \qquad (1.5)$$

$$Ag^{+}(aq) + Br^{-}(aq) \longrightarrow AgBr(s) \quad \textbf{(cream)} \qquad (1.5)$$

15 SUMMARY SHEET



Mind map of the main families of organic compounds studied in Upper Sixth.

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